SUPPLEMENT: DOPE: GENERATIVE COMPRESSION OF PREDICTIVE STRUCTURE

Anonymous
Affiliation withheld for review.

ABSTRACT

This supplement holds the method details, the release-gate specification and its conformance test, the data and selection protocol, the statistics, every five-seed contrast and level behind the main paper, the controls, the fidelity and empirical-privacy panels, the ablations, the baseline coverage, the per-lineage tables, and the extended related work. A blank retention cell is an unavailable fit or an auditor that is uninformative on one of the three sample seeds; it is never a zero. BeyondArena is a separate fit check and is never pooled with the PMLB panel.

A METHOD DETAILS

A.1 ARTIFACT CONTENTS AND SAMPLING

Equation (1) of the main paper is the conditional mean; the joint draw is separate. The profile reported in the main paper samples with the symbolic kernel stored in the artifact; the separate neural-joint sampler is unused. The stored dependence is one choice from the compile search: independence, a Chow–Liu tree, a triangular autoregression, a sparse Gaussian copula, a factor model, a truncated vine, or a two-component mixture, and the reported artifacts span several of these families.

Training screens the inputs by absolute rank correlation with the target and keeps at most twelve. The artifact stores a marginal and a transform for each feature, the selected dependence, the residual, a noise model, and a per-feature missing rate. A draw samples a latent vector from the stored dependence, maps each coordinate through the standard normal distribution function, inverts the marginal, and inverts the feature transform; the residual then supplies $\hat{y}$. When the training target is not binary, the stored noise is heteroscedastic: a linear fit of the log absolute residual on up to four screened features, with the scale clamped between the lower and upper deciles of those absolute residuals. The sampled target adds Gaussian noise of that scale and is clamped to the unit interval of the training transform. Each feature is then replaced, independently, by a missing mark with its stored probability. Before encoding, the compiler quantizes the marginals, the dependence, the residual, and the noise model. The charged byte count of a DOPE artifact is the encoded model file plus `projection.json`; a baseline is charged the files its sampler reads, as listed in Appendix C.3.

A.2 TRAINING FREEZE, COMPILE SEARCH, AND THE REPORTED PROFILE

Training maps the screened inputs through the width-16 $\tanh$ layer and runs AdamW on the regression mean-squared error, with gradient clipping at 5. The hidden weights and biases are then frozen, and the intercept, the linear skip, and the hidden readout are refit on the frozen features. Coefficients whose absolute value is below 10^{-5} are dropped from the skip.

The compile search records the proxy $0.45u + 0.25d + 0.30j$ and keeps the candidate with the smallest normalized gate shortfall, then the smallest artifact, then the lowest candidate id. It reads the fit rows only. A candidate record carries no release certificate, and no global DOPE family has been selected.

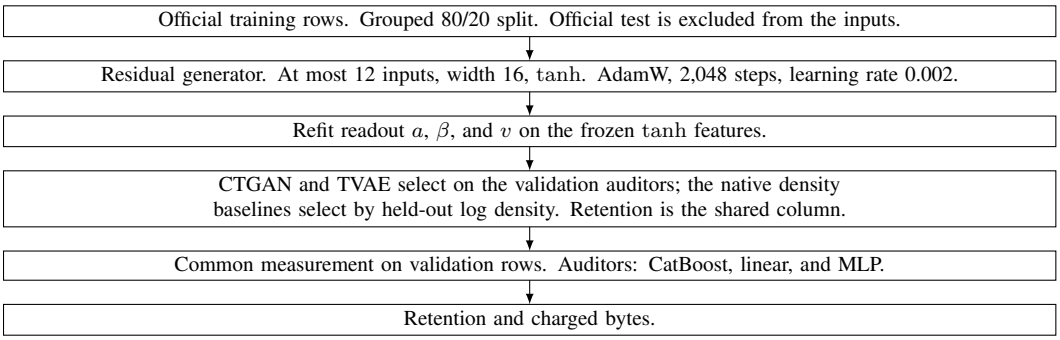


Figure 1: **Pipeline of the reported profile.** The official test is outside the inputs. No lineage denominator and no Holm family.

The reported research profile, `features12_steps2048`, is one of four bounded research profiles (twelve or twenty-four screened inputs, 512 or 2,048 steps) that were compared by validation retention at fit seed 11 and sample seeds 101, 211, and 307. Three committed publications under `research/benchmark/results/` record that comparison: a six-lineage discovery panel (`s3-matched-discovery-validation.json`), six disjoint confirmation lineages (`s3-matched-confirmation-validation.json`), and all hundred lineages under CatBoost, the linear auditor, and the MLP at n and $4n$ (`dope-s3-population-validation.json`). Each keeps all four profiles. The scope statements of the first two end with "production family unselected"; the third sets the field `global_family_selected` to false, and its report describes descriptive aggregates with no production family selection. None of the three states a selecting rule. The first committed record that shows the reported profile alone is a later CTGAN and TVAE comparison figure, and no contemporaneous record states the rule, the auditor, or the rows that singled it out. Earlier versions of this paper said the profile was chosen on the discovery panel and checked on the confirmation lineages, and called it pre-registered; no record supports either statement, so this version withdraws both. We classify every lineage and all three auditors as selection-exposed. The twelve discovery and confirmation lineages cover all four simulated families and three lineages outside them, so the source-family sensitivity cannot serve as a held-out-source result for this choice. The untuned trainer default, used for any other profile, sets the hidden width between 4 and 12 and trains for 96 steps; the main paper reports it as a separate arm. Other neural shapes (micro-TVAEs, TinyMAT, joint TVAE, a masked transformer, and a TabSyn candidate) are outside this measurement, and the direct-rank TabDDPM candidate is a research comparator with no claim on the 10,240-byte tier.

A.3 Dataset sketch and router vector

An index can store a fixed sketch of a training table, which a separate dataset-action router consumes; neither enters the generator reported in the main paper. `embed-dataset` min-max normalizes each training column onto the unit interval, leaves a missing entry missing, and never reads the official test. Each column becomes twelve content summaries: a missing rate, moments, quantiles, a unique-value share, a tail ratio, and the absolute association with the target. Columns are sorted by those summaries, variance first. A ten-number header records the task, the log row count, the log column count, and the shape of the target. Each of the twelve summaries is then reduced across columns to a six-number distribution (minimum, lower quartile, median, upper quartile, maximum, and mean), which is the 12×6 block. The sketch then keeps the first 64 columns and stores each column's twelve summaries, padding a shorter table with zeros, and absolute correlations among the kept columns collapse to one further six-number distribution. The width is therefore $10 + 12 \times 6 + 64 \times 12 + 6 = 856$. The router vector appends 24 candidate states, each a hidden state of width 128 plus 10 action predictions, to that sketch, for $856 + 24 \times (128 + 10) = 4{,}168$ dimensions. A neighbour found with the sketch is still fit with the reported profile and still charged in bytes.

B RELEASE GATE

B.1 MFS-v2, THE HISTORICAL FORMULA

MFS-v2 is the release score that historical receipts pin. With component $c_i \in [0, 1]$, weight $w_i > 0$, $\epsilon = 10^{-6}$, and $W = \sum_i w_i$,

$$\text{MFS-v2} = 100 \exp\left(\frac{1}{W} \sum_i w_i \ln(\epsilon + c_i) \right)$$

when the hard-gate map is non-empty, every hard gate passed, and every positive-weight component is finite and inside $[0, 1]$; otherwise the scalar is null. Its weights are utility transfer 0.30, driver fidelity 0.20, distribution fidelity 0.20, structure fidelity 0.15, coverage realism 0.10, and compactness 0.05. Privacy has soft weight zero, and a membership failure, an attribute failure, or a copied row is a gate that utility cannot offset. Compactness is $\text{clamp}(1 - \text{bytes}/10240, 0, 1)$, and an artifact past $10{,}240$ bytes fails the byte gate.

B.2 WHAT THE COMPONENTS MEASURE

Utility transfer is a one-sided lower bound on retention, clamped into $[0, 1]$; it is distinct from the median retention in the main paper. Driver fidelity is an equal mixture of feature-importance rank agreement and top-feature-set overlap between the real and the synthetic table. Distribution fidelity is half the minimum marginal fidelity, plus a quarter of the sliced-Wasserstein fidelity and a quarter of the maximum-mean-discrepancy fidelity. Structure fidelity is the minimum dependence fidelity. The driver term stored beside the panel is the same equal mixture, applied to absolute correlations with the target; it differs from the permutation-importance auditor in the v2 definition, and this paper reports no component value from it.

B.3 MFS-v3 AND REPRESENTATION CLOSENESS

MFS-v3 adds representation closeness to the geometric mean, as in equation (2) of the main paper. v3 utility transfer is the same one-sided bound on the Kumo large retention: the model is conditioned on the synthetic rows and scored on real holdout rows. Mitra and TabICL must be measured on the same mapped tables for the gate. Their retentions stay out of that bound, and a CatBoost retention cannot fill this component. Real fit rows, real holdout rows, and synthetic rows share one map, fit on the real fit rows only: numeric columns go to the open unit interval by an empirical mid-rank quantile, categorical levels are pre-encoded to fixed codes in that interval, and column names become HMAC digests. The map withholds the minimum and the maximum, and a second fit on the synthetic rows is a protocol failure.

Representation closeness asks whether Kumo large places the mapped synthetic table near the mapped real table. The distance is half the Euclidean length between the two unit-length vectors, so it lies in $[0, 1]$, and closeness is one minus that distance divided by the distance to a matched null, clamped to $[0, 1]$. The synthetic table must also sit closer than an independent-marginal table of the same size under the same map. The matched null is chosen among three independent-marginal tables drawn on the shared map, seeded by the first eight bytes, little-endian, of the SHA digest of the dataset, the method, the configuration, the sample seed, and the candidate index 0, 1, or 2; candidates are ordered by ordinary linear mean squared error on the mapped holdout, then by index, and the middle candidate is the matched null. Those losses are fixed before any encoder runs. Mitra and TabICL must show the same sign as Kumo large on the gap against the independent-marginal table; a sign flip, a zero match distance, or a missing vector makes the component null. Neither is averaged into representation closeness. The licensed encoder set is Kumo large (OpenMDW and Apache), Mitra (Apache), and TabICL (BSD); Foundation, TabPFN, TabDPT, the project's Hyperion, and target-aware blocks are refused. TabDPT is refused because its pretraining tables overlap this panel; whether the pretraining corpora of the three admitted encoders overlap PMLB is unknown, so their retentions are reported as transfer measurements rather than as independent evidence.

A recipient keeps the generator and a hash manifest: a salt identifier, the normalizer name, the encoder pin, one digest per column, a dtype code, and the byte length. The manifest carries no row,

Table 1: **MFS-v3 contract fields as implemented.** Each row is read from `production/kpi-contract-v3.json` by the scorer. No Holm family.

Scalar	$100 \exp(\sum_i w_i \log \max(s_i, \epsilon) / \sum_i w_i)$, clipped to $[0, 100]$
Weights	compactness = 0.05, coverage_realism = 0.05, distribution_fidelity = 0.15, driver_fidelity = 0.15, representation_closeness = 0.15, structure_fidelity = 0.1, utility_transfer = 0.35
Epsilon	1×10^{-6}
Bytes	$b \in \mathbb{Z},\ 0 < b \leq 10240$
Prerequisites	Real-vs-real control; required sizes; artifact sampling; locked metric implementations.
Matched null	Median linear MSE of three independent-marginal candidates; candidate-index tie break.
Near-copy floor	$q = 0.01$ of nearest-other distinct real-row squared distances; linear interpolation.

Table 2: **Invented applications of the bounded MFS-v3 formula.** Arithmetic illustrations only. No Holm family.

Invented input	Scalar
All seven components equal to one	100
Compactness component set to zero	50.12
Byte gate failed	null

header, category string, extremum, or encoder weight. A public sidecar that echoes a source string or an extremum fails the cleartext gate, and an exact copy of a real row fails its own gate. The near-copy floor is the 0.01 quantile, with linear interpolation, of squared Euclidean distances from each distinct real fit row to the nearest other distinct real fit row on the shared map; a synthetic row below that floor, an undefined floor, or a charge above $10{,}240$ bytes makes the scalar null.

B.4 Executable conformance

The release score uses the contract's weighted geometric mean with a component floor, clipped to $[0, 100]$; the components lie in $[0, 1]$ and a failed gate yields a null scalar. The matched null is the median linear-MSE candidate among three deterministic independent-marginal tables. This is a median selector; a learnability band is outside this implementation. The near-copy floor is the declared quantile of nearest-other squared distances among distinct real fit rows on the shared map, with linear interpolation, and undefined floors fail the gate. The scorer tests in `research/benchmark/tests/test_mfs_v3.py` verify exactly 100 for a perfect profile, the byte boundary, and each prerequisite; the panel tests verify median null selection and the quantile floor. Table 1 lists the implemented fields, and Table 2 applies the bounded formula to invented components.

B.5 Clean-license transfer on the density block

The density block was also scored by Kumo large, Mitra, and TabICL on the shared map, with the three sample seeds of the main panel. These three are pretrained in-context models: the synthetic rows replace the real rows as their context, and no model is trained or fine-tuned on them, so their rows measure context substitution, not a newly trained foundation model. These auditors are separate from the CatBoost, linear, and MLP rows, and their Holm adjustment is one family per auditor of size three (that auditor against the three density comparators). Kumo large averages 16 inverse-transformed quantile forecasts. The receipt scores 0 of 388 method–lineage cells (97 lineages $\times$ 4 methods), so the scalar is null. Cleartext is unscanned on 388 method–lineage cells.

Table 3: **Kumo large, Mitra, and TabICL on the density block at $4n$.** Each cell is a retention; the denominator is the paired informative lineages in that row. Holm family size is three per auditor: that auditor's DOPE against the Gaussian copula, Chow–Liu, and independent marginals. W/T/L counts lineages; a tie is a difference of zero.

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
Kumo large	Gaussian copula	97	**0.863** [0.809, 0.880]	0.747 [0.668, 0.778]	0.059 [0.0098, 0.116]	62/0/35	5.70×10^{-4}
Kumo large	Chow–Liu	97	**0.863** [0.809, 0.880]	0.562 [0.501, 0.629]	0.247 [0.195, 0.306]	85/0/12	1.34×10^{-11}
Kumo large	Indep. marginals	97	**0.863** [0.809, 0.880]	-0.000819 [-0.00189, 7.58e-05]	0.874 [0.834, 0.888]	94/0/3	7.46×10^{-17}
Mitra	Gaussian copula	97	**0.858** [0.803, 0.887]	0.713 [0.585, 0.771]	0.066 [0.023, 0.135]	65/0/32	1.34×10^{-4}
Mitra	Chow–Liu	97	**0.858** [0.803, 0.887]	0.583 [0.527, 0.643]	0.242 [0.182, 0.293]	85/0/12	2.71×10^{-11}
Mitra	Indep. marginals	97	**0.858** [0.803, 0.887]	-0.00907 [-0.014, -0.00377]	0.890 [0.840, 0.909]	94/0/3	6.01×10^{-17}
TabICL	Gaussian copula	97	**0.871** [0.835, 0.893]	0.754 [0.672, 0.787]	0.079 [0.00947, 0.116]	62/0/35	1.61×10^{-4}
TabICL	Chow–Liu	97	**0.871** [0.835, 0.893]	0.587 [0.484, 0.650]	0.267 [0.200, 0.302]	87/0/10	7.12×10^{-12}
TabICL	Indep. marginals	97	**0.871** [0.835, 0.893]	-0.00446 [-0.012, 0.0012]	0.891 [0.846, 0.919]	92/0/5	5.83×10^{-17}

Holm family: 3 tests per auditor; DOPE versus Gaussian copula, Chow–Liu, and independent marginals.

Table 4: **Pinned representation vectors.** Each row names the activation that the density-panel receipt reads as that auditor's table vector. No Holm family.

Auditor	Pinned vector
Kumo large	`row_project_output_mean_labeled_rows`
Mitra	`final_layer_norm_input_mean_self_query`
TabICL	`tf_icl_output_mean_self_query`

C Data, selection, cost, and availability

C.1 Data lock and partitions

The data lock admits 104 MIT-licensed entries and prepares 100. 4 lineages have no split object because the same source-row digest appears in two partitions (`check_nonoverlap` raises on that overlap), and another 3,755 catalog entries lack a redistribution grant and are excluded. The prepared cut is a grouped $80/20$ split of the official training rows, seed 1729. Worker directories hold `train.csv` and `validation.csv` only; the official test is held elsewhere and its contents are unread. The fit view has 72 to 800 rows; sources above the quota use a hybrid coreset that keeps the observed feature and target range. Of these prepared lineages, 20 names begin with feynman, which the PMLB catalog describes as synthetic physics, 14 begin with strogatz, which that catalog describes as simulated dynamics, 25 contain the Friedman-style fri marker, 6 contain bng and are Bayesian-network generators, and 35 are the remainder. This paper does not re-verify which of the remainder are observational measurements. Distinct lineage identifiers are not a claim of independent source families. The legacy lineage-bootstrap intervals resample lineages independently. The printed family-cluster intervals use a two-stage bootstrap. Lineages in the four simulated name families (feynman, strogatz, fri, and bng) form one cluster per family, and every other lineage is a singleton cluster. Each draw resamples clusters with replacement, then resamples lineages with replacement inside each drawn cluster. A verified parent map is absent from the ledger, so those intervals put remainder lineages that share a parent in separate clusters, including the cpu_small lineages. Wilcoxon and TOST retain independent-lineage assumptions.

The fit view is not the full PMLB table. Frozen catalog metadata draws each official training and test file by quota stratification on the target distribution and high-variance feature quantile bins, aiming at 1000 training rows and 1000 test rows, with at most 2000 rows in total and at least 101 rows. A source smaller than that quota keeps every row and still reserves a test share of at least 0.2. Of the 100 prepared lineages, 60 were already inside the quota. The other lineages use a hybrid coreset that keeps the observed feature and target range; the bin counts are 1 lineage at 11 bins, and 39 lineages at 50 bins. The generator then groups identical official-training rows and assigns those groups with seed 1729 so that about 80% of the rows become the fit view and the rest become the validation holdout. Duplicate rows are not split across that cut. On these lineages the fit view has 72 to 800 rows. The bound keeps every lineage on one worker view and leaves the official test sealed. It is not a second draw at scoring time.

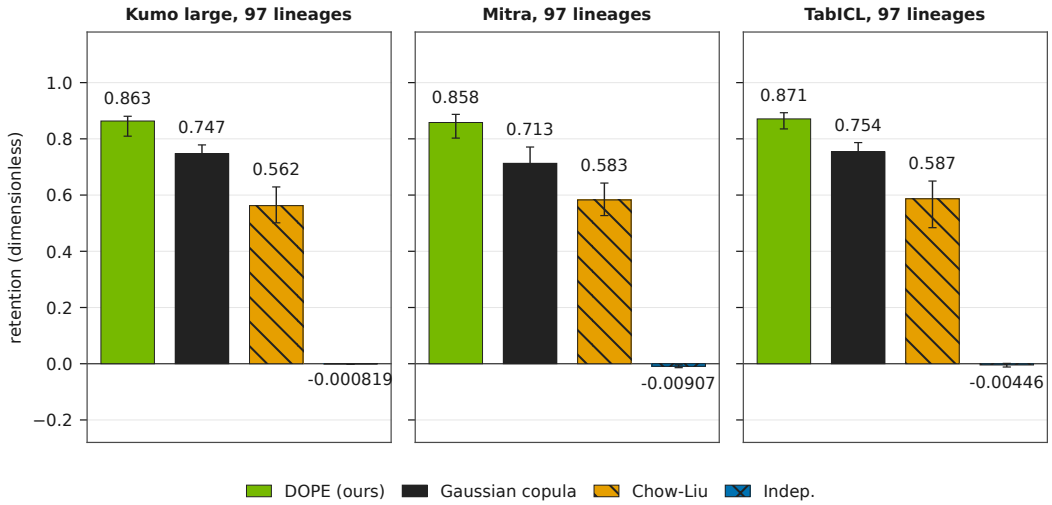


Figure 2: **Kumo large, Mitra, and TabICL retention at $4n$.** Each bar is the lineage median and its lineage-bootstrap interval; the denominator is the paired informative lineages in that panel. Holm family size is three per auditor: that auditor's DOPE against the Gaussian copula, Chow–Liu, and independent marginals.

Table 5: **Test-split provenance.** File-name counts only. No retention denominator and no Holm family.

Item	Record
Catalog SHA-256	`ab9fda8d2dea46067b70a42812e9d3c1d9dc6ba025780100df34377e81aa1120`
Eligible / prepared / overlap exclusions	104 / 100 / 4
Unknown-rights catalog entries held out	3755
Split	`official_test_grouped_training_80_20`, seed 1729
Overlap lineage ids	`590210721973399a`, `9fe58a656fdb458a`, `d06a584f07d0acd5`, `d0ba629ed07b5d63`
Final evaluation authorized	false
S3 worker directories / `train.csv` / `validation.csv`	100 / 100 / 100
`test.csv` under S3 workers	0
BeyondArena workers / worker `test.csv`	5 / 0
Evaluator `test.csv` files, contents not read	5

C.2 Selection rule for each method

Table 6 states what chose each configuration and which rows it read. Only DOPE's compile search and the density baselines' fixed or density-selected configurations avoid the scoring auditors; CTGAN and TVAE select on the same validation rows and the same linear and MLP auditors that score retention, and DOPE's research profile was compared with three alternatives on the validation rows of every evaluated lineage under all three auditors (Appendix A.2). The sealed test run removes all of these overlaps for its estimate. TabSyn's five-seed panel has two incomplete parts: six sample cells of one seed-23 lineage reached the sample time cap and stay unscored (final, without a retry), and one other lineage has no stored seed-11 samples. Every lineage keeps at least four complete fit seeds, so all enter the five-seed comparison.

C.3 Compute ledger and budget protocol

Table 7 lists fit time, peak memory, GPU models, GPU hours, and charged bytes for receipts that store a fit timer. Peak RAM is resident bytes in mebibytes, peak VRAM is `peak_gpu_used_mib`, and GPU hours sum the walls on which the receipt recorded a GPU process. No receipt stores a separate CPU timer. Sample-timer medians (maxima) are 186.7 (197.0) for CTGAN, 186.1 (196.6) for TVAE, and 342.4 (511.7) for Forest-Flow. Gaussian copula, Chow–Liu, and independent-marginal attempts store one unsplit `wall_seconds`, with medians 1.71, 0.168, and 0.162 and maxima 39.9, 1.29, and 0.429. The logged replay medians 4.56 and 14.4 are replay walls, distinct from the scored runs. Forest-Flow medians use the 12 native-grid operation walls; 953 sums `fit_seconds` and 8,801 sums receipt operations. Byte charges count the files a sampler reads: the DOPE model plus `projection.json`; every TabSyn file except intermediate

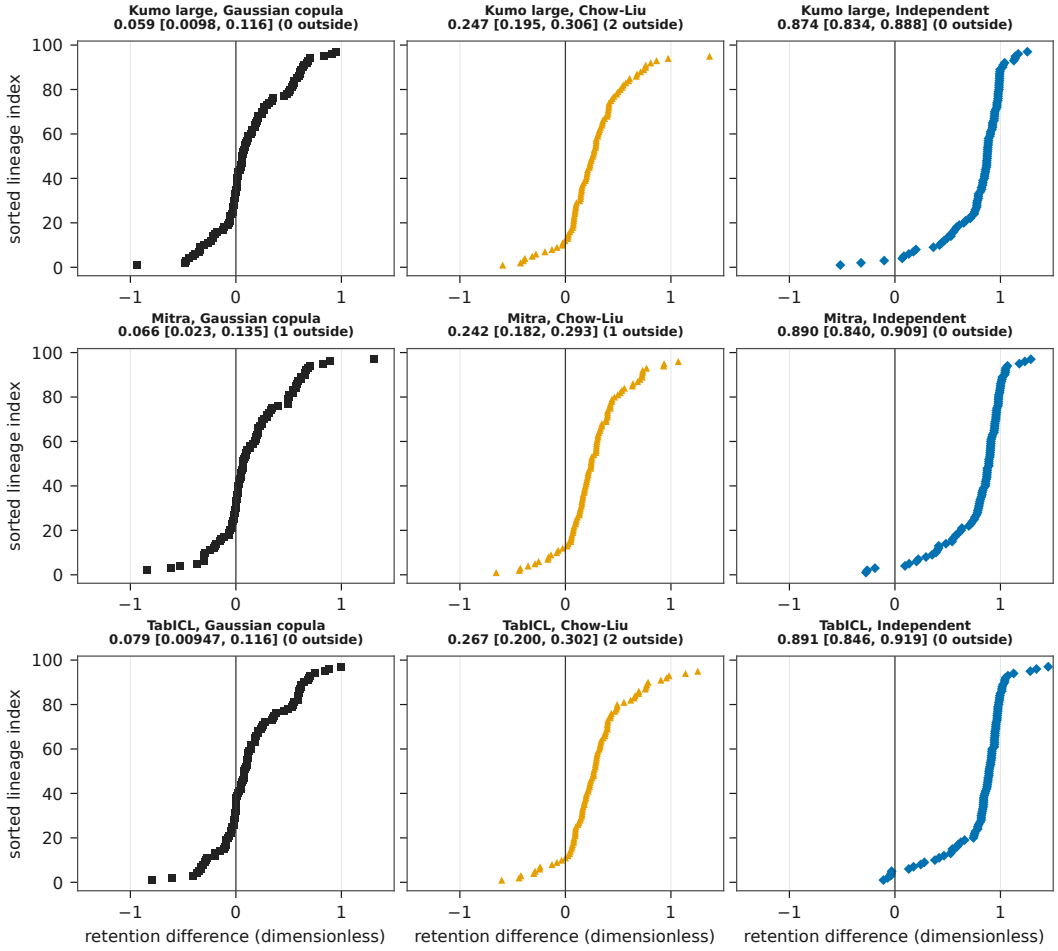


Figure 3: **Paired retention differences for Kumo large, Mitra, and TabICL, sorted by lineage.** The denominator is the paired informative lineages in that panel. Holm family size is three per auditor: that auditor's DOPE against the Gaussian copula, Chow–Liu, and independent marginals.

diffusion-epoch checkpoints, which the sampler never reads; and the stored per-lineage charge for every other baseline.

The frozen Forest-Flow budget keeps the 600-second fit cap and every timed-out attempt receipt; a timeout cell has missing quality metrics and counts as no DOPE win. The October 8 disposition addendum was written after 75 observed timeouts and is therefore no retroactive pre-registration. predeclare_v2 instead fixes a common budget (`B1`: 1,800 seconds per fit on one 16-core slot and one GPU) for every method and every lineage, applied non-selectively; `docs/whitepaper/BASELINE_PROTOCOL.md` links the frozen checkpoint and every disposition.

C.4 Public data and artifact availability

A credential-free path to the source data is

```
python3 -m research.benchmark.fetch_pmlb --list
python3 -m research.benchmark.fetch_pmlb DATASET_ID OUTPUT_DIRECTORY
```

It authenticates PMLB data against an upstream Git LFS SHA pin at a fixed public revision. The public manifest lists every prepared lineage's PMLB identifier and the historical train, validation, and test split hashes from the committed data lock. The historical staging used a hybrid coreset and

Table 6: **Selection rule for each method.** The scored column is retention on the validation cut. No Holm family.

Method	Criterion	Data used	Budget
DOPE	Profile compared with three alternatives by validation retention at fit seed 11; the deciding rule has no contemporaneous record. Compile search keeps the smallest gate shortfall, then the smallest artifact.	Profile comparison: validation cut of every lineage, all three auditors. Compile search reads fit rows only.	At most twelve inputs, width 16, 2,048 steps, five fit seeds.
DOPE untuned default	No selection.	Fit rows only.	Trainer default, five fit seeds.
Gaussian copula	Held-out mean log density. Copulas 0.14.1 Gaussian multivariate model.	Validation cut, distinct from the retention auditors.	One library configuration.
CTGAN	Mean of linear and MLP training-mean skill.	The validation rows and the two auditors used at scoring.	Author implementation; stopped runs omitted.
TVAE	Same skill as CTGAN.	Same rows and auditors as CTGAN.	Author implementation; stopped runs omitted.
Chow–Liu	Native: held-out mean log density over a fixed trial set. Default: no search.	Native: validation cut. Default: fit view, eight bins, Laplace smoothing 1.0. Written for this comparison.	Fixed trial set.
Independent marginals	Native: held-out mean log density over a fixed trial set. Default: no search.	Native: validation cut. Default: fit-view marginals. Written for this comparison.	Fixed trial set.
ARF	Author default; separately, the native objective.	Author default reads no validation rows; the native arm reads the validation cut.	Five fit seeds.
TabSyn	No selection.	Author architecture with a scaled schedule.	200 VAE and at most 1,000 diffusion epochs (the authors' early stop can end a fit sooner), five fit seeds from 11 to 71; six seed-23 cells unscored, one lineage without seed-11 samples.
Forest-Flow	Author regression objective on a frozen grid.	6 confirmation lineages.	Fit cap 600 seconds.

Table 7: **Fit time, memory, GPU, and charged bytes where a fit timer was stored.** The denominator is the receipts that stored a timer. No Holm family.

Method	Fit s	RAM / VRAM	Hardware	GPU h	Bytes
DOPE	7.25(13.5)	920.7/ 1067	one NVIDIA GeForce RTX 4090 and two NVIDIA GeForce RTX 3090 GPUs	0.193	1,895.5(758–22,233)
CTGAN	213.5(268.1)	1557.8/ 1101	NVIDIA GeForce RTX 4090	2.52	741,834(528,075–1,428,695)
TVAE	214.2(253.2)	1252.9/ 1069	NVIDIA GeForce RTX 4090	2.45	317,141.5(157,578–741,659)
Forest-Flow	185.9(380.6)	6112.0/ 3181	NVIDIA GeForce RTX 4090	2.44	325,848,913.5(115,202,253–1,085,155,062)

target-stratified selection whose exact row indices were never captured, so the public fetch reproduces the source datasets and the split identities but cannot rebuild the exact historical partitions; publishing the fit and validation partitions themselves (PMLB is MIT-licensed) closes that gap and is scheduled with the artifact release. The originally scored model files and synthetic rows have no published bundle yet; the committed validation ledgers, the original metadata, and the compressed numeric logs, authenticated by the input pins and rebuilt by `just paper`, reproduce every reduction and figure.

C.5 Reproducibility and hashes

Regenerate every `generated/` file and the figure PDFs with `just paper` from the commit that contains these files. The same recipe writes the main PDF, the anonymous main PDF, the supplement PDF, and the anonymous supplement PDF. That recipe reads the committed validation ledgers, original fit metadata, and compressed numeric replay logs. The replay cost, loss curves, and BeyondArena extracts have file and field mappings in `generated/original-field-map.json`. Those original metadata snapshots were captured after the historical runs; they leave historical custody and the scored artifacts in place. Pinned LICENSE extracts are reread when their local store is available. Official test files are sealed until the single pre-registered run. The catalog hash is ab9fda8d2dea46067b70a42812e9d3c1 d9dc6ba025780100df34377e81aa1120. The grouped training split uses seed 1729. The historical DOPE and legacy comparison fits use fit seed 11 and sample seeds 101, 211, and 307; the five-seed DOPE refit and its pins are in

research/benchmark/review_fixes/predeclare_v2.json. Retained panels document the additional ARF and Forest-Flow fit seeds and the measured TabSyn and TabDDPM subsets separately. python3 research/benchmark/verify_paper_numbers.py exits nonzero when a displayed macro disagrees with those ledgers. The remaining baselines, the shadow-model and artifact attacks, and the full ablation grid enter the tables only after their runs close. The fidelity table is a separate empirical ledger on the grouped validation split. Empirical resemblance is no differential-privacy guarantee and no HIPAA de-identification.

PMLB is MIT. Every dataset entry in the data lock records SPDX MIT and the note that the source license is MIT (PMLB). CTGAN and TVAE share one pinned LICENSE file: Business Source License 1.1, licensor DataCebo, Inc., licensed work CTGAN. The Gaussian copula uses the pinned Copulas LICENSE: Business Source License 1.1, licensor DataCebo, Inc., licensed work Copulas. TabSyn is Apache License 2.0. TabDDPM is MIT License, Copyright (c) 2022 Akim Kotelnikov. Forest-Diffusion is MIT License, Copyright (c) 2023 Samsung Electronics Co., Ltd. All Rights Reserved. ARF is MIT License, Copyright (c) 2023 Kristin Blesch, Marvin Wright. Chow–Liu and the independent marginals are implementations written for this comparison, under this repository's license: MIT License, Copyright (c) 2026 Dope contributors. The SDV repository LICENSE is absent from the pinned snapshot, so this paragraph leaves that license unrecorded.

The rows below are the SHA-256 digests of each ledger and lock the paper reads, plus the commit and the file hash of the harness that measured the grouped-validation fidelity ledger. Regenerating the macros never rewrites a ledger hash.

Table 8: Ledger and lock digests. No Holm family.

Ledger or lock	SHA-256
Code commit of the measured harness	17f3294d139e19f0ec831605f8750368eebc30bb
Harness SHA-256	a952062c0f83805f6442a440a5ae15293a843349 d423eaa106a9c56a20d14c0f
research/benchmark/results/paper-original-metadata-v1/index.json	6d73d5f7b37849526a98a451b41a49e045ef21e9 11058d920940c5ea989f7fd9
docs/whitepaper/generated/original-field-map.json	5b164d434414d3389f639d3645f40b11a999b32c 56daa2cfe36d25b2205082b6
docs/whitepaper/scripts/paper_receipts.py	a2d6d45e3ff06a38ed61ab029fadfd14529978bc fc89618aa9ceb3fbe57238ee
research/benchmark/expanded_validation_metrics.py	a952062c0f83805f6442a440a5ae15293a843349 d423eaa106a9c56a20d14c0f
research/benchmark/methods.lock.json	899c355cb55769710ed369fea6b082589b318588 60fb74929bf4190de28aca33
research/benchmark/results/expanded-validation-diagnostics.json	cd20ac0d07a1dcab7c5e46db195adb463172b517 4387bb71a054b2e5f340beeb
research/benchmark/results/s3-coreset-procedure.json	73251e43cf7838687427569931bf6127f42e9323 74af33137c57fd8fe8544704
research/benchmark/results/s3-lineage-record.json	bbd0852c49f595315104261efa4eaa6f50db257e 323b8c002aac6737a2dbe7d7
research/benchmark/results/density-matched-population-validation.json	5264b88a40ad5efb21d9b119789caeb13e71e911 a17f1d8b57f1d8e8ac31732a
research/benchmark/results/sdv-matched-population-validation.json	4dc367eb78159a0386882d23ae1f99b0aa2cc9f4 25d1aae499b6f3453222a44e
research/benchmark/results/s3-matched-forest-confirmation-validation.json	97a25be902954cad16c5b5802d0e0ba468dea104 221efa2465434c94e6774a1f
research/benchmark/results/s3-data.lock.json	857b61324d0a5b7dbbb98e0c147fcc79c5bb8616 e7ca61afd6785469c17f038f
research/benchmark/results/arf-native-closure-watch-v1.receipt.json	7e8eeb1ba94ca747367e98d5a016d2ed3ed4c40c 5a98ca76d7a5fe27ce9c173c
research/benchmark/results/beyondarena-s3-inventory.json	504166764631dbdf5e02d116334af5cc587731fc f3efe130d62240e6dd512d64
docs/whitepaper/generated/fit-trace.json	d8c31b9e4eb59cb1a087cf5dcdde4a91a73eca00 f3704fbff647a7da6ce65522
docs/whitepaper/generated/loss-curves.json	0124915497c98e560ef99c1e5ecf7ad876f91744 39ff9c763cf1cf906b4aebe1
docs/whitepaper/generated/replay-cost.json	faf77e7ce1ed1d22127701bf68bbfe2f8d183f89 cbd3475dd80094888c36f086
docs/whitepaper/generated/provenance.json	dea64ea61eaacac0e287298bba3ba76aa7900819 aa00feb3a5d66cdcb4fb8a17
docs/whitepaper/generated/beyond-fit.json	a1447f908bc9c51876d345597039607c734b507b f85aa8e08db0639e80a1f036

Table 9: **Informativeness threshold and auditor counts.** The denominator for each median is the informative count beside it. No Holm family.

Threshold	DOPE	Gaussian	Chow–Liu	Indep.
0	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.001	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.01	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.05	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.1	0.942 (96)	0.668 (98)	0.584 (98)	-0.012 (98)

Threshold	CatBoost	Linear	MLP
0.001	97	93	75
0.01	97	92	75
0.05	97	90	73

D STATISTICS

D.1 RETENTION ALGEBRA AND VARIANCE

With r the retention, the ratio rearranges to

$$L_{\text{synthetic}} = L_{\text{real}} + (1 - r)\big(L_{\text{null}} - L_{\text{real}}\big).$$

Invented losses 1, 0.01, and retention 0.94 give a synthetic-trained loss of 0.0694, which is 6.94 times the real-trained mean squared error. The CatBoost median in this paper is not this example. Let $D = L_{\text{null}} - L_{\text{real}}$, so $R = (L_{\text{null}} - L_{\text{synthetic}})/D$ when $D \neq 0$. The partial derivatives are

$$\frac{\partial R}{\partial L_{\text{null}}} = \frac{L_{\text{synthetic}} - L_{\text{real}}}{D^2}, \qquad \frac{\partial R}{\partial L_{\text{synthetic}}} = -\frac{1}{D}, \qquad \frac{\partial R}{\partial L_{\text{real}}} = \frac{R}{D}.$$

If the three losses have covariance Σ, the delta-method variance of one lineage's retention is $\nabla R^{\top} \Sigma \nabla R$, which diverges as D approaches zero. The informativeness rule drops a lineage whose real improvement is below 0.01 times the absolute null loss as a stability cut; the tables report bootstrap intervals rather than this standard error.

D.2 INFORMATIVENESS THRESHOLD

The primary threshold is 0.01 times the absolute null loss. Table 9 lists the CatBoost median by threshold and the informative counts; each column uses its own informative count, so the Chow–Liu column differs from the density-block paired median of 0.586 in Table 23. Thresholds 0, 0.001, and 0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942. Reading the threshold counts from top to bottom, CatBoost is 97, 97, 97; linear is 93, 92, 90; and the MLP is 75, 75, 73.

D.3 BOOTSTRAP, FAMILIES, AND EQUIVALENCE

The v2 analyses follow `research/benchmark/review_fixes/predeclare_v2.json`, committed before any five-seed result existed. A fit value is the median of its three sample-seed retentions; a lineage value is the median of its complete fits (at least three for a five-seed arm); the panel level is the median of lineage values. Paired contrasts use the Hodges–Lehmann pseudo-median of lineage differences, which is the location the two-sided Wilcoxon signed-rank test addresses, and report the median difference beside it. Intervals come from a nested percentile bootstrap with $10{,}000$ draws: the four simulated name prefixes are clusters and every other lineage is its own cluster; a draw resamples clusters, then lineages inside each drawn cluster, then fit seeds inside each drawn lineage, separately for each arm. Holm families are declared per size: the primary family of 12 tests, the density family of 9, a deep-generator family of 12, an other-generator family of 9, and a control family of 6; a member without data keeps its slot. The fit-seed-11 tables of supplement E keep the first version's per-size families (nine density, six neural, six ARF, and three Forest-Flow tests), which each caption names. Equivalence requires the 90% nested interval of the Hodges–Lehmann difference to lie inside ± 0.02; a 20% trimmed-mean (Yuen) test is a sensitivity. The v1 contract tested the mean paired difference with two one-sided t

tests at ± 0.02 and unadjusted $\alpha = 0.05$; paired retention differences have heavy tails when the denominator is small, so that estimand is kept only as a continuity column below. The Wilcoxon, Yuen, and v1 mean tests treat lineages as independent; only the bootstrap intervals resample the family clusters, and every verdict reads those intervals.

A Friedman test and a Nemenyi critical difference compare the four density methods on their common complete-informative set, with rank 1 for the higher retention; on CatBoost the Holm sign-test values are 2.15×10^{-15}, 6.72×10^{-24}, and 4.95×10^{-27}. A Bayesian signed-rank procedure would be an alternative to these frequentist tests and is not computed here.

E FIVE-SEED PANEL

Table 10 gives the retention level of every arm, auditor, and size, and Table 11 every DOPE-minus-comparator contrast with its family-adjusted p-value and both equivalence rules. Table 12 gives the seed-exchangeability diagnostics: how often seed 11 is the best of five, the estimate from seeds 23 to 71 alone, and the binary-by-seed diagnostic on the lineages common to every cell. The historical-binary refit reproduces the samples of the historical seed-11 population byte for byte on 86 of 98 lineages. The later binary is the one an earlier four-seed run used; it writes the same seed-11 artifacts as the historical binary on 100 of 100 lineages, and its samples match the refit at seeds 23 to 71 on 400 of 400 fits. Every retention cell in these tables, for DOPE and for every baseline including Forest-Flow, was scored again from its stored synthetic sample file by one pinned evaluator with auditor seed 1729; a cell whose sample file was missing stays out. Each statistic draws its own bootstrap stream, so two rows built from identical samples, such as the seeds-23-to-71 sensitivity and the later-binary four-seed arm, can differ in the third decimal of an interval. The fit-seed-11 privacy and detector panels in Appendix G come from separate earlier runs, which record the Forest-Flow cells as unavailable. The historical density, SDV, Forest-Flow, and fit-seed-11 DOPE receipts had used the sample seed as the auditor seed, while the ARF, TabSyn, and control receipts had used 1729; the uniform re-score removes that difference, so the five-seed tables and the historical block tables below differ slightly for this reason as well as for the extra fit seeds. TabSyn's later fit seeds change only TabSyn's own measurements; ARF's cells and Hodges–Lehmann differences are unchanged. ARF's Holm-adjusted p-values still differ from the version that scored TabSyn at fit seed 11 only, because TabSyn's contrasts share ARF's pre-declared Holm family and a Holm adjustment depends on every p-value in its family.

Table 10: **Retention level of every arm.** Median of lineage values; a five-seed arm needs three complete fits per lineage. No Holm family.

Arm	Size	Auditor	Lineages	Fits	Median	95% nested interval
DOPE (ours)	n	CatBoost	99	495	0.824	[0.725, 0.942]
DOPE (ours)	n	Linear	94	470	0.986	[0.964, 0.993]
DOPE (ours)	n	MLP	88	440	0.969	[0.895, 0.992]
DOPE (ours)	$4n$	CatBoost	99	495	0.898	[0.804, 0.972]
DOPE (ours)	$4n$	Linear	94	470	0.991	[0.972, 0.997]
DOPE (ours)	$4n$	MLP	88	440	1.133	[1.029, 1.246]
DOPE, untuned default	n	CatBoost	99	495	0.636	[0.387, 0.825]
DOPE, untuned default	n	Linear	94	470	0.986	[0.974, 0.992]
DOPE, untuned default	n	MLP	88	440	0.887	[0.801, 0.957]
DOPE, untuned default	$4n$	CatBoost	99	495	0.704	[0.424, 0.877]
DOPE, untuned default	$4n$	Linear	94	470	0.993	[0.983, 0.998]
DOPE, untuned default	$4n$	MLP	88	440	0.991	[0.960, 1.058]
TabSyn (scaled schedule)	n	CatBoost	99	493	0.863	[0.751, 0.928]
TabSyn (scaled schedule)	n	Linear	94	469	0.932	[0.791, 0.983]
TabSyn (scaled schedule)	n	MLP	88	439	0.884	[0.734, 0.976]
TabSyn (scaled schedule)	$4n$	CatBoost	99	493	0.924	[0.814, 0.965]
TabSyn (scaled schedule)	$4n$	Linear	94	469	0.952	[0.827, 0.987]
TabSyn (scaled schedule)	$4n$	MLP	88	439	1.013	[0.954, 1.142]
ARF (author default)	n	CatBoost	99	495	0.718	[0.570, 0.844]
ARF (author default)	n	Linear	94	470	0.903	[0.794, 0.965]
ARF (author default)	n	MLP	88	440	0.828	[0.631, 0.886]
ARF (author default)	$4n$	CatBoost	99	495	0.822	[0.689, 0.894]
ARF (author default)	$4n$	Linear	94	470	0.938	[0.846, 0.982]
ARF (author default)	$4n$	MLP	88	440	0.991	[0.886, 1.090]
ARF (native selected)	n	CatBoost	99	495	0.671	[0.551, 0.827]
ARF (native selected)	n	Linear	94	470	0.894	[0.742, 0.966]
ARF (native selected)	n	MLP	88	440	0.805	[0.583, 0.897]
ARF (native selected)	$4n$	CatBoost	99	495	0.804	[0.647, 0.885]
ARF (native selected)	$4n$	Linear	94	470	0.920	[0.828, 0.977]

Arm	Size	Auditor	Lineages	Fits	Median	95% nested interval
ARF (native selected)	$4n$	MLP	88	440	0.966	[0.843, 1.027]
Gaussian copula	n	CatBoost	99	99	0.610	[0.310, 0.778]
Gaussian copula	n	Linear	94	94	0.961	[0.915, 0.990]
Gaussian copula	n	MLP	88	88	0.800	[0.618, 0.890]
Gaussian copula	$4n$	CatBoost	99	99	0.669	[0.369, 0.867]
Gaussian copula	$4n$	Linear	94	94	0.987	[0.963, 0.996]
Gaussian copula	$4n$	MLP	88	88	0.978	[0.892, 1.072]
Chow-Liu tree	n	CatBoost	99	99	0.466	[0.246, 0.578]
Chow-Liu tree	n	Linear	94	94	0.636	[0.267, 0.756]
Chow-Liu tree	n	MLP	88	88	0.454	[0.180, 0.631]
Chow-Liu tree	$4n$	CatBoost	99	99	0.576	[0.294, 0.638]
Chow-Liu tree	$4n$	Linear	94	94	0.659	[0.282, 0.763]
Chow-Liu tree	$4n$	MLP	88	88	0.660	[0.347, 0.804]
Independent marginals	n	CatBoost	99	99	−0.046	[−0.083, −0.027]
Independent marginals	n	Linear	94	94	−0.050	[−0.082, −0.015]
Independent marginals	n	MLP	88	88	−0.215	[−0.484, −0.069]
Independent marginals	$4n$	CatBoost	99	99	−0.012	[−0.040, −0.002]
Independent marginals	$4n$	Linear	94	94	0.002	[−0.029, 0.019]
Independent marginals	$4n$	MLP	88	88	−0.049	[−0.161, −0.018]
TVAE (native selected)	n	CatBoost	21	21	0.606	[0.412, 0.739]
TVAE (native selected)	n	Linear	21	21	0.764	[0.552, 0.862]
TVAE (native selected)	n	MLP	20	20	0.721	[0.254, 0.796]
TVAE (native selected)	$4n$	CatBoost	21	21	0.674	[0.452, 0.776]
TVAE (native selected)	$4n$	Linear	21	21	0.796	[0.581, 0.875]
TVAE (native selected)	$4n$	MLP	20	20	0.812	[0.644, 0.877]
CTGAN (native selected)	n	CatBoost	21	21	−0.074	[−0.477, 0.118]
CTGAN (native selected)	n	Linear	21	21	−0.056	[−0.384, 0.118]
CTGAN (native selected)	n	MLP	20	20	−0.247	[−1.072, 0.104]
CTGAN (native selected)	$4n$	CatBoost	21	21	−0.080	[−0.282, 0.223]
CTGAN (native selected)	$4n$	Linear	21	21	−0.033	[−0.454, 0.124]
CTGAN (native selected)	$4n$	MLP	20	20	0.033	[−0.338, 0.328]
Forest-Flow (historical)	n	CatBoost	6	6	0.977	[0.807, 0.999]
Forest-Flow (historical)	n	Linear	5	5	0.981	[0.754, 1.083]
Forest-Flow (historical)	n	MLP	6	6	0.885	[0.578, 1.000]
Forest-Flow (historical)	$4n$	CatBoost	6	6	1.011	[0.905, 1.041]
Forest-Flow (historical)	$4n$	Linear	5	5	0.985	[0.977, 1.001]
Forest-Flow (historical)	$4n$	MLP	6	6	1.054	[0.929, 2.732]
Predictor-only control	n	CatBoost	99	495	0.596	[0.340, 0.726]
Predictor-only control	n	Linear	94	470	0.970	[0.916, 0.994]
Predictor-only control	n	MLP	88	440	0.849	[0.729, 0.913]
Predictor-only control	$4n$	CatBoost	99	495	0.631	[0.375, 0.784]
Predictor-only control	$4n$	Linear	94	470	0.981	[0.955, 0.997]
Predictor-only control	$4n$	MLP	88	440	0.947	[0.883, 1.018]
Real rows resampled	$4n$	CatBoost	99	99	1.018	[1.005, 1.028]
Real rows resampled	$4n$	Linear	94	94	0.998	[0.991, 1.000]
Real rows resampled	$4n$	MLP	88	88	1.174	[1.051, 1.234]
Chow-Liu tree (default)	n	CatBoost	99	99	0.564	[0.455, 0.677]
Chow-Liu tree (default)	n	Linear	94	94	0.661	[0.608, 0.787]
Chow-Liu tree (default)	n	MLP	88	88	0.523	[0.393, 0.684]
Chow-Liu tree (default)	$4n$	CatBoost	99	99	0.636	[0.545, 0.754]
Chow-Liu tree (default)	$4n$	Linear	94	94	0.727	[0.640, 0.810]
Chow-Liu tree (default)	$4n$	MLP	88	88	0.755	[0.649, 0.949]
CTGAN (author default)	n	CatBoost	17	17	−0.284	[−0.562, −0.054]
CTGAN (author default)	n	Linear	17	17	−0.322	[−1.564, −0.070]
CTGAN (author default)	n	MLP	16	16	−0.399	[−0.817, −0.147]
CTGAN (author default)	$4n$	CatBoost	17	17	−0.237	[−0.509, −0.022]
CTGAN (author default)	$4n$	Linear	17	17	−0.294	[−1.606, −0.031]
CTGAN (author default)	$4n$	MLP	16	16	−0.170	[−0.658, −0.003]
DOPE, later binary, seed 11	n	CatBoost	99	99	0.883	[0.820, 0.970]
DOPE, later binary, seed 11	n	Linear	94	94	0.979	[0.964, 0.994]
DOPE, later binary, seed 11	n	MLP	88	88	0.988	[0.924, 0.995]
DOPE, later binary, seed 11	$4n$	CatBoost	99	99	0.944	[0.893, 0.985]
DOPE, later binary, seed 11	$4n$	Linear	94	94	0.994	[0.972, 0.998]
DOPE, later binary, seed 11	$4n$	MLP	88	88	1.175	[1.044, 1.349]
DOPE, later binary, seeds 23-71	n	CatBoost	99	396	0.791	[0.686, 0.930]
DOPE, later binary, seeds 23-71	n	Linear	94	376	0.984	[0.960, 0.992]
DOPE, later binary, seeds 23-71	n	MLP	88	352	0.958	[0.883, 0.990]
DOPE, later binary, seeds 23-71	$4n$	CatBoost	99	396	0.862	[0.772, 0.960]
DOPE, later binary, seeds 23-71	$4n$	Linear	94	376	0.989	[0.970, 0.996]
DOPE, later binary, seeds 23-71	$4n$	MLP	88	352	1.125	[1.026, 1.229]
DOPE, historical seed 11	n	CatBoost	97	97	0.880	[0.822, 0.971]
DOPE, historical seed 11	n	Linear	92	92	0.984	[0.968, 0.994]
DOPE, historical seed 11	n	MLP	87	87	0.988	[0.923, 0.996]
DOPE, historical seed 11	$4n$	CatBoost	97	97	0.949	[0.893, 0.987]
DOPE, historical seed 11	$4n$	Linear	92	92	0.994	[0.972, 0.997]
DOPE, historical seed 11	$4n$	MLP	87	87	1.174	[1.045, 1.349]
DOPE historical, 12 inputs, 512 steps	n	CatBoost	97	97	0.750	[0.415, 0.923]
DOPE historical, 12 inputs, 512 steps	n	Linear	92	92	0.982	[0.957, 0.990]
DOPE historical, 12 inputs, 512 steps	n	MLP	87	87	0.960	[0.831, 0.988]

Arm	Size	Auditor	Lineages	Fits	Median	95% nested interval
DOPE historical, 12 inputs, 512 steps	$4n$	CatBoost	97	97	0.801	$[0.445, 0.958]$
DOPE historical, 12 inputs, 512 steps	$4n$	Linear	92	92	0.984	$[0.966, 0.994]$
DOPE historical, 12 inputs, 512 steps	$4n$	MLP	87	87	1.046	$[0.998, 1.201]$
DOPE historical, 24 inputs, 512 steps	n	CatBoost	93	93	0.762	$[0.416, 0.924]$
DOPE historical, 24 inputs, 512 steps	n	Linear	88	88	0.982	$[0.964, 0.991]$
DOPE historical, 24 inputs, 512 steps	n	MLP	84	84	0.964	$[0.868, 0.993]$
DOPE historical, 24 inputs, 512 steps	$4n$	CatBoost	93	93	0.802	$[0.457, 0.958]$
DOPE historical, 24 inputs, 512 steps	$4n$	Linear	88	88	0.986	$[0.968, 0.996]$
DOPE historical, 24 inputs, 512 steps	$4n$	MLP	84	84	1.035	$[0.999, 1.204]$
DOPE historical, 24 inputs, 2048 steps	n	CatBoost	93	93	0.884	$[0.747, 0.971]$
DOPE historical, 24 inputs, 2048 steps	n	Linear	88	88	0.977	$[0.953, 0.995]$
DOPE historical, 24 inputs, 2048 steps	n	MLP	84	84	0.990	$[0.949, 0.999]$
DOPE historical, 24 inputs, 2048 steps	$4n$	CatBoost	93	93	0.931	$[0.814, 0.985]$
DOPE historical, 24 inputs, 2048 steps	$4n$	Linear	88	88	0.990	$[0.967, 0.997]$
DOPE historical, 24 inputs, 2048 steps	$4n$	MLP	84	84	1.124	$[1.045, 1.266]$
Forest-Flow default (historical)	n	CatBoost	6	6	0.977	$[0.807, 0.999]$
Forest-Flow default (historical)	n	Linear	5	5	0.981	$[0.754, 1.083]$
Forest-Flow default (historical)	n	MLP	6	6	0.885	$[0.578, 1.000]$
Forest-Flow default (historical)	$4n$	CatBoost	6	6	1.011	$[0.905, 1.041]$
Forest-Flow default (historical)	$4n$	Linear	5	5	0.985	$[0.977, 1.001]$
Forest-Flow default (historical)	$4n$	MLP	6	6	1.054	$[0.929, 2.732]$
Gaussian copula (default)	n	CatBoost	99	99	0.565	$[0.260, 0.809]$
Gaussian copula (default)	n	Linear	94	94	0.921	$[0.742, 0.962]$
Gaussian copula (default)	n	MLP	88	88	0.786	$[0.573, 0.931]$
Gaussian copula (default)	$4n$	CatBoost	99	99	0.605	$[0.305, 0.879]$
Gaussian copula (default)	$4n$	Linear	94	94	0.956	$[0.878, 0.983]$
Gaussian copula (default)	$4n$	MLP	88	88	0.935	$[0.851, 1.033]$
Independent marginals (default)	n	CatBoost	99	99	-0.044	$[-0.077, -0.028]$
Independent marginals (default)	n	Linear	94	94	-0.049	$[-0.092, -0.005]$
Independent marginals (default)	n	MLP	88	88	-0.184	$[-0.449, -0.053]$
Independent marginals (default)	$4n$	CatBoost	99	99	-0.016	$[-0.040, -0.006]$
Independent marginals (default)	$4n$	Linear	94	94	0.001	$[-0.027, 0.021]$
Independent marginals (default)	$4n$	MLP	88	88	-0.047	$[-0.132, -0.018]$
Predictor-only, split 2027	n	CatBoost	99	99	0.530	$[0.322, 0.729]$
Predictor-only, split 2027	n	Linear	97	97	0.990	$[0.951, 0.998]$
Predictor-only, split 2027	n	MLP	84	84	0.828	$[0.731, 0.910]$
Predictor-only, split 2027	$4n$	CatBoost	99	99	0.600	$[0.365, 0.778]$
Predictor-only, split 2027	$4n$	Linear	97	97	0.998	$[0.987, 1.001]$
Predictor-only, split 2027	$4n$	MLP	84	84	0.950	$[0.883, 1.031]$
Predictor-only, split 2999	n	CatBoost	100	100	0.570	$[0.339, 0.722]$
Predictor-only, split 2999	n	Linear	92	92	0.993	$[0.972, 0.999]$
Predictor-only, split 2999	n	MLP	86	86	0.844	$[0.764, 0.937]$
Predictor-only, split 2999	$4n$	CatBoost	100	100	0.647	$[0.368, 0.777]$
Predictor-only, split 2999	$4n$	Linear	92	92	0.998	$[0.979, 1.002]$
Predictor-only, split 2999	$4n$	MLP	86	86	0.963	$[0.882, 1.065]$
Predictor-only, split 4099	n	CatBoost	100	100	0.562	$[0.308, 0.716]$
Predictor-only, split 4099	n	Linear	95	95	0.987	$[0.961, 0.995]$
Predictor-only, split 4099	n	MLP	86	86	0.837	$[0.732, 0.942]$
Predictor-only, split 4099	$4n$	CatBoost	100	100	0.640	$[0.362, 0.777]$
Predictor-only, split 4099	$4n$	Linear	95	95	0.993	$[0.974, 0.999]$
Predictor-only, split 4099	$4n$	MLP	86	86	0.965	$[0.917, 1.021]$
Predictor-only, split 8191	n	CatBoost	100	100	0.611	$[0.337, 0.752]$
Predictor-only, split 8191	n	Linear	96	96	0.978	$[0.960, 0.996]$
Predictor-only, split 8191	n	MLP	85	85	0.810	$[0.727, 0.903]$
Predictor-only, split 8191	$4n$	CatBoost	100	100	0.677	$[0.359, 0.787]$
Predictor-only, split 8191	$4n$	Linear	96	96	0.996	$[0.969, 0.999]$
Predictor-only, split 8191	$4n$	MLP	85	85	0.951	$[0.884, 1.018]$
TVAE (author default)	n	CatBoost	18	18	0.300	$[0.072, 0.547]$
TVAE (author default)	n	Linear	18	18	0.354	$[-0.480, 0.657]$
TVAE (author default)	n	MLP	17	17	0.179	$[-1.050, 0.569]$
TVAE (author default)	$4n$	CatBoost	18	18	0.283	$[0.175, 0.596]$
TVAE (author default)	$4n$	Linear	18	18	0.429	$[0.156, 0.682]$
TVAE (author default)	$4n$	MLP	17	17	0.333	$[-0.417, 0.612]$

Table 11: **All DOPE-minus-comparator contrasts.** Nested bootstrap 95% intervals. The Holm family size is in parentheses. The families are pre-declared per size in predeclare_v2, each by three auditors: primary (author-default ARF, TabSyn, TabDDPM, and Forest-Flow under the common budget), density (Gaussian copula, Chow–Liu, and independent marginals), deep generator (author-default CTGAN and TVAE, TabDiff, GReaT), secondary (native-selected ARF, synthpop, SMOTE), and controls (predictor-only control and resampled real rows). HL eq. is the pre-declared equivalence rule (90% interval of the Hodges–Lehmann difference inside ± 0.02); Yuen eq. is the trimmed-mean sensitivity.

Size	Auditor	Comparator	n	Median diff.	Hodges–Lehmann	W/T/L	Holm p (family)	HL eq.	Yuen eq.
n	CatBoost	TabSyn (scaled schedule)	99	0.004 [−0.054, 0.058]	0.002 [−0.057, 0.056]	53/0/46	1.000 (12)	no	no
n	Linear	TabSyn (scaled schedule)	94	0.053 [0.006, 0.168]	0.098 [0.014, 0.244]	74/0/20	3.03×10^{-8} (12)	no	no
n	MLP	TabSyn (scaled schedule)	88	0.061 [0.003, 0.160]	0.100 [0.019, 0.221]	64/0/24	1.60×10^{-5} (12)	no	no
n	CatBoost	ARF (author default)	99	0.084 [0.004, 0.165]	0.090 [0.006, 0.190]	71/0/28	0.001 (12)	no	no
n	Linear	ARF (author default)	94	0.065 [0.003, 0.190]	0.090 [0.008, 0.251]	74/0/20	3.53×10^{-6} (12)	no	no
n	MLP	ARF (author default)	88	0.111 [0.050, 0.226]	0.144 [0.061, 0.332]	74/0/14	5.54×10^{-8} (12)	no	no
n	CatBoost	ARF (native selected)	99	0.083 [0.009, 0.201]	0.109 [0.025, 0.220]	71/0/28	1.71×10^{-4} (9)	no	no
n	Linear	ARF (native selected)	94	0.077 [0.010, 0.198]	0.104 [0.019, 0.256]	76/0/18	1.10×10^{-7} (9)	no	no
n	MLP	ARF (native selected)	88	0.142 [0.063, 0.274]	0.175 [0.081, 0.363]	76/0/12	3.06×10^{-10} (9)	no	no
n	CatBoost	Gaussian copula	99	0.196 [0.090, 0.243]	0.202 [0.110, 0.266]	88/0/11	1.41×10^{-14} (9)	no	no
n	Linear	Gaussian copula	94	0.008 [−0.005, 0.038]	0.020 [−0.005, 0.049]	59/0/35	0.010 (9)	no	no
n	MLP	Gaussian copula	88	0.132 [0.090, 0.185]	0.157 [0.097, 0.242]	73/0/15	2.68×10^{-9} (9)	no	no
n	CatBoost	Chow-Liu tree	99	0.313 [0.240, 0.513]	0.369 [0.268, 0.660]	87/0/12	1.24×10^{-15} (9)	no	no
n	Linear	Chow-Liu tree	94	0.342 [0.226, 0.624]	0.481 [0.282, 0.819]	91/0/3	1.97×10^{-15} (9)	no	no
n	MLP	Chow-Liu tree	88	0.489 [0.340, 0.755]	0.620 [0.406, 1.079]	83/0/5	4.89×10^{-14} (9)	no	no
n	CatBoost	Independent marginals	99	0.866 [0.767, 0.982]	0.849 [0.745, 0.964]	99/0/0	5.13×10^{-17} (9)	no	no
n	Linear	Independent marginals	94	1.002 [0.974, 1.065]	1.023 [0.975, 1.097]	94/0/0	3.05×10^{-16} (9)	no	no
n	MLP	Independent marginals	88	1.164 [1.039, 1.473]	1.265 [1.078, 1.549]	87/0/1	1.17×10^{-14} (9)	no	no
n	CatBoost	TVAE (native selected)	21	0.195 [0.089, 0.395]	0.221 [0.115, 0.347]	19/0/2	—	no	no
n	Linear	TVAE (native selected)	21	0.250 [0.128, 0.427]	0.284 [0.158, 0.549]	20/0/1	—	no	no
n	MLP	TVAE (native selected)	20	0.221 [0.109, 0.507]	0.254 [0.118, 0.596]	17/0/3	—	no	no
n	CatBoost	CTGAN (native selected)	21	0.785 [0.706, 1.342]	0.880 [0.731, 1.204]	21/0/0	—	no	no
n	Linear	CTGAN (native selected)	21	1.052 [0.849, 1.342]	1.104 [0.833, 1.732]	21/0/0	—	no	no
n	MLP	CTGAN (native selected)	20	1.135 [0.772, 1.776]	1.295 [0.789, 2.228]	19/0/1	—	no	no
n	CatBoost	Forest-Flow (historical)	6	−0.045 [−0.463, 0.020]	−0.206 [−0.385, 0.003]	1/0/5	—	no	no
n	Linear	Forest-Flow (historical)	5	0.024 [−0.005, 0.074]	0.037 [−0.034, 0.074]	5/0/0	—	no	no
n	MLP	Forest-Flow (historical)	6	0.005 [−0.041, 0.123]	0.010 [−0.086, 0.123]	3/0/3	—	no	no
n	CatBoost	DOPE, untuned default	99	0.101 [0.035, 0.155]	0.127 [0.052, 0.186]	89/0/10	—	no	no
n	Linear	DOPE, untuned default	94	−0.002 [−0.016, 0.003]	−0.007 [−0.027, 0.002]	42/0/52	—	no	yes
n	MLP	DOPE, untuned default	88	0.034 [0.003, 0.069]	0.050 [0.008, 0.082]	63/0/25	—	no	no
n	CatBoost	Predictor-only control	99	0.215 [0.128, 0.271]	0.214 [0.148, 0.269]	93/0/6	4.36×10^{-15} (6)	no	no
n	Linear	Predictor-only control	94	0.003 [−0.014, 0.041]	0.016 [−0.022, 0.054]	55/0/39	0.142 (6)	no	no
n	MLP	Predictor-only control	88	0.106 [0.044, 0.168]	0.134 [0.067, 0.195]	73/0/15	1.96×10^{-8} (6)	no	no
n	CatBoost	Chow-Liu tree (default)	99	0.239 [0.175, 0.313]	0.261 [0.185, 0.352]	86/0/13	—	no	no
n	Linear	Chow-Liu tree (default)	94	0.282 [0.177, 0.346]	0.327 [0.202, 0.481]	90/0/4	—	no	no

Size	Auditor	Comparator	n	Median diff.	Hodges–Lehmann	W/T/L	Holm p (family)	HL eq.	Yuen eq.
n	MLP	Chow-Liu tree (default)	88	0.389 [0.242, 0.502]	0.490 [0.311, 0.769]	82/0/6	—	no	no
n	CatBoost	CTGAN (author default)	17	1.093 [0.787, 1.446]	1.108 [0.861, 1.707]	17/0/0	1.53×10^{-4} (12)	no	no
n	Linear	CTGAN (author default)	17	1.314 [0.981, 2.569]	1.610 [1.151, 2.510]	17/0/0	1.53×10^{-4} (12)	no	no
n	MLP	CTGAN (author default)	16	1.314 [0.891, 1.790]	1.378 [1.030, 2.528]	15/0/1	0.001 (12)	no	no
n	CatBoost	DOPE, later binary, seed 11	99	−0.020 [−0.058, −0.003]	−0.033 [−0.074, −0.009]	14/10/75	—	no	no
n	Linear	DOPE, later binary, seed 11	94	0.000 [−0.001, 0.003]	0.002 [−0.006, 0.013]	40/19/35	—	yes	yes
n	MLP	DOPE, later binary, seed 11	88	−0.003 [−0.010, 0.000]	−0.007 [−0.021, 0.000]	25/16/47	—	yes	yes
n	CatBoost	DOPE, later binary, seeds 23-71	99	0.008 [0.000, 0.011]	0.010 [0.000, 0.020]	79/0/20	—	yes	yes
n	Linear	DOPE, later binary, seeds 23-71	94	0.000 [−0.002, 0.002]	0.000 [−0.005, 0.006]	44/0/50	—	yes	yes
n	MLP	DOPE, later binary, seeds 23-71	88	0.003 [−0.001, 0.006]	0.003 [−0.005, 0.012]	54/0/34	—	yes	yes
n	CatBoost	DOPE, historical seed 11	97	−0.022 [−0.060, −0.003]	−0.034 [−0.076, −0.009]	15/8/74	—	no	no
n	Linear	DOPE, historical seed 11	92	0.000 [−0.002, 0.002]	0.001 [−0.006, 0.011]	38/16/38	—	yes	yes
n	MLP	DOPE, historical seed 11	87	−0.003 [−0.011, 0.000]	−0.007 [−0.023, 0.000]	25/15/47	—	yes	yes
n	CatBoost	DOPE historical, 12 inputs, 512 steps	97	0.032 [0.006, 0.092]	0.060 [0.012, 0.143]	72/0/25	—	no	no
n	Linear	DOPE historical, 12 inputs, 512 steps	92	0.002 [−0.006, 0.007]	0.001 [−0.010, 0.012]	49/0/43	—	yes	yes
n	MLP	DOPE historical, 12 inputs, 512 steps	87	0.007 [−0.004, 0.016]	0.015 [−0.005, 0.038]	50/0/37	—	no	no
n	CatBoost	DOPE historical, 24 inputs, 512 steps	93	0.032 [0.006, 0.087]	0.053 [0.012, 0.126]	69/0/24	—	no	no
n	Linear	DOPE historical, 24 inputs, 512 steps	88	0.003 [−0.005, 0.008]	0.003 [−0.009, 0.012]	48/0/40	—	yes	yes
n	MLP	DOPE historical, 24 inputs, 512 steps	84	0.002 [−0.015, 0.014]	0.010 [−0.015, 0.029]	45/0/39	—	no	no
n	CatBoost	DOPE historical, 24 inputs, 2048 steps	93	−0.013 [−0.030, −0.001]	−0.019 [−0.041, −0.006]	24/6/63	—	no	no
n	Linear	DOPE historical, 24 inputs, 2048 steps	88	0.000 [−0.005, 0.001]	0.000 [−0.012, 0.012]	35/14/39	—	yes	yes
n	MLP	DOPE historical, 24 inputs, 2048 steps	84	−0.005 [−0.020, 0.000]	−0.014 [−0.040, −0.003]	21/13/50	—	no	no
n	CatBoost	Forest-Flow default (historical)	6	−0.045 [−0.463, 0.020]	−0.206 [−0.366, 0.003]	1/0/5	—	no	no
n	Linear	Forest-Flow default (historical)	5	0.024 [−0.005, 0.074]	0.037 [−0.043, 0.074]	5/0/0	—	no	no
n	MLP	Forest-Flow default (historical)	6	0.005 [−0.041, 0.123]	0.010 [−0.074, 0.123]	3/0/3	—	no	no
n	CatBoost	Gaussian copula (default)	99	0.222 [0.092, 0.283]	0.229 [0.121, 0.303]	89/0/10	—	no	no
n	Linear	Gaussian copula (default)	94	0.048 [0.013, 0.150]	0.104 [0.025, 0.224]	71/0/23	—	no	no
n	MLP	Gaussian copula (default)	88	0.129 [0.069, 0.231]	0.178 [0.085, 0.309]	74/0/14	—	no	no
n	CatBoost	Independent marginals (default)	99	0.875 [0.757, 0.981]	0.842 [0.739, 0.962]	99/0/0	—	no	no
n	Linear	Independent marginals (default)	94	1.011 [0.969, 1.061]	1.017 [0.965, 1.089]	93/0/1	—	no	no
n	MLP	Independent marginals (default)	88	1.121 [1.039, 1.441]	1.229 [1.061, 1.449]	86/0/2	—	no	no
n	CatBoost	Predictor-only, split 2027	98	0.214 [0.147, 0.281]	0.230 [0.160, 0.301]	92/0/6	—	no	no
n	Linear	Predictor-only, split 2027	93	−0.006 [−0.032, 0.009]	−0.010 [−0.043, 0.012]	40/0/53	—	no	no
n	MLP	Predictor-only, split 2027	78	0.104 [0.020, 0.162]	0.128 [0.040, 0.198]	61/0/17	—	no	no
n	CatBoost	Predictor-only, split 2999	99	0.200 [0.111, 0.263]	0.216 [0.137, 0.287]	86/0/13	—	no	no
n	Linear	Predictor-only, split 2999	89	−0.007 [−0.023, 0.008]	−0.006 [−0.034, 0.015]	35/0/54	—	no	yes
n	MLP	Predictor-only, split 2999	82	0.074 [0.011, 0.159]	0.121 [0.030, 0.191]	61/0/21	—	no	no
n	CatBoost	Predictor-only, split 4099	99	0.223 [0.126, 0.275]	0.226 [0.152, 0.299]	86/0/13	—	no	no
n	Linear	Predictor-only, split 4099	91	−0.005 [−0.016, 0.012]	−0.003 [−0.033, 0.025]	40/0/51	—	no	yes

Size	Auditor	Comparator	n	Median diff.	Hodges–Lehmann	W/T/L	Holm p (family)	HL eq.	Yuen eq.
n	MLP	Predictor-only, split 4099	84	0.098 [0.005, 0.159]	0.119 [0.041, 0.197]	58/0/26	—	no	no
n	CatBoost	Predictor-only, split 8191	99	0.179 [0.091, 0.252]	0.199 [0.121, 0.263]	85/0/14	—	no	no
n	Linear	Predictor-only, split 8191	91	0.004 [−0.010, 0.027]	0.007 [−0.016, 0.037]	50/0/41	—	no	yes
n	MLP	Predictor-only, split 8191	81	0.101 [0.024, 0.170]	0.133 [0.044, 0.201]	67/0/14	—	no	no
n	CatBoost	TVAE (author default)	18	0.452 [0.329, 0.778]	0.504 [0.365, 0.747]	18/0/0	9.16×10^{-5} (12)	no	no
n	Linear	TVAE (author default)	18	0.593 [0.326, 1.897]	0.628 [0.432, 12.069]	18/0/0	9.16×10^{-5} (12)	no	no
n	MLP	TVAE (author default)	17	0.632 [0.270, 1.242]	0.748 [0.378, 1.925]	16/0/1	2.44×10^{-4} (12)	no	no
$4n$	CatBoost	TabSyn (scaled schedule)	99	0.006 [−0.054, 0.045]	−0.003 [−0.053, 0.053]	53/0/46	1.000 (12)	no	no
$4n$	Linear	TabSyn (scaled schedule)	94	0.037 [0.002, 0.136]	0.069 [0.005, 0.210]	74/0/20	8.23×10^{-7} (12)	no	no
$4n$	MLP	TabSyn (scaled schedule)	88	0.005 [−0.006, 0.154]	0.054 [−0.006, 0.277]	52/0/36	0.109 (12)	no	no
$4n$	CatBoost	ARF (author default)	99	0.035 [−0.021, 0.130]	0.048 [−0.034, 0.149]	60/0/39	0.229 (12)	no	no
$4n$	Linear	ARF (author default)	94	0.045 [−0.002, 0.155]	0.062 [−0.011, 0.193]	70/0/24	2.87×10^{-4} (12)	no	no
$4n$	MLP	ARF (author default)	88	0.117 [0.020, 0.237]	0.137 [0.041, 0.397]	70/0/18	3.26×10^{-7} (12)	no	no
$4n$	CatBoost	ARF (native selected)	99	0.051 [−0.005, 0.162]	0.083 [0.006, 0.191]	66/0/33	0.004 (9)	no	no
$4n$	Linear	ARF (native selected)	94	0.073 [0.003, 0.164]	0.083 [0.006, 0.217]	73/0/21	1.89×10^{-6} (9)	no	no
$4n$	MLP	ARF (native selected)	88	0.106 [0.034, 0.306]	0.191 [0.060, 0.431]	71/0/17	2.48×10^{-8} (9)	no	no
$4n$	CatBoost	Gaussian copula	99	0.164 [0.062, 0.242]	0.191 [0.089, 0.273]	83/0/16	1.46×10^{-13} (9)	no	no
$4n$	Linear	Gaussian copula	94	0.002 [−0.014, 0.015]	0.007 [−0.023, 0.030]	53/0/41	0.183 (9)	no	yes
$4n$	MLP	Gaussian copula	88	0.170 [0.047, 0.264]	0.234 [0.126, 0.390]	77/0/11	9.26×10^{-13} (9)	no	no
$4n$	CatBoost	Chow-Liu tree	99	0.270 [0.201, 0.517]	0.356 [0.242, 0.728]	91/0/8	5.59×10^{-15} (9)	no	no
$4n$	Linear	Chow-Liu tree	94	0.333 [0.209, 0.659]	0.495 [0.266, 0.816]	91/0/3	1.91×10^{-15} (9)	no	no
$4n$	MLP	Chow-Liu tree	88	0.566 [0.353, 0.795]	0.658 [0.435, 1.166]	88/0/0	2.24×10^{-15} (9)	no	no
$4n$	CatBoost	Independent marginals	99	0.913 [0.826, 0.978]	0.881 [0.790, 0.964]	99/0/0	5.13×10^{-17} (9)	no	no
$4n$	Linear	Independent marginals	94	0.982 [0.937, 1.014]	0.982 [0.926, 1.008]	94/0/0	3.05×10^{-16} (9)	no	no
$4n$	MLP	Independent marginals	88	1.220 [1.094, 1.467]	1.327 [1.143, 1.629]	88/0/0	2.24×10^{-15} (9)	no	no
$4n$	CatBoost	TVAE (native selected)	21	0.201 [0.080, 0.351]	0.201 [0.097, 0.339]	19/0/2	—	no	no
$4n$	Linear	TVAE (native selected)	21	0.213 [0.130, 0.414]	0.262 [0.145, 0.473]	20/0/1	—	no	no
$4n$	MLP	TVAE (native selected)	20	0.286 [0.141, 0.694]	0.426 [0.189, 1.231]	19/0/1	—	no	no
$4n$	CatBoost	CTGAN (native selected)	21	0.792 [0.705, 1.302]	0.912 [0.731, 1.298]	21/0/0	—	no	no
$4n$	Linear	CTGAN (native selected)	21	0.983 [0.843, 1.426]	1.066 [0.812, 1.695]	21/0/0	—	no	no
$4n$	MLP	CTGAN (native selected)	20	1.086 [0.707, 2.268]	1.439 [0.788, 2.272]	20/0/0	—	no	no
$4n$	CatBoost	Forest-Flow (historical)	6	−0.035 [−0.477, −0.004]	−0.231 [−0.382, −0.005]	0/0/6	—	no	no
$4n$	Linear	Forest-Flow (historical)	5	0.000 [−0.060, 0.037]	0.002 [−0.051, 0.117]	3/0/2	—	no	no
$4n$	MLP	Forest-Flow (historical)	6	−0.027 [−0.411, −0.001]	−0.080 [−0.562, 0.004]	1/0/5	—	no	no
$4n$	CatBoost	DOPE, untuned default	99	0.110 [0.033, 0.166]	0.134 [0.053, 0.208]	87/0/12	—	no	no
$4n$	Linear	DOPE, untuned default	94	−0.001 [−0.017, 0.006]	−0.003 [−0.026, 0.005]	43/0/51	—	no	yes
$4n$	MLP	DOPE, untuned default	88	0.080 [0.028, 0.132]	0.138 [0.052, 0.251]	69/0/19	—	no	no
$4n$	CatBoost	Predictor-only control	99	0.194 [0.069, 0.252]	0.197 [0.112, 0.267]	92/0/7	2.80×10^{-14} (6)	no	no
$4n$	Linear	Predictor-only control	94	0.001 [−0.015, 0.031]	0.008 [−0.027, 0.046]	51/0/43	0.230 (6)	no	yes

Size	Auditor	Comparator	n	Median diff.	Hodges–Lehmann	W/T/L	Holm p (family)	HL eq.	Yuen eq.
$4n$	MLP	Predictor-only control	88	0.165 $[0.077, 0.248]$	0.227 $[0.118, 0.378]$	79/0/9	4.73×10^{-13} (6)	no	no
$4n$	CatBoost	Real rows resampled	99	-0.132 $[-0.222, -0.050]$	-0.179 $[-0.276, -0.074]$	10/0/89	2.02×10^{-13} (6)	no	no
$4n$	Linear	Real rows resampled	94	-0.008 $[-0.023, 0.003]$	-0.008 $[-0.046, 0.008]$	33/0/61	0.050 (6)	no	yes
$4n$	MLP	Real rows resampled	88	-0.032 $[-0.067, 0.000]$	-0.037 $[-0.101, 0.070]$	25/0/63	0.046 (6)	no	no
$4n$	CatBoost	Chow-Liu tree (default)	99	0.211 $[0.096, 0.283]$	0.230 $[0.125, 0.348]$	82/0/17	—	no	no
$4n$	Linear	Chow-Liu tree (default)	94	0.259 $[0.136, 0.355]$	0.317 $[0.183, 0.466]$	87/0/7	—	no	no
$4n$	MLP	Chow-Liu tree (default)	88	0.394 $[0.209, 0.570]$	0.489 $[0.284, 0.799]$	84/0/4	—	no	no
$4n$	CatBoost	CTGAN (author default)	17	0.978 $[0.810, 1.489]$	1.097 $[0.863, 1.760]$	17/0/0	1.53×10^{-4} (12)	no	no
$4n$	Linear	CTGAN (author default)	17	1.277 $[0.987, 2.628]$	1.463 $[1.093, 2.148]$	17/0/0	1.53×10^{-4} (12)	no	no
$4n$	MLP	CTGAN (author default)	16	1.454 $[1.030, 2.301]$	1.541 $[1.148, 2.552]$	16/0/0	2.14×10^{-4} (12)	no	no
$4n$	CatBoost	DOPE, later binary, seed 11	99	-0.021 $[-0.061, 0.000]$	-0.037 $[-0.079, -0.008]$	19/15/65	—	no	no
$4n$	Linear	DOPE, later binary, seed 11	94	0.000 $[-0.002, 0.000]$	0.000 $[-0.010, 0.007]$	39/16/39	—	yes	yes
$4n$	MLP	DOPE, later binary, seed 11	88	-0.012 $[-0.018, 0.000]$	-0.020 $[-0.042, -0.002]$	24/12/52	—	no	no
$4n$	CatBoost	DOPE, later binary, seeds 23-71	99	0.007 $[0.000, 0.012]$	0.011 $[0.000, 0.021]$	76/0/23	—	yes	yes
$4n$	Linear	DOPE, later binary, seeds 23-71	94	0.000 $[-0.001, 0.003]$	0.000 $[-0.003, 0.007]$	45/0/49	—	yes	yes
$4n$	MLP	DOPE, later binary, seeds 23-71	88	0.005 $[0.000, 0.008]$	0.007 $[-0.004, 0.019]$	56/0/32	—	yes	yes
$4n$	CatBoost	DOPE, historical seed 11	97	-0.028 $[-0.061, 0.000]$	-0.037 $[-0.081, -0.008]$	20/12/65	—	no	no
$4n$	Linear	DOPE, historical seed 11	92	0.000 $[-0.003, 0.001]$	0.000 $[-0.011, 0.006]$	38/14/40	—	yes	yes
$4n$	MLP	DOPE, historical seed 11	87	-0.012 $[-0.021, 0.000]$	-0.020 $[-0.042, -0.001]$	24/11/52	—	no	no
$4n$	CatBoost	DOPE historical, 12 inputs, 512 steps	97	0.036 $[0.008, 0.107]$	0.061 $[0.013, 0.162]$	73/0/24	—	no	no
$4n$	Linear	DOPE historical, 12 inputs, 512 steps	92	0.001 $[-0.005, 0.007]$	0.003 $[-0.008, 0.013]$	49/0/43	—	yes	yes
$4n$	MLP	DOPE historical, 12 inputs, 512 steps	87	0.025 $[0.004, 0.071]$	0.058 $[0.015, 0.142]$	62/0/25	—	no	no
$4n$	CatBoost	DOPE historical, 24 inputs, 512 steps	93	0.033 $[0.006, 0.075]$	0.051 $[0.013, 0.137]$	70/0/23	—	no	no
$4n$	Linear	DOPE historical, 24 inputs, 512 steps	88	0.001 $[-0.005, 0.009]$	0.003 $[-0.010, 0.016]$	48/0/40	—	yes	yes
$4n$	MLP	DOPE historical, 24 inputs, 512 steps	84	0.024 $[0.002, 0.064]$	0.047 $[0.005, 0.124]$	57/0/27	—	no	no
$4n$	CatBoost	DOPE historical, 24 inputs, 2048 steps	93	-0.012 $[-0.033, 0.000]$	-0.020 $[-0.045, -0.004]$	27/11/55	—	no	no
$4n$	Linear	DOPE historical, 24 inputs, 2048 steps	88	0.000 $[-0.006, 0.004]$	0.000 $[-0.015, 0.012]$	38/11/39	—	yes	yes
$4n$	MLP	DOPE historical, 24 inputs, 2048 steps	84	-0.011 $[-0.033, 0.000]$	-0.023 $[-0.050, 0.000]$	24/9/51	—	no	no
$4n$	CatBoost	Forest-Flow default (historical)	6	-0.035 $[-0.477, -0.004]$	-0.231 $[-0.408, -0.005]$	0/0/6	—	no	no
$4n$	Linear	Forest-Flow default (historical)	5	0.000 $[-0.033, 0.037]$	0.002 $[-0.051, 0.117]$	3/0/2	—	no	no
$4n$	MLP	Forest-Flow default (historical)	6	-0.027 $[-0.411, -0.001]$	-0.080 $[-0.562, 0.004]$	1/0/5	—	no	no
$4n$	CatBoost	Gaussian copula (default)	99	0.201 $[0.062, 0.264]$	0.215 $[0.091, 0.302]$	84/0/15	—	no	no
$4n$	Linear	Gaussian copula (default)	94	0.034 $[0.003, 0.079]$	0.061 $[0.008, 0.169]$	68/0/26	—	no	no
$4n$	MLP	Gaussian copula (default)	88	0.185 $[0.100, 0.351]$	0.299 $[0.144, 0.618]$	78/0/10	—	no	no
$4n$	CatBoost	Independent marginals (default)	99	0.917 $[0.819, 0.975]$	0.884 $[0.790, 0.966]$	99/0/0	—	no	no
$4n$	Linear	Independent marginals (default)	94	0.978 $[0.947, 1.012]$	0.982 $[0.924, 1.008]$	92/0/2	—	no	no
$4n$	MLP	Independent marginals (default)	88	1.188 $[1.056, 1.411]$	1.305 $[1.121, 1.580]$	88/0/0	—	no	no
$4n$	CatBoost	Predictor-only, split 2027	98	0.215 $[0.097, 0.296]$	0.229 $[0.141, 0.300]$	92/0/6	—	no	no
$4n$	Linear	Predictor-only, split 2027	93	-0.009 $[-0.034, 0.002]$	-0.015 $[-0.060, 0.001]$	35/0/58	—	no	no

Table 12: **Seed exchangeability and the binary-by-seed diagnostic, CatBoost.** The count of lineages where seed 11 is the best of five is descriptive: a seed acts on every lineage at once, so the lineages are not independent trials and no p-value is given. The 2×2 cells are medians on the lineages common to all cells, with the median on each cell's own lineages beside it. No Holm family.

Diagnostic	Count	Value
Seed 11 highest of five (CatBoost, $4n$)	47 of 99	19.8 if seeds were exchangeable; descriptive, no p-value
Seeds 23–71 only, CatBoost, n	99 lineages	$0.791\ [0.692, 0.930]$
Seeds 23–71 only, Linear, n	94 lineages	$0.984\ [0.961, 0.992]$
Seeds 23–71 only, MLP, n	88 lineages	$0.958\ [0.875, 0.990]$
Seeds 23–71 only, CatBoost, $4n$	99 lineages	$0.862\ [0.771, 0.958]$
Seeds 23–71 only, Linear, $4n$	94 lineages	$0.989\ [0.970, 0.996]$
Seeds 23–71 only, MLP, $4n$	88 lineages	$1.125\ [1.025, 1.235]$
2×2 cell: historical binary, seed 11	97 common (99 own)	0.949 (own lineages 0.944)
2×2 cell: historical binary, seeds 23–71	97 common (99 own)	0.862 (own lineages 0.862)
2×2 cell: later binary, seed 11	97 common (99 own)	0.949 (own lineages 0.944)
2×2 cell: later binary, seeds 23–71	97 common (99 own)	0.862 (own lineages 0.862)
2×2 cell: historical seed-11 population	97 common (97 own)	0.949 (own lineages 0.949)

Size	Auditor	Comparator	n	Median diff.	Hodges–Lehmann	W/T/L	Holm p (family)	HL eq.	Yuen eq.
$4n$	MLP	Predictor-only, split 2027	78	0.103 $[0.034, 0.205]$	0.194 $[0.067, 0.362]$	60/0/18	—	no	no
$4n$	CatBoost	Predictor-only, split 2999	99	0.177 $[0.062, 0.266]$	0.198 $[0.107, 0.280]$	83/0/16	—	no	no
$4n$	Linear	Predictor-only, split 2999	89	-0.007 $[-0.031, 0.009]$	-0.010 $[-0.060, 0.013]$	34/0/55	—	no	no
$4n$	MLP	Predictor-only, split 2999	82	0.144 $[0.045, 0.263]$	0.202 $[0.075, 0.413]$	64/0/18	—	no	no
$4n$	CatBoost	Predictor-only, split 4099	99	0.188 $[0.088, 0.262]$	0.209 $[0.130, 0.291]$	89/0/10	—	no	no
$4n$	Linear	Predictor-only, split 4099	91	-0.003 $[-0.018, 0.014]$	-0.002 $[-0.028, 0.025]$	40/0/51	—	no	yes
$4n$	MLP	Predictor-only, split 4099	84	0.113 $[0.011, 0.199]$	0.154 $[0.041, 0.290]$	61/0/23	—	no	no
$4n$	CatBoost	Predictor-only, split 8191	99	0.172 $[0.086, 0.245]$	0.190 $[0.099, 0.269]$	85/0/14	—	no	no
$4n$	Linear	Predictor-only, split 8191	91	-0.004 $[-0.020, 0.011]$	-0.004 $[-0.031, 0.022]$	39/0/52	—	no	yes
$4n$	MLP	Predictor-only, split 8191	81	0.122 $[0.040, 0.219]$	0.178 $[0.064, 0.319]$	67/0/14	—	no	no
$4n$	CatBoost	TVAE (author default)	18	0.470 $[0.299, 0.785]$	0.505 $[0.357, 0.739]$	18/0/0	9.16×10^{-5} (12)	no	no
$4n$	Linear	TVAE (author default)	18	0.562 $[0.321, 0.832]$	0.575 $[0.393, 7.242]$	18/0/0	9.16×10^{-5} (12)	no	no
$4n$	MLP	TVAE (author default)	17	0.761 $[0.375, 1.880]$	0.932 $[0.557, 2.021]$	17/0/0	1.53×10^{-4} (12)	no	no

Table 13 repeats the first version's source-family sensitivity on the five-seed panel under the inherited family rule: the excluding-simulated rows drop the four simulated families, and inside one family the family is fixed while its lineages, and each lineage's fit seeds, are resampled.

Table 13: **Source-family sensitivity, five-seed panel.** A DOPE row is DOPE's median retention within the scope; a comparator row is the Hodges–Lehmann DOPE-minus-comparator difference on the lineages both arms complete there. The excluding-simulated rows drop the four simulated families; inside one family the family is fixed and its lineages are resampled, and fit seeds are resampled inside each lineage separately for each arm. A one-seed comparator contributes its single fit. Linear and MLP contrasts are in the family file. These rows are a sensitivity outside every Holm family.

Scope	Size	Auditor	Row	n	Estimate	95% nested CI	W/T/L
excl. simulated	n	CatBoost	DOPE (ours)	34	0.839	$[0.728, 0.913]$	—
excl. simulated	n	CatBoost	− ARF (author default)	34	0.055	$[-0.036, 0.117]$	20/0/14

Scope	Size	Auditor	Row	n	Estimate	95% nested CI	W/T/L
excl. simulated	n	CatBoost	− TabSyn (scaled schedule)	34	-0.012	$[-0.079, 0.057]$	16/0/18
excl. simulated	n	CatBoost	− Gaussian copula	34	0.140	$[0.076, 0.201]$	30/0/4
excl. simulated	n	CatBoost	− Chow-Liu tree	34	0.471	$[0.346, 0.656]$	33/0/1
excl. simulated	n	CatBoost	− Independent marginals	34	0.866	$[0.788, 0.923]$	34/0/0
excl. simulated	n	Linear	DOPE (ours)	32	0.992	$[0.961, 0.996]$	—
excl. simulated	n	MLP	DOPE (ours)	31	0.946	$[0.892, 0.994]$	—
excl. simulated	$4n$	CatBoost	DOPE (ours)	34	0.900	$[0.785, 0.968]$	—
excl. simulated	$4n$	CatBoost	− ARF (author default)	34	0.015	$[-0.069, 0.081]$	17/0/17
excl. simulated	$4n$	CatBoost	− TabSyn (scaled schedule)	34	-0.013	$[-0.078, 0.051]$	17/0/17
excl. simulated	$4n$	CatBoost	− Gaussian copula	34	0.142	$[0.073, 0.203]$	25/0/9
excl. simulated	$4n$	CatBoost	− Chow-Liu tree	34	0.488	$[0.330, 0.696]$	34/0/0
excl. simulated	$4n$	CatBoost	− Independent marginals	34	0.876	$[0.782, 0.950]$	34/0/0
excl. simulated	$4n$	Linear	DOPE (ours)	32	0.992	$[0.970, 1.000]$	—
excl. simulated	$4n$	MLP	DOPE (ours)	31	1.160	$[1.019, 1.257]$	—
feynman	n	CatBoost	DOPE (ours)	20	0.951	$[0.927, 0.983]$	—
feynman	n	CatBoost	− ARF (author default)	20	0.064	$[0.026, 0.101]$	18/0/2
feynman	n	CatBoost	− TabSyn (scaled schedule)	20	0.007	$[-0.019, 0.024]$	14/0/6
feynman	n	CatBoost	− Gaussian copula	20	0.170	$[0.112, 0.236]$	20/0/0
feynman	n	CatBoost	− Chow-Liu tree	20	0.325	$[0.236, 0.540]$	20/0/0
feynman	n	CatBoost	− Independent marginals	20	0.994	$[0.958, 1.021]$	20/0/0
feynman	n	Linear	DOPE (ours)	20	0.990	$[0.984, 0.999]$	—
feynman	n	MLP	DOPE (ours)	20	0.971	$[0.944, 0.994]$	—
feynman	$4n$	CatBoost	DOPE (ours)	20	0.980	$[0.954, 0.993]$	—
feynman	$4n$	CatBoost	− ARF (author default)	20	0.048	$[0.013, 0.089]$	18/0/2
feynman	$4n$	CatBoost	− TabSyn (scaled schedule)	20	-0.004	$[-0.038, 0.014]$	12/0/8
feynman	$4n$	CatBoost	− Gaussian copula	20	0.159	$[0.106, 0.210]$	20/0/0
feynman	$4n$	CatBoost	− Chow-Liu tree	20	0.297	$[0.214, 0.490]$	20/0/0
feynman	$4n$	CatBoost	− Independent marginals	20	0.976	$[0.945, 0.988]$	20/0/0
feynman	$4n$	Linear	DOPE (ours)	20	0.994	$[0.984, 0.998]$	—
feynman	$4n$	MLP	DOPE (ours)	20	1.018	$[0.997, 1.136]$	—
strogatz	n	CatBoost	DOPE (ours)	14	0.775	$[0.493, 0.975]$	—
strogatz	n	CatBoost	− ARF (author default)	14	0.006	$[-0.126, 0.191]$	8/0/6
strogatz	n	CatBoost	− TabSyn (scaled schedule)	14	-0.048	$[-0.147, 0.118]$	5/0/9
strogatz	n	CatBoost	− Gaussian copula	14	0.239	$[0.063, 0.546]$	10/0/4
strogatz	n	CatBoost	− Chow-Liu tree	14	1.127	$[0.386, 2.814]$	11/0/3
strogatz	n	CatBoost	− Independent marginals	14	0.844	$[0.654, 1.033]$	14/0/0
strogatz	n	Linear	DOPE (ours)	13	0.968	$[0.818, 0.996]$	—
strogatz	n	MLP	DOPE (ours)	11	0.974	$[0.895, 1.008]$	—
strogatz	$4n$	CatBoost	DOPE (ours)	14	0.836	$[0.541, 0.975]$	—
strogatz	$4n$	CatBoost	− ARF (author default)	14	-0.034	$[-0.132, 0.097]$	5/0/9
strogatz	$4n$	CatBoost	− TabSyn (scaled schedule)	14	-0.023	$[-0.133, 0.131]$	6/0/8
strogatz	$4n$	CatBoost	− Gaussian copula	14	0.287	$[0.044, 0.624]$	10/0/4
strogatz	$4n$	CatBoost	− Chow-Liu tree	14	1.534	$[0.364, 2.959]$	12/0/2
strogatz	$4n$	CatBoost	− Independent marginals	14	0.877	$[0.672, 1.084]$	14/0/0
strogatz	$4n$	Linear	DOPE (ours)	13	0.942	$[0.778, 0.997]$	—
strogatz	$4n$	MLP	DOPE (ours)	11	1.384	$[1.019, 1.859]$	—
fri	n	CatBoost	DOPE (ours)	25	0.632	$[0.553, 0.786]$	—
fri	n	CatBoost	− ARF (author default)	25	0.282	$[0.134, 0.440]$	20/0/5
fri	n	CatBoost	− TabSyn (scaled schedule)	25	-0.041	$[-0.169, 0.120]$	12/0/13
fri	n	CatBoost	− Gaussian copula	25	0.324	$[0.236, 0.400]$	24/0/1
fri	n	CatBoost	− Chow-Liu tree	25	0.223	$[0.124, 0.318]$	19/0/6
fri	n	CatBoost	− Independent marginals	25	0.682	$[0.599, 0.771]$	25/0/0
fri	n	Linear	DOPE (ours)	24	0.973	$[0.904, 0.999]$	—
fri	n	MLP	DOPE (ours)	21	0.817	$[0.636, 0.983]$	—
fri	$4n$	CatBoost	DOPE (ours)	25	0.774	$[0.613, 0.847]$	—
fri	$4n$	CatBoost	− ARF (author default)	25	0.245	$[0.079, 0.411]$	17/0/8
fri	$4n$	CatBoost	− TabSyn (scaled schedule)	25	-0.032	$[-0.164, 0.115]$	12/0/13
fri	$4n$	CatBoost	− Gaussian copula	25	0.338	$[0.246, 0.421]$	25/0/0
fri	$4n$	CatBoost	− Chow-Liu tree	25	0.203	$[0.099, 0.299]$	21/0/4
fri	$4n$	CatBoost	− Independent marginals	25	0.757	$[0.664, 0.852]$	25/0/0
fri	$4n$	Linear	DOPE (ours)	24	0.996	$[0.981, 1.022]$	—
fri	$4n$	MLP	DOPE (ours)	21	1.245	$[0.973, 1.584]$	—
bng	n	CatBoost	DOPE (ours)	6	0.923	$[0.095, 0.990]$	—
bng	n	CatBoost	− ARF (author default)	6	0.110	$[-0.322, 0.197]$	5/0/1
bng	n	CatBoost	− TabSyn (scaled schedule)	6	0.230	$[0.043, 0.940]$	6/0/0
bng	n	CatBoost	− Gaussian copula	6	0.082	$[-0.038, 0.316]$	4/0/2
bng	n	CatBoost	− Chow-Liu tree	6	0.159	$[-0.007, 0.445]$	4/0/2
bng	n	CatBoost	− Independent marginals	6	0.944	$[0.664, 1.184]$	6/0/0
bng	n	Linear	DOPE (ours)	5	0.984	$[0.855, 0.999]$	—
bng	n	MLP	DOPE (ours)	5	1.149	$[0.975, 1.624]$	—
bng	$4n$	CatBoost	DOPE (ours)	6	0.962	$[0.246, 1.018]$	—
bng	$4n$	CatBoost	− ARF (author default)	6	-0.042	$[-0.461, 0.103]$	3/0/3
bng	$4n$	CatBoost	− TabSyn (scaled schedule)	6	0.242	$[0.053, 1.313]$	6/0/0
bng	$4n$	CatBoost	− Gaussian copula	6	-0.024	$[-0.112, 0.068]$	3/0/3
bng	$4n$	CatBoost	− Chow-Liu tree	6	0.077	$[-0.035, 0.493]$	4/0/2
bng	$4n$	CatBoost	− Independent marginals	6	0.980	$[0.568, 1.118]$	6/0/0
bng	$4n$	Linear	DOPE (ours)	5	0.983	$[0.883, 1.003]$	—
bng	$4n$	MLP	DOPE (ours)	5	1.134	$[1.026, 3.107]$	—

Table 14 gives the pre-declared all-five-fits sensitivity of the five-seed arms: a lineage enters a row only when all five of its fit seeds completed, and a DOPE-minus row pairs the lineages where both arms did.

Table 14: **All-five-fits sensitivity, CatBoost.** The pre-declared lineage sensitivity of the five-seed arms: a lineage enters a row only when all five fit seeds have a complete fit. A level row is the median lineage value of one arm; a DOPE-minus row is the Hodges–Lehmann difference on lineages where both arms have all five fits. Intervals are the panel's nested bootstrap. Linear and MLP rows are in the all-five file. These rows are a sensitivity outside every Holm family.

Size	Row	Lineages	Fits	Estimate	95% nested CI	W/T/L
n	ARF (author default)	99	495	0.718	$[0.571, 0.845]$	—
n	ARF (native selected)	99	495	0.671	$[0.554, 0.826]$	—
n	DOPE (ours)	99	495	0.824	$[0.724, 0.942]$	—
n	DOPE, untuned default	99	495	0.636	$[0.387, 0.819]$	—
n	TabSyn (scaled schedule)	97	485	0.865	$[0.751, 0.929]$	—
n	Predictor-only control	99	495	0.596	$[0.340, 0.725]$	—
n	DOPE − ARF (author default)	99	—	0.090	$[0.004, 0.186]$	71/0/28
n	DOPE − ARF (native selected)	99	—	0.109	$[0.022, 0.219]$	71/0/28
n	DOPE − DOPE, untuned default	99	—	0.127	$[0.053, 0.186]$	89/0/10
n	DOPE − TabSyn (scaled schedule)	97	—	-0.003	$[-0.064, 0.051]$	51/0/46
n	DOPE − Predictor-only control	99	—	0.214	$[0.148, 0.267]$	93/0/6
$4n$	ARF (author default)	99	495	0.822	$[0.689, 0.894]$	—
$4n$	ARF (native selected)	99	495	0.804	$[0.651, 0.887]$	—
$4n$	DOPE (ours)	99	495	0.898	$[0.803, 0.970]$	—
$4n$	DOPE, untuned default	99	495	0.704	$[0.425, 0.875]$	—
$4n$	TabSyn (scaled schedule)	97	485	0.931	$[0.815, 0.969]$	—
$4n$	Predictor-only control	99	495	0.631	$[0.364, 0.784]$	—
$4n$	DOPE − ARF (author default)	99	—	0.048	$[-0.031, 0.144]$	60/0/39
$4n$	DOPE − ARF (native selected)	99	—	0.083	$[0.005, 0.190]$	66/0/33
$4n$	DOPE − DOPE, untuned default	99	—	0.134	$[0.052, 0.207]$	87/0/12
$4n$	DOPE − TabSyn (scaled schedule)	97	—	-0.008	$[-0.059, 0.048]$	51/0/46
$4n$	DOPE − Predictor-only control	99	—	0.197	$[0.111, 0.268]$	92/0/7

E.1 FIT-SEED-11 REVIEW PANELS

The tables below are the fit-seed-11 review panels that preceded the five-seed refit. A method with two configurations keeps one row per configuration. Their intervals are the family-cluster bootstrap of the median, their equivalence column is the v1 mean-difference test, and their Holm families are split by size: nine density tests, six neural tests, six ARF tests, and three Forest-Flow tests per size. Forest-Flow cells store no null loss, so its loss-path columns in Table 20 are empty while its informative count matches the size table. The sensitivity column of that table is the capped cohort, a different denominator from the headline.

Table 15: **DOPE minus comparator at both sizes, fit seed 11.** Median paired retention difference with its family-cluster bootstrap 95% interval, the mean difference, and the v1 mean-difference equivalence test at ± 0.02. W/T/L counts lineages. The Wilcoxon and equivalence tests treat lineages as independent. Holm families are separate per size: density 9 (DOPE against the Gaussian copula, Chow–Liu, and independent marginals), neural 6 (DOPE against CTGAN and TVAE), ARF 6 (DOPE against author-default and native ARF), and Forest-Flow 3 (DOPE against Forest-Flow) tests, each under the three auditors.

Size	Auditor	Comparator	Configuration	n	Median diff.	Hierarchical CI	Mean	TOST	W/T/L	Holm p
$1n$	CatBoost	Gaussian copula	native selected	97	0.2178	[0.1028, 0.3340]	0.2483	no	90/0/7	0.000
$1n$	CatBoost	Chow-Liu	native selected	97	0.3635	[0.2986, 0.5314]	7.7460	no	93/0/4	0.000
$1n$	CatBoost	Independent marginals	native selected	97	0.9238	[0.8575, 1.0178]	1.1580	no	97/0/0	0.000
$1n$	Linear	Gaussian copula	native selected	92	0.0088	[-0.0118, 0.0329]	0.2382	no	53/0/39	0.071
$1n$	Linear	Chow-Liu	native selected	92	0.3309	[0.2275, 0.6482]	1.5301	no	86/0/6	0.000
$1n$	Linear	Independent marginals	native selected	92	0.9955	[0.9691, 1.0693]	1.1054	no	92/0/0	0.000
$1n$	MLP	Gaussian copula	native selected	75	0.1453	[0.0479, 0.2445]	0.1971	no	64/0/11	0.000
$1n$	MLP	Chow-Liu	native selected	75	0.5339	[0.3343, 0.8171]	1.1319	no	75/0/0	0.000
$1n$	MLP	Independent marginals	native selected	75	1.0462	[0.9834, 1.2772]	1.3698	no	75/0/0	0.000
$4n$	CatBoost	Gaussian copula	native selected	97	0.2237	[0.0723, 0.3530]	0.2613	no	86/0/11	0.000
$4n$	CatBoost	Chow-Liu	native selected	97	0.3370	[0.2823, 0.5048]	8.0937	no	94/0/3	0.000
$4n$	CatBoost	Independent marginals	native selected	97	0.9556	[0.8911, 0.9875]	1.2876	no	96/0/1	0.000
$4n$	Linear	Gaussian copula	native selected	92	0.0037	[-0.0144, 0.0138]	0.1832	no	49/0/43	0.373
$4n$	Linear	Chow-Liu	native selected	92	0.3399	[0.1947, 0.6595]	1.4415	no	86/0/6	0.000
$4n$	Linear	Independent marginals	native selected	92	0.9879	[0.9427, 1.0142]	1.0440	no	92/0/0	0.000
$4n$	MLP	Gaussian copula	native selected	75	0.1850	[0.0811, 0.2624]	0.3295	no	65/0/10	0.000
$4n$	MLP	Chow-Liu	native selected	75	0.5301	[0.3183, 0.8832]	0.9903	no	74/0/1	0.000
$4n$	MLP	Independent marginals	native selected	75	1.1043	[1.0367, 1.3283]	1.3100	no	75/0/0	0.000
$1n$	CatBoost	CTGAN	native selected	21	0.8816	[0.7166, 1.4415]	1.0872	no	21/0/0	0.000
$1n$	CatBoost	TVAE	native selected	21	0.2381	[0.1133, 0.4819]	0.2867	no	19/0/2	0.000

Size	Auditor	Comparator	Configuration	n	Median diff.	Hierarchical CI	Mean	TOST	W/T/L	Holm p
$1n$	Linear	CTGAN	native selected	21	1.0516	[0.8722, 1.3740]	1.4244	no	21/0/0	0.000
$1n$	Linear	TVAE	native selected	21	0.2510	[0.1501, 0.4434]	1.3010	no	20/0/1	0.000
$1n$	MLP	CTGAN	native selected	19	0.9336	[0.6818, 2.0568]	2.1780	no	19/0/0	0.000
$1n$	MLP	TVAE	native selected	19	0.2725	[0.1706, 0.7092]	0.5985	no	18/0/1	0.000
$4n$	CatBoost	CTGAN	native selected	21	0.8807	[0.7020, 1.3326]	1.0962	no	21/0/0	0.000
$4n$	CatBoost	TVAE	native selected	21	0.2361	[0.0868, 0.4471]	0.2673	no	19/0/2	0.000
$4n$	Linear	CTGAN	native selected	21	1.0485	[0.8718, 1.4642]	1.3489	no	21/0/0	0.000
$4n$	Linear	TVAE	native selected	21	0.2629	[0.1298, 0.4183]	1.0124	no	20/0/1	0.000
$4n$	MLP	CTGAN	native selected	19	1.0788	[0.8605, 2.0296]	1.8412	no	19/0/0	0.000
$4n$	MLP	TVAE	native selected	19	0.2842	[0.1227, 0.8120]	0.7771	no	17/0/2	0.000
$1n$	CatBoost	ARF	author default	97	0.1086	[0.0026, 0.2394]	0.1144	no	73/0/24	0.000
$1n$	CatBoost	ARF	native selected	97	0.1010	[0.0236, 0.2404]	0.1354	no	76/0/21	0.000
$1n$	Linear	ARF	author default	92	0.0409	[0.0049, 0.1752]	0.1786	no	64/0/28	0.000
$1n$	Linear	ARF	native selected	92	0.0649	[0.0059, 0.2012]	0.2123	no	68/0/24	0.000
$1n$	MLP	ARF	author default	75	0.1371	[0.0311, 0.2860]	0.2632	no	61/0/14	0.000
$1n$	MLP	ARF	native selected	75	0.1423	[0.0464, 0.2643]	0.3097	no	64/0/11	0.000
$4n$	CatBoost	ARF	author default	97	0.0767	[0.0103, 0.1957]	0.2059	no	69/0/28	0.000
$4n$	CatBoost	ARF	native selected	97	0.0954	[0.0166, 0.2137]	0.1737	no	70/0/27	0.000
$4n$	Linear	ARF	author default	92	0.0274	[-0.0020, 0.1195]	0.0924	no	61/0/31	0.000
$4n$	Linear	ARF	native selected	92	0.0369	[0.0008, 0.1639]	0.1249	no	65/0/27	0.000
$4n$	MLP	ARF	author default	75	0.1153	[0.0340, 0.3709]	0.2894	no	57/0/18	0.000
$4n$	MLP	ARF	native selected	75	0.1429	[0.0283, 0.3673]	0.3089	no	58/0/17	0.000
$1n$	CatBoost	Forest-Flow	native selected	6	-0.0198	[-0.4355, 0.0060]	-0.0917	no	1/0/5	0.469
$1n$	Linear	Forest-Flow	native selected	5	0.0275	[-0.1396, 0.0723]	-0.0040	no	3/0/2	1.000
$1n$	MLP	Forest-Flow	native selected	4	0.0196	[-0.0405, 0.2906]	0.0723	no	2/0/2	1.000
$4n$	CatBoost	Forest-Flow	native selected	6	-0.0269	[-0.4550, 0.0080]	-0.1089	no	1/0/5	0.375
$4n$	Linear	Forest-Flow	native selected	5	0.0115	[-0.2599, 0.3016]	0.0118	no	3/0/2	0.812
$4n$	MLP	Forest-Flow	native selected	4	-0.0317	[-0.2275, -0.0102]	-0.0753	no	0/0/4	0.375

Table 16: **TabSyn and DOPE retention levels, fit seed 11.** TabSyn runs the scaled schedule (200 VAE and 1,000 diffusion epochs). A lineage median needs three informative sample seeds at size n or $4n$; own-lineage rows use every informative TabSyn lineage, and matched rows use the lineages shared with informative DOPE features12_steps2048. Intervals are the family-cluster bootstrap. Lineages absent from the admitted cohort: 02a45777900441ce; absence is no win. No Holm family.

Method	Configuration	Cohort	Size	Auditor	n lineages	Median	Hierarchical CI
TabSyn	scaled schedule	own lineages	$1n$	CatBoost	98	0.8557	[0.6955, 0.9459]
TabSyn	scaled schedule	own lineages	$1n$	Linear	93	0.9107	[0.7705, 0.9890]
TabSyn	scaled schedule	own lineages	$1n$	MLP	87	0.9170	[0.7414, 0.9844]
TabSyn	scaled schedule	own lineages	$4n$	CatBoost	98	0.8966	[0.8188, 0.9779]
TabSyn	scaled schedule	own lineages	$4n$	Linear	93	0.9585	[0.8297, 0.9897]
TabSyn	scaled schedule	own lineages	$4n$	MLP	87	1.0314	[0.9689, 1.1865]
TabSyn	scaled schedule	matched lineages	$1n$	CatBoost	96	0.8599	[0.7076, 0.9459]
TabSyn	scaled schedule	matched lineages	$1n$	Linear	91	0.9280	[0.7705, 0.9896]
TabSyn	scaled schedule	matched lineages	$1n$	MLP	72	0.9438	[0.8201, 0.9857]
TabSyn	scaled schedule	matched lineages	$4n$	CatBoost	96	0.9009	[0.8161, 0.9788]
TabSyn	scaled schedule	matched lineages	$4n$	Linear	91	0.9595	[0.8310, 0.9897]
TabSyn	scaled schedule	matched lineages	$4n$	MLP	72	1.0355	[0.9992, 1.1968]
DOPE	DOPE profile	matched lineages	$1n$	CatBoost	96	0.8852	[0.8224, 0.9690]
DOPE	DOPE profile	matched lineages	$1n$	Linear	91	0.9812	[0.9659, 0.9942]
DOPE	DOPE profile	matched lineages	$1n$	MLP	72	0.9732	[0.9490, 0.9898]
DOPE	DOPE profile	matched lineages	$4n$	CatBoost	96	0.9386	[0.8922, 0.9831]
DOPE	DOPE profile	matched lineages	$4n$	Linear	91	0.9940	[0.9718, 0.9967]
DOPE	DOPE profile	matched lineages	$4n$	MLP	72	1.0684	[1.0048, 1.2714]

Table 17: **DOPE minus TabSyn, fit seed 11.** Median paired retention difference on the matched lineages with its family-cluster bootstrap interval, the mean difference, the v1 mean-difference equivalence test at ± 0.02, and W/T/L. The Holm family has three tests per size (DOPE against TabSyn under each of the three auditors), separate from every other comparator block; the Wilcoxon and equivalence tests treat lineages as independent.

Size	Auditor	Comparator	Configuration	n	Median diff.	Hierarchical CI	Mean	TOST	W/T/L	Holm p
$1n$	CatBoost	TabSyn	scaled schedule	96	0.0122	[-0.0033, 0.0851]	0.0338	no	59/0/37	0.042
$1n$	Linear	TabSyn	scaled schedule	91	0.0348	[-0.0008, 0.1265]	58.8632	no	61/0/30	0.000
$1n$	MLP	TabSyn	scaled schedule	72	0.0333	[-0.0078, 0.1272]	0.2127	no	43/0/29	0.011
$4n$	CatBoost	TabSyn	scaled schedule	96	0.0157	[-0.0033, 0.0890]	0.0737	no	56/0/40	0.056
$4n$	Linear	TabSyn	scaled schedule	91	0.0199	[-0.0018, 0.1364]	68.9447	no	60/0/31	0.000
$4n$	MLP	TabSyn	scaled schedule	72	-0.0041	[-0.0384, 0.0738]	0.1739	no	33/0/39	0.682

Table 18: **v1 mean-difference equivalence, fit seed 11.** Mean DOPE-minus-comparator retention difference, its 90% t interval, and the larger of the two one-sided p-values against ± 0.02. The test treats lineages as independent, and outliers move the mean; it is kept for continuity with the first version. Holm families as in the paired table; the equivalence p is unadjusted.

Size	Auditor	Comparator	Configuration	Mean diff.	90% CI	TOST p	Equivalent at ± 0.02
$1n$	CatBoost	Gaussian copula	native selected	0.2483	[0.1779, 0.3188]	1.000	no
$1n$	CatBoost	Chow-Liu	native selected	7.7460	[-3.7576, 19.2497]	0.866	no
$1n$	CatBoost	Independent marginals	native selected	1.1580	[0.7097, 1.6062]	1.000	no
$1n$	Linear	Gaussian copula	native selected	0.2382	[0.0337, 0.4427]	0.960	no
$1n$	Linear	Chow-Liu	native selected	1.5301	[0.7702, 2.2899]	0.999	no
$1n$	Linear	Independent marginals	native selected	1.1054	[0.9906, 1.2202]	1.000	no
$1n$	MLP	Gaussian copula	native selected	0.1971	[0.1217, 0.2725]	1.000	no
$1n$	MLP	Chow-Liu	native selected	1.1319	[0.7996, 1.4642]	1.000	no
$1n$	MLP	Independent marginals	native selected	1.3698	[1.1741, 1.5655]	1.000	no
$4n$	CatBoost	Gaussian copula	native selected	0.2613	[0.2007, 0.3220]	1.000	no
$4n$	CatBoost	Chow-Liu	native selected	8.0937	[-3.9560, 20.1434]	0.866	no
$4n$	CatBoost	Independent marginals	native selected	1.2876	[0.6528, 1.9224]	0.999	no
$4n$	Linear	Gaussian copula	native selected	0.1832	[0.0184, 0.3480]	0.948	no
$4n$	Linear	Chow-Liu	native selected	1.4415	[0.7544, 2.1285]	1.000	no
$4n$	Linear	Independent marginals	native selected	1.0440	[0.9496, 1.1384]	1.000	no
$4n$	MLP	Gaussian copula	native selected	0.3295	[0.2309, 0.4282]	1.000	no
$4n$	MLP	Chow-Liu	native selected	0.9903	[0.6981, 1.2825]	1.000	no
$4n$	MLP	Independent marginals	native selected	1.3100	[1.2007, 1.4194]	1.000	no
$1n$	CatBoost	CTGAN	native selected	1.0872	[0.8392, 1.3352]	1.000	no
$1n$	CatBoost	TVAE	native selected	0.2867	[0.1907, 0.3827]	1.000	no
$1n$	Linear	CTGAN	native selected	1.4244	[0.9705, 1.8783]	1.000	no
$1n$	Linear	TVAE	native selected	1.3010	[-0.2842, 2.8862]	0.911	no
$1n$	MLP	CTGAN	native selected	2.1780	[1.0282, 3.3279]	0.998	no
$1n$	MLP	TVAE	native selected	0.5985	[0.2704, 0.9265]	0.997	no
$4n$	CatBoost	CTGAN	native selected	1.0962	[0.8335, 1.3588]	1.000	no
$4n$	CatBoost	TVAE	native selected	0.2673	[0.1704, 0.3642]	1.000	no
$4n$	Linear	CTGAN	native selected	1.3489	[0.9775, 1.7204]	1.000	no
$4n$	Linear	TVAE	native selected	1.0124	[-0.1398, 2.1646]	0.923	no
$4n$	MLP	CTGAN	native selected	1.8412	[1.1388, 2.5436]	1.000	no
$4n$	MLP	TVAE	native selected	0.7771	[0.2190, 1.3353]	0.985	no
$1n$	CatBoost	ARF	author default	0.1144	[0.0426, 0.1862]	0.984	no
$1n$	CatBoost	ARF	native selected	0.1354	[0.0651, 0.2057]	0.996	no
$1n$	Linear	ARF	author default	0.1786	[0.0606, 0.2965]	0.986	no
$1n$	Linear	ARF	native selected	0.2123	[0.0869, 0.3378]	0.994	no
$1n$	MLP	ARF	author default	0.2632	[0.1518, 0.3746]	1.000	no
$1n$	MLP	ARF	native selected	0.3097	[0.1406, 0.4789]	0.997	no
$4n$	CatBoost	ARF	author default	0.2059	[0.0480, 0.3638]	0.973	no
$4n$	CatBoost	ARF	native selected	0.1737	[0.1098, 0.2376]	1.000	no
$4n$	Linear	ARF	author default	0.0924	[0.0269, 0.1579]	0.965	no
$4n$	Linear	ARF	native selected	0.1249	[0.0554, 0.1943]	0.993	no
$4n$	MLP	ARF	author default	0.2894	[0.1767, 0.4022]	1.000	no
$4n$	MLP	ARF	native selected	0.3089	[0.1650, 0.4527]	0.999	no
$1n$	CatBoost	Forest-Flow	native selected	-0.0917	[-0.2333, 0.0500]	0.823	no
$1n$	Linear	Forest-Flow	native selected	-0.0040	[-0.0820, 0.0740]	0.342	no
$1n$	MLP	Forest-Flow	native selected	0.0723	[-0.1050, 0.2497]	0.731	no
$4n$	CatBoost	Forest-Flow	native selected	-0.1089	[-0.2579, 0.0401]	0.859	no
$4n$	Linear	Forest-Flow	native selected	0.0118	[-0.1779, 0.2015]	0.466	no
$4n$	MLP	Forest-Flow	native selected	-0.0753	[-0.1953, 0.0447]	0.821	no

Table 19: **Source-family sensitivity, fit seed 11.** A DOPE row is the median DOPE retention within one source family; a comparator row is the median DOPE-minus-comparator difference on that family, under CatBoost. The excluding-simulated rows drop the four simulated families. Intervals are the family-cluster bootstrap. These rows are a sensitivity, printed without Holm p-values.

Family	Size	Auditor	Contrast	n	Median	Hierarchical CI
feynman	$1n$	CatBoost	DOPE	20	0.9703	[0.9454, 0.9906]
feynman	$1n$	CatBoost	Gaussian copula	20	0.2002	[0.1097, 0.2549]
feynman	$1n$	CatBoost	Chow-Liu	20	0.3253	[0.2200, 0.4252]
feynman	$1n$	CatBoost	Independent marginals	20	1.0201	[0.9776, 1.0254]
feynman	$1n$	Linear	DOPE	20	0.9938	[0.9806, 0.9975]
feynman	$1n$	MLP	DOPE	19	0.9713	[0.9609, 0.9885]
feynman	$4n$	CatBoost	DOPE	20	0.9894	[0.9713, 0.9944]
feynman	$4n$	CatBoost	Gaussian copula	20	0.1894	[0.0993, 0.2840]
feynman	$4n$	CatBoost	Chow-Liu	20	0.2829	[0.2346, 0.3576]
feynman	$4n$	CatBoost	Independent marginals	20	0.9813	[0.9777, 0.9947]
feynman	$4n$	Linear	DOPE	20	0.9941	[0.9854, 0.9982]
feynman	$4n$	MLP	DOPE	19	0.9965	[0.9860, 1.0067]
strogatz	$1n$	CatBoost	DOPE	14	0.7819	[0.4900, 0.9793]
strogatz	$1n$	CatBoost	Gaussian copula	14	0.1925	[-0.0086, 0.3450]
strogatz	$1n$	CatBoost	Chow-Liu	14	0.8451	[0.2765, 2.3318]
strogatz	$1n$	CatBoost	Independent marginals	14	0.8779	[0.6389, 1.0237]

Family	Size	Auditor	Contrast	n	Median	Hierarchical CI
strogatz	$1n$	Linear	DOPE	13	0.9687	[0.8345, 0.9960]
strogatz	$1n$	MLP	DOPE	9	0.9967	[0.8086, 1.0923]
strogatz	$4n$	CatBoost	DOPE	14	0.8854	[0.5626, 0.9845]
strogatz	$4n$	CatBoost	Gaussian copula	14	0.1303	[-0.0086, 0.3896]
strogatz	$4n$	CatBoost	Chow-Liu	14	0.8739	[0.2276, 2.9286]
strogatz	$4n$	CatBoost	Independent marginals	14	0.9465	[0.6883, 1.0851]
strogatz	$4n$	Linear	DOPE	13	0.9718	[0.7502, 0.9969]
strogatz	$4n$	MLP	DOPE	9	1.0815	[0.9862, 1.4876]
fri	$1n$	CatBoost	DOPE	24	0.8292	[0.7550, 0.8791]
fri	$1n$	CatBoost	Gaussian copula	24	0.4997	[0.3340, 0.5624]
fri	$1n$	CatBoost	Chow-Liu	24	0.3401	[0.2105, 0.4416]
fri	$1n$	CatBoost	Independent marginals	24	0.8660	[0.7905, 0.8976]
fri	$1n$	Linear	DOPE	23	0.9675	[0.8787, 1.0009]
fri	$1n$	MLP	DOPE	15	0.9889	[0.8894, 1.0310]
fri	$4n$	CatBoost	DOPE	24	0.9054	[0.8485, 0.9276]
fri	$4n$	CatBoost	Gaussian copula	24	0.5583	[0.3530, 0.5913]
fri	$4n$	CatBoost	Chow-Liu	24	0.2964	[0.1981, 0.4561]
fri	$4n$	CatBoost	Independent marginals	24	0.9152	[0.8394, 0.9537]
fri	$4n$	Linear	DOPE	23	0.9960	[0.9668, 1.0359]
fri	$4n$	MLP	DOPE	15	1.5792	[1.0250, 1.6797]
bng	$1n$	CatBoost	DOPE	6	0.9118	[0.0978, 1.1020]
bng	$1n$	CatBoost	Gaussian copula	6	0.1041	[0.0472, 0.1614]
bng	$1n$	CatBoost	Chow-Liu	6	0.1781	[0.0312, 0.6353]
bng	$1n$	CatBoost	Independent marginals	6	1.0364	[0.5939, 1.3518]
bng	$1n$	Linear	DOPE	5	0.9718	[0.9305, 1.0382]
bng	$1n$	MLP	DOPE	4	1.0475	[0.9751, 1.6088]
bng	$4n$	CatBoost	DOPE	6	0.9334	[0.2662, 1.0918]
bng	$4n$	CatBoost	Gaussian copula	6	0.0387	[-0.0122, 0.1742]
bng	$4n$	CatBoost	Chow-Liu	6	0.0761	[0.0430, 0.5724]
bng	$4n$	CatBoost	Independent marginals	6	0.9060	[0.4007, 1.2632]
bng	$4n$	Linear	DOPE	5	0.9941	[0.9173, 1.0458]
bng	$4n$	MLP	DOPE	4	1.2962	[0.9871, 1.7049]
other	$1n$	CatBoost	DOPE	33	0.9020	[0.8417, 0.9725]
other	$1n$	CatBoost	Gaussian copula	33	0.1651	[0.0633, 0.2597]
other	$1n$	CatBoost	Chow-Liu	33	0.4339	[0.2997, 0.6052]
other	$1n$	CatBoost	Independent marginals	33	0.9217	[0.8303, 0.9973]
other	$1n$	Linear	DOPE	31	0.9812	[0.9713, 0.9952]
other	$1n$	MLP	DOPE	28	0.9322	[0.8545, 0.9864]
other	$4n$	CatBoost	DOPE	33	0.9373	[0.8908, 0.9878]
other	$4n$	CatBoost	Gaussian copula	33	0.1912	[0.0762, 0.2495]
other	$4n$	CatBoost	Chow-Liu	33	0.4514	[0.3214, 0.6687]
other	$4n$	CatBoost	Independent marginals	33	0.9293	[0.8699, 1.0026]
other	$4n$	Linear	DOPE	31	0.9951	[0.9662, 1.0017]
other	$4n$	MLP	DOPE	28	1.0955	[1.0357, 1.1953]
excluding simulated	$1n$	CatBoost	Gaussian copula	33	0.1651	[0.0698, 0.2637]
excluding simulated	$1n$	CatBoost	Chow-Liu	33	0.4339	[0.2997, 0.6052]
excluding simulated	$1n$	CatBoost	Independent marginals	33	0.9217	[0.8575, 0.9973]
excluding simulated	$4n$	CatBoost	Gaussian copula	33	0.1912	[0.0762, 0.2495]
excluding simulated	$4n$	CatBoost	Chow-Liu	33	0.4514	[0.2957, 0.6687]
excluding simulated	$4n$	CatBoost	Independent marginals	33	0.9293	[0.8699, 1.0026]

Table 20: **Informative-lineage denominators, fit seed 11, CatBoost.** Informative is the lineage count behind each retention level. Complete-loss rows have three finite stored null, TRTR, and TSTR losses, and the noninformative, undefined, capped, and sensitivity counts are taken among them. No Holm family.

Method	Configuration	Size	Informative	Complete-loss rows	Non-informative	Undefined	Capped	Sensitivity n	Capped median
DOPE	DOPE profile	$1n$	97	98	1	0	1	98	0.8862
DOPE	DOPE profile	$4n$	97	98	1	0	1	98	0.9359
Gaussian copula	native selected	$1n$	99	100	1	0	1	100	0.5996
Gaussian copula	native selected	$4n$	99	100	1	0	1	100	0.6658
Chow-Liu	native selected	$1n$	99	100	1	0	12	100	0.4809
Chow-Liu	native selected	$4n$	99	100	1	0	11	100	0.5782
Independent marginals	native selected	$1n$	99	100	1	0	3	100	-0.0440
Independent marginals	native selected	$4n$	99	100	1	0	1	100	-0.0120
CTGAN	native selected	$1n$	21	22	1	0	2	22	-0.0889
CTGAN	native selected	$4n$	21	22	1	0	2	22	-0.0801
TVAE	native selected	$1n$	21	22	1	0	0	22	0.6094
TVAE	native selected	$4n$	21	22	1	0	0	22	0.6582
ARF	author default	$1n$	99	100	1	0	1	100	0.7505
ARF	author default	$4n$	99	100	1	0	1	100	0.8314
ARF	native selected	$1n$	99	100	1	0	1	100	0.7410
ARF	native selected	$4n$	99	100	1	0	1	100	0.8237
Forest-Flow	native selected	$1n$	6	0	0	0	0	0	—

Method	Configuration	Size	Informative	Complete-loss rows	Non-informative	Undefined	Capped	Sensitivity n	Capped median
Forest-Flow	native selected	$4n$	6	0	0	0	0	0	—

Table 21: **CatBoost auditor RMSE, fit seed 11.** Median over regression lineages of the validation RMSE of the auditor trained on synthetic rows and of the auditor trained on the real fit rows. RMSE is in each lineage's original target units: the square root of the median sample-seed mean squared error, which the evaluator computes on the target min–max projected and clipped to $[0, 1]$ by the training split, times that split's target range. A median therefore mixes lineage scales and is no cross-method ranking. No Holm family.

Method	Configuration	Size	Regression lineages	Synthetic RMSE	Real TRTR RMSE
DOPE	DOPE profile	$1n$	98	0.0887	0.0522
DOPE	DOPE profile	$4n$	98	0.0786	0.0522
Gaussian copula	native selected	$1n$	100	0.1166	0.0529
Gaussian copula	native selected	$4n$	100	0.1124	0.0529
Chow-Liu	native selected	$1n$	100	0.1359	0.0529
Chow-Liu	native selected	$4n$	100	0.1280	0.0529
Independent marginals	native selected	$1n$	100	0.1796	0.0529
Independent marginals	native selected	$4n$	100	0.1760	0.0529
CTGAN	native selected	$1n$	22	0.1862	0.0678
CTGAN	native selected	$4n$	22	0.1780	0.0678
TVAE	native selected	$1n$	22	0.1314	0.0678
TVAE	native selected	$4n$	22	0.1257	0.0678
ARF	author default	$1n$	100	0.0971	0.0529
ARF	author default	$4n$	100	0.0866	0.0529
ARF	native selected	$1n$	100	0.1019	0.0529
ARF	native selected	$4n$	100	0.0958	0.0529
Forest-Flow	native selected	$1n$	0	—	—
Forest-Flow	native selected	$4n$	0	—	—

E.2 Historical block tables

The density, neural, and Forest-Flow block tables below are the fit-seed-11 receipts behind the first version of this paper; their intervals are lineage-bootstrap intervals of the median, and they are kept for continuity. The paired median difference is distinct from the difference of two medians. The two DOPE artifacts above the cap in that population are 4544_GeographicalOriginalofMusic at 22,233 bytes and 588_fri_c4_1000_100 at 19,019 bytes, and 678_visualizing_environmental has a fit whose CatBoost group is uninformative, so the CatBoost median uses 97 lineages while the byte count uses 100. The displayed byte range runs from 758 to 22,233. CatBoost reads RMSE 0.053 for the real-trained auditor and 0.081 for the synthetic-trained one, with training-mean skill 0.881 and 0.792; RMSE is the square root of the stored mean squared error, and conventional held-out R^2 is unstored for these rows. Table 22 collects the first version's headline contrasts. On its 6 Forest-Flow lineages the CatBoost medians are 0.940 for DOPE, whose artifacts have a median of 2,067 bytes, and 1.003 [0.926, 1.026] for Forest-Flow. The Nemenyi critical differences for the four density methods are 0.476 under CatBoost and 0.489 under the linear auditor.

F Controls

The real-row control draws the published training rows with replacement until the table has four times the training row count. The predictor-only control fits a ridge regression on the training rows, samples each feature independently from its training marginal, and adds Gaussian residual noise; neither control uses the DOPE encoding, and the predictor-only gap bundles DOPE's dependence model with DOPE's nonlinear residual. The paired column is DOPE minus the control. In the fit-seed-11 control table below, the size-n Holm family is the three predictor-only auditor tests, because the real-row control is defined at $4n$ only, and the size-$4n$ family has six tests; the five-seed panel keeps all six declared control slots at each size. Extra predictor fit seeds are reported and are outside the Holm family, and split-seed rows are a separate sensitivity.

Table 22: **First-version headline contrasts at $4n$, fit seed 11.** Paired median DOPE-minus-baseline retention difference with W/T/L over lineages; a tie is a difference of zero. The ARF rows use the family-cluster interval of the review panel and its size-$4n$ Holm family of six: DOPE against default and native ARF under each of the three auditors. The density, neural, and Forest-Flow rows show each block's highest-median baseline with a lineage-bootstrap interval and the block's Holm family: nine (DOPE against the Gaussian copula, Chow–Liu, and independent marginals under each auditor), six (DOPE against CTGAN and TVAE under each auditor), and three (DOPE against Forest-Flow under each auditor). The five-seed panel supersedes these rows; they are the traceability record of the first version.

Block	Auditor	Baseline	n	Difference	W/T/L	Holm p
Full panel	CatBoost	Author-default ARF	97	0.0767 [0.0103, 0.1957]	69/0/28	1.31×10^{-5}
Full panel	Linear	Author-default ARF	92	0.0274 [-0.0020, 0.1195]	61/0/31	8.53×10^{-5}
Full panel	MLP	Author-default ARF	75	0.1153 [0.0340, 0.3709]	57/0/18	4.56×10^{-8}
Density	CatBoost	Gaussian copula	97	0.224 [0.149, 0.290]	86/0/11	1.35×10^{-13}
Density	Linear	Gaussian copula	92	0.00366 [-0.00438, 0.012]	49/0/43	0.373
Density	MLP	Gaussian copula	75	0.185 [0.128, 0.228]	65/0/10	4.96×10^{-11}
Neural	CatBoost	TVAE	21	0.236 [0.087, 0.431]	19/0/2	1.57×10^{-4}
Neural	Linear	TVAE	21	0.263 [0.166, 0.410]	20/0/1	1.57×10^{-4}
Neural	MLP	TVAE	19	0.284 [0.153, 0.734]	17/0/2	1.57×10^{-4}
Forest	CatBoost	Forest-Flow	6	-0.027 [-0.308, 0.00804]	1/0/5	0.375
Forest	Linear	Forest-Flow	5	0.012 [-0.260, 0.302]	3/0/2	0.812
Forest	MLP	Forest-Flow	4	-0.032 [-0.227, -0.010]	0/0/4	0.375

Table 23: **Density block at $4n$, fit seed 11.** Each cell is a retention; the denominator is the paired informative lineages in that row. Holm family size is nine: DOPE against the Gaussian copula, Chow–Liu, and independent marginals under each of the three auditors. W/T/L counts lineages; a tie is a difference of zero.

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
CatBoost	Gaussian copula	97	**0.940** [0.914, 0.976]	0.656 [0.431, 0.803]	0.224 [0.149, 0.290]	86/0/11	1.35×10^{-13}
CatBoost	Chow–Liu	97	**0.940** [0.914, 0.976]	0.586 [0.465, 0.637]	0.337 [0.282, 0.451]	94/0/3	4.85×10^{-16}
CatBoost	Indep. marginals	97	**0.940** [0.914, 0.976]	-0.012 [-0.020, -0.00649]	0.956 [0.918, 0.980]	96/0/1	1.17×10^{-16}
Linear	Gaussian copula	92	0.994 [0.981, 0.996]	0.989 [0.968, 0.994]	0.00366 [-0.00438, 0.012]	49/0/43	0.373
Linear	Chow–Liu	92	**0.994** [0.981, 0.996]	0.661 [0.356, 0.738]	0.340 [0.244, 0.425]	86/0/6	7.12×10^{-15}
Linear	Indep. marginals	92	**0.994** [0.981, 0.996]	0.00206 [-0.023, 0.00583]	0.988 [0.975, 1.001]	92/0/0	5.70×10^{-16}
MLP	Gaussian copula	75	**1.061** [1.024, 1.125]	0.928 [0.865, 0.953]	0.185 [0.128, 0.228]	65/0/10	4.96×10^{-11}
MLP	Chow–Liu	75	**1.061** [1.024, 1.125]	0.692 [0.582, 0.754]	0.530 [0.350, 0.651]	74/0/1	2.11×10^{-13}
MLP	Indep. marginals	75	**1.061** [1.024, 1.125]	-0.028 [-0.047, -0.013]	1.104 [1.047, 1.179]	75/0/0	2.11×10^{-13}

Table 29: **DOPE minus each control, fit seed 11.** A paired row is the median DOPE-minus-control retention difference with its family-cluster bootstrap 95% interval, the mean difference, the v1 mean-difference equivalence test at ± 0.02, and W/T/L over lineages; the Wilcoxon and equivalence tests treat lineages as independent. Predictor fit-seed and split-seed rows give the control's own CatBoost retention level. The size-n Holm family is the three predictor-only tests (DOPE against the predictor-only control under each auditor), because the real-row control exists at $4n$ only; the size-$4n$ family has six tests (DOPE against the predictor-only control and the resampled real rows under each auditor). Fit-seed and split-seed rows are outside both Holm families.

Size	Auditor	Control	n	Median diff. or level	Hierarchical CI	Mean	TOST	W/T/L	Holm p
$1n$	CatBoost	Predictor-only control	97	0.2528	[0.1449, 0.3903]	0.2286	no	87/0/10	0.000
$1n$	Linear	Predictor-only control	92	-0.0026	[-0.0176, 0.0137]	0.0156	no	42/0/50	0.523
$1n$	MLP	Predictor-only control	73	0.0739	[-0.0009, 0.1498]	0.1331	no	52/0/21	0.000
$4n$	CatBoost	Real rows resampled	97	-0.0806	[-0.1276, -0.0290]	-0.1922	no	14/0/83	0.000
$4n$	CatBoost	Predictor-only control	97	0.2680	[0.0784, 0.3687]	0.2265	no	84/0/13	0.000
$4n$	Linear	Real rows resampled	92	-0.0058	[-0.0203, 0.0028]	0.0028	no	34/0/58	0.147
$4n$	Linear	Predictor-only control	92	-0.0051	[-0.0181, 0.0077]	-0.0023	no	39/0/53	0.229
$4n$	MLP	Real rows resampled	73	-0.0522	[-0.1495, 0.0057]	-0.0256	no	23/0/50	0.087
$4n$	MLP	Predictor-only control	73	0.1296	[0.0356, 0.1867]	0.2230	no	55/0/18	0.000
$1n$	CatBoost	predictor fit 11	99	0.5962	[0.3397, 0.7479]	—	—	—	—
$4n$	CatBoost	predictor fit 11	99	0.6374	[0.3756, 0.7905]	—	—	—	—
$1n$	CatBoost	predictor fit 23	99	0.5664	[0.3364, 0.7239]	—	—	—	—
$4n$	CatBoost	predictor fit 23	99	0.6363	[0.3854, 0.7705]	—	—	—	—
$1n$	CatBoost	predictor fit 37	99	0.5497	[0.3611, 0.7305]	—	—	—	—
$4n$	CatBoost	predictor fit 37	99	0.6263	[0.3547, 0.7705]	—	—	—	—
$1n$	CatBoost	predictor fit 53	99	0.6074	[0.3341, 0.7333]	—	—	—	—
$4n$	CatBoost	predictor fit 53	99	0.6164	[0.3525, 0.7823]	—	—	—	—
$1n$	CatBoost	predictor fit 71	99	0.5933	[0.3386, 0.7163]	—	—	—	—
$4n$	CatBoost	predictor fit 71	99	0.6722	[0.3677, 0.8007]	—	—	—	—
$1n$	CatBoost	split 2027	99	0.5300	[0.3215, 0.7287]	—	—	—	—
$4n$	CatBoost	split 2027	99	0.5997	[0.3654, 0.7784]	—	—	—	—

Size	Auditor	Control	n	Median diff. or level	Hierarchical CI	Mean	TOST	W/T/L	Holm p
$1n$	CatBoost	split 2999	100	0.5705	[0.3371, 0.7224]	—	—	—	—
$4n$	CatBoost	split 2999	100	0.6475	[0.3684, 0.7774]	—	—	—	—
$1n$	CatBoost	split 4099	100	0.5623	[0.3075, 0.7163]	—	—	—	—
$4n$	CatBoost	split 4099	100	0.6399	[0.3621, 0.7742]	—	—	—	—
$1n$	CatBoost	split 8191	100	0.6108	[0.3366, 0.7559]	—	—	—	—
$4n$	CatBoost	split 8191	100	0.6767	[0.3594, 0.7694]	—	—	—	—

G FIDELITY AND EMPIRICAL PRIVACY

Table 33 gives the alpha-precision, beta-recall, DCR-share, and DOMIAS columns of the grouped-validation fidelity ledger of the fit-seed-11 population. On the density cells of the same ledger, DOPE has median logistic two-sample AUC 0.517 (95% lineage-bootstrap interval 0.504 to 0.532, $n = 98$), mean marginal Kolmogorov–Smirnov distance 0.114 (0.103 to 0.135, $n = 98$), and pairwise correlation fidelity 0.953 (0.945 to 0.961, $n = 98$); each count is the lineages with a complete three-seed fidelity group, and Table 32 prints the same medians with family-cluster intervals. A logistic detector near one half can still miss nonlinear differences, which is why the main paper reports a CatBoost detector. The DCR column of that ledger is the fraction of synthetic queries whose nearest reference is in the fit view, with ties scored one half. Under an independent draw, the unbalanced share sits near the fit-view fraction of the pooled references. The grouped split keeps about four fifths of those rows in the fit view, so the baseline is near four fifths, not one half. The DOMIAS column is the isolated kernel density-ratio equation from the pinned author expression, evaluated on this numeric projection, which differs from the original DOMIAS training pipeline.

Table 31 is the review privacy panel behind Table 3 of the main paper: for every method on the fit-seed-11 samples, a CatBoost two-sample detector with at most 200 rows per class, the median distance to the closest fit row and to the closest validation row, the nearest-neighbour distance ratio against the fit rows, and a distance-based membership AUC. The fit reference holds about four times as many rows as the validation reference, so the distance to the fit rows is smaller by construction. None of these numbers is a differential-privacy guarantee or a HIPAA de-identification claim. Shadow-model attacks, attribute inference, and an attack on the shipped artifact are defined in predeclare_v2 and are scheduled. Table 30 compares every arm's marginal, pairwise, and logistic-detector fidelity on the five-seed cells, and Table 32 adds the stored fidelity of the fit-seed-11 population ledgers.

Table 30: **Fidelity of every arm at $4n$, five-seed cells.** Median over lineages with the 95% nested bootstrap interval of the mean marginal Kolmogorov–Smirnov distance (lower is closer), the pairwise correlation fidelity (higher is closer), and a logistic two-sample AUC (one half is indistinguishable); n counts lineages with a complete three-seed group. A five-seed arm needs three complete fits per lineage. No Holm family.

Arm	n	KS distance	Pair correlation	Logistic C2ST
DOPE (ours)	100	0.117 $[0.089, 0.165]$	0.953 $[0.930, 0.964]$	0.531 $[0.516, 0.548]$
DOPE, untuned default	100	0.116 $[0.090, 0.165]$	0.952 $[0.932, 0.962]$	0.532 $[0.512, 0.547]$
TabSyn (scaled schedule)	100	0.126 $[0.087, 0.141]$	0.958 $[0.946, 0.962]$	0.538 $[0.518, 0.567]$
ARF (author default)	100	0.103 $[0.084, 0.140]$	0.962 $[0.948, 0.965]$	0.521 $[0.505, 0.538]$
ARF (native selected)	100	0.108 $[0.084, 0.138]$	0.962 $[0.946, 0.964]$	0.521 $[0.506, 0.539]$
Gaussian copula	100	0.134 $[0.114, 0.211]$	0.963 $[0.950, 0.969]$	0.520 $[0.502, 0.563]$
Chow-Liu tree	100	0.127 $[0.085, 0.229]$	0.928 $[0.891, 0.945]$	0.555 $[0.516, 0.625]$
Independent marginals	100	0.130 $[0.081, 0.267]$	0.899 $[0.826, 0.934]$	0.517 $[0.494, 0.568]$
TVAE (native selected)	22	0.138 $[0.117, 0.243]$	0.926 $[0.893, 0.945]$	0.582 $[0.574, 0.629]$
CTGAN (native selected)	22	0.230 $[0.184, 0.301]$	0.912 $[0.837, 0.943]$	0.770 $[0.667, 0.858]$
Forest-Flow (historical)	6	0.123 $[0.056, 0.206]$	0.969 $[0.932, 0.981]$	0.492 $[0.462, 0.540]$
Predictor-only control	100	0.105 $[0.084, 0.141]$	0.945 $[0.882, 0.961]$	0.510 $[0.497, 0.527]$
Real rows resampled	100	0.080 $[0.069, 0.106]$	0.964 $[0.954, 0.970]$	0.503 $[0.482, 0.530]$

Table 24: **Neural block at $4n$, fit seed 11.** The denominator is the paired informative lineages in the cell; CatBoost uses 21. Holm family size is six: DOPE against CTGAN and TVAE under each of the three auditors. W/T/L counts lineages; a tie is a difference of zero.

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
CatBoost	CTGAN	21	**0.953** [0.896, 0.990]	-0.081 [-0.202, 0.103]	0.881 [0.702, 1.150]	21/0/0	5.72×10^{-6}
CatBoost	TVAE	21	**0.953** [0.896, 0.990]	0.669 [0.539, 0.787]	0.236 [0.087, 0.431]	19/0/2	1.57×10^{-4}
Linear	CTGAN	21	**0.999** [0.996, 1.015]	-0.033 [-0.225, 0.080]	1.049 [0.916, 1.236]	21/0/0	5.72×10^{-6}
Linear	TVAE	21	**0.999** [0.996, 1.015]	0.796 [0.605, 0.867]	0.263 [0.166, 0.410]	20/0/1	1.57×10^{-4}
MLP	CTGAN	19	**1.026** [0.998, 1.263]	-0.149 [-0.471, 0.083]	1.079 [0.877, 2.030]	19/0/0	1.53×10^{-5}
MLP	TVAE	19	**1.026** [0.998, 1.263]	0.753 [0.478, 0.852]	0.284 [0.153, 0.734]	17/0/2	1.57×10^{-4}

Table 25: **Forest-Flow confirmation at $4n$, CatBoost, fit seed 11.** The denominator is 6 lineages. Holm family size is three: DOPE against Forest-Flow under each of the three auditors.

Lineage	DOPE bytes	DOPE	Forest-Flow bytes	Forest-Flow
1193_BNG_lowbwt	2,436	0.970	416,921,273	1.004
503_wind	3,846	0.996	637,056,689	1.016
582_fri_c1_500_25	5,670	0.909	1,085,155,062	0.883
596_fri_c2_250_5	1,698	0.807	234,776,554	0.968
feynman_I_12_1	1,032	0.991	123,784,681	1.001
strogatz_vdp1	984	0.581	115,202,253	1.036

Table 31: **Holdout privacy and CatBoost detection, fit seed 11.** Each value is the median over lineages of each lineage's median over three sample seeds, with the family-cluster bootstrap interval. Own rows use each method's lineages whose synthetic row count equals fit rows times size; matched-density rows use the lineages shared by DOPE, the Gaussian copula, Chow–Liu, and independent marginals. Fit DCR and NNDR use the fit rows as the reference set and holdout DCR the validation rows, about a quarter as many. Distance MIA AUC is an empirical membership attack, and C2ST CatBoost AUC is the CatBoost detector. These are empirical attack scores, not formal differential privacy and not HIPAA de-identification. Unavailable cells are no wins: CTGAN 468 (sample file unresolved); DOPE 12 (sample file unresolved); Forest-Flow 36 (sample file absent); TVAE 468 (sample file unresolved). No Holm family.

Method	Configuration	Size	Metric	n lineages	Median	Hierarchical CI
ARF	author default	$1n$	DCR validation	100	0.1095	[0.0532, 0.1639]
ARF	author default	$1n$	DCR fit	100	0.0826	[0.0332, 0.1257]
ARF	author default	$1n$	NNDR fit	100	0.8983	[0.8617, 0.9278]
ARF	author default	$1n$	Distance MIA AUC	100	0.5452	[0.5298, 0.5580]
ARF	author default	$1n$	C2ST CatBoost AUC	100	0.6159	[0.5751, 0.6928]
ARF	author default	$4n$	DCR validation	100	0.1088	[0.0517, 0.1617]
ARF	author default	$4n$	DCR fit	100	0.0820	[0.0331, 0.1253]
ARF	author default	$4n$	NNDR fit	100	0.9069	[0.8639, 0.9282]
ARF	author default	$4n$	Distance MIA AUC	100	0.5563	[0.5358, 0.5754]
ARF	author default	$4n$	C2ST CatBoost AUC	100	0.6120	[0.5696, 0.6877]
ARF	native selected	$1n$	DCR validation	100	0.1108	[0.0544, 0.1637]
ARF	native selected	$1n$	DCR fit	100	0.0823	[0.0361, 0.1316]
ARF	native selected	$1n$	NNDR fit	100	0.9079	[0.8687, 0.9271]
ARF	native selected	$1n$	Distance MIA AUC	100	0.5422	[0.5309, 0.5576]
ARF	native selected	$1n$	C2ST CatBoost AUC	100	0.6100	[0.5789, 0.7089]
ARF	native selected	$4n$	DCR validation	100	0.1109	[0.0547, 0.1635]
ARF	native selected	$4n$	DCR fit	100	0.0825	[0.0389, 0.1295]
ARF	native selected	$4n$	NNDR fit	100	0.9107	[0.8647, 0.9281]
ARF	native selected	$4n$	Distance MIA AUC	100	0.5408	[0.5303, 0.5562]
ARF	native selected	$4n$	C2ST CatBoost AUC	100	0.6091	[0.5766, 0.7075]
CTGAN	native selected	$1n$	DCR validation	22	0.1863	[0.1163, 0.2214]
CTGAN	native selected	$1n$	DCR fit	22	0.1538	[0.0903, 0.1899]
CTGAN	native selected	$1n$	NNDR fit	22	0.9395	[0.9106, 0.9610]
CTGAN	native selected	$1n$	Distance MIA AUC	22	0.4804	[0.4514, 0.5101]
CTGAN	native selected	$1n$	C2ST CatBoost AUC	21	0.9522	[0.8778, 0.9841]
CTGAN	native selected	$4n$	DCR validation	22	0.1869	[0.1157, 0.2225]
CTGAN	native selected	$4n$	DCR fit	22	0.1529	[0.0867, 0.1906]
CTGAN	native selected	$4n$	NNDR fit	22	0.9413	[0.9117, 0.9581]
CTGAN	native selected	$4n$	Distance MIA AUC	22	0.5024	[0.4721, 0.5168]
CTGAN	native selected	$4n$	C2ST CatBoost AUC	22	0.9535	[0.8487, 0.9777]
Chow-Liu	native selected	$1n$	DCR validation	100	0.1334	[0.0976, 0.2006]
Chow-Liu	native selected	$1n$	DCR fit	100	0.1078	[0.0786, 0.1671]
Chow-Liu	native selected	$1n$	NNDR fit	100	0.9228	[0.8889, 0.9638]
Chow-Liu	native selected	$1n$	Distance MIA AUC	100	0.5077	[0.4962, 0.5202]
Chow-Liu	native selected	$1n$	C2ST CatBoost AUC	100	0.6868	[0.6025, 0.8986]
Chow-Liu	native selected	$4n$	DCR validation	100	0.1291	[0.1053, 0.1901]
Chow-Liu	native selected	$4n$	DCR fit	100	0.1080	[0.0858, 0.1661]
Chow-Liu	native selected	$4n$	NNDR fit	100	0.9238	[0.8883, 0.9644]
Chow-Liu	native selected	$4n$	Distance MIA AUC	100	0.5196	[0.5037, 0.5350]

Method	Configuration	Size	Metric	n lineages	Median	Hierarchical CI
Chow-Liu	native selected	$4n$	C2ST CatBoost AUC	99	0.7061	[0.5910, 0.9369]
DOPE	DOPE profile	$1n$	DCR validation	98	0.1054	[0.0529, 0.1560]
DOPE	DOPE profile	$1n$	DCR fit	98	0.0800	[0.0423, 0.1254]
DOPE	DOPE profile	$1n$	NNDR fit	98	0.9040	[0.8465, 0.9368]
DOPE	DOPE profile	$1n$	Distance MIA AUC	98	0.5132	[0.5012, 0.5249]
DOPE	DOPE profile	$1n$	C2ST CatBoost AUC	98	0.6344	[0.5705, 0.8368]
DOPE	DOPE profile	$4n$	DCR validation	98	0.1086	[0.0528, 0.1521]
DOPE	DOPE profile	$4n$	DCR fit	98	0.0797	[0.0421, 0.1280]
DOPE	DOPE profile	$4n$	NNDR fit	98	0.9053	[0.8482, 0.9360]
DOPE	DOPE profile	$4n$	Distance MIA AUC	98	0.5052	[0.4921, 0.5160]
DOPE	DOPE profile	$4n$	C2ST CatBoost AUC	98	0.6405	[0.5830, 0.8312]
Gaussian copula	native selected	$1n$	DCR validation	100	0.1094	[0.0780, 0.1543]
Gaussian copula	native selected	$1n$	DCR fit	100	0.0868	[0.0621, 0.1268]
Gaussian copula	native selected	$1n$	NNDR fit	100	0.9177	[0.8814, 0.9603]
Gaussian copula	native selected	$1n$	Distance MIA AUC	100	0.5050	[0.4960, 0.5293]
Gaussian copula	native selected	$1n$	C2ST CatBoost AUC	100	0.7343	[0.6789, 0.8981]
Gaussian copula	native selected	$4n$	DCR validation	100	0.1074	[0.0774, 0.1563]
Gaussian copula	native selected	$4n$	DCR fit	100	0.0863	[0.0627, 0.1280]
Gaussian copula	native selected	$4n$	NNDR fit	100	0.9201	[0.8867, 0.9610]
Gaussian copula	native selected	$4n$	Distance MIA AUC	100	0.5022	[0.4851, 0.5158]
Gaussian copula	native selected	$4n$	C2ST CatBoost AUC	99	0.7346	[0.6902, 0.8689]
TVAE	native selected	$1n$	DCR validation	22	0.1122	[0.0853, 0.1943]
TVAE	native selected	$1n$	DCR fit	22	0.0815	[0.0533, 0.1535]
TVAE	native selected	$1n$	NNDR fit	22	0.9056	[0.8479, 0.9264]
TVAE	native selected	$1n$	Distance MIA AUC	22	0.5312	[0.4992, 0.5585]
TVAE	native selected	$1n$	C2ST CatBoost AUC	22	0.7479	[0.7221, 0.8859]
TVAE	native selected	$4n$	DCR validation	22	0.1118	[0.0855, 0.1938]
TVAE	native selected	$4n$	DCR fit	22	0.0799	[0.0535, 0.1516]
TVAE	native selected	$4n$	NNDR fit	22	0.9058	[0.8489, 0.9312]
TVAE	native selected	$4n$	Distance MIA AUC	22	0.5315	[0.4915, 0.5424]
TVAE	native selected	$4n$	C2ST CatBoost AUC	22	0.7309	[0.7032, 0.8893]
Independent marginals	native selected	$1n$	DCR validation	100	0.1360	[0.1010, 0.1877]
Independent marginals	native selected	$1n$	DCR fit	100	0.1132	[0.0890, 0.1587]
Independent marginals	native selected	$1n$	NNDR fit	100	0.9457	[0.9120, 0.9757]
Independent marginals	native selected	$1n$	Distance MIA AUC	100	0.4982	[0.4883, 0.5096]
Independent marginals	native selected	$1n$	C2ST CatBoost AUC	99	0.8694	[0.7602, 0.9749]
Independent marginals	native selected	$4n$	DCR validation	100	0.1319	[0.0995, 0.1848]
Independent marginals	native selected	$4n$	DCR fit	100	0.1134	[0.0886, 0.1580]
Independent marginals	native selected	$4n$	NNDR fit	100	0.9453	[0.9141, 0.9702]
Independent marginals	native selected	$4n$	Distance MIA AUC	100	0.5081	[0.4906, 0.5180]
Independent marginals	native selected	$4n$	C2ST CatBoost AUC	100	0.8578	[0.7553, 0.9727]
DOPE	matched density	$1n$	DCR validation	98	0.1054	[0.0529, 0.1560]
Gaussian copula	matched density	$1n$	DCR validation	98	0.1094	[0.0770, 0.1543]
Chow-Liu	matched density	$1n$	DCR validation	98	0.1304	[0.0976, 0.1913]
Independent marginals	matched density	$1n$	DCR validation	98	0.1360	[0.1000, 0.1872]
DOPE	matched density	$1n$	DCR fit	98	0.0800	[0.0423, 0.1254]
Gaussian copula	matched density	$1n$	DCR fit	98	0.0859	[0.0617, 0.1268]
Chow-Liu	matched density	$1n$	DCR fit	98	0.1075	[0.0786, 0.1613]
Independent marginals	matched density	$1n$	DCR fit	98	0.1128	[0.0890, 0.1578]
DOPE	matched density	$1n$	NNDR fit	98	0.9040	[0.8495, 0.9368]
Gaussian copula	matched density	$1n$	NNDR fit	98	0.9166	[0.8814, 0.9603]
Chow-Liu	matched density	$1n$	NNDR fit	98	0.9219	[0.8888, 0.9638]
Independent marginals	matched density	$1n$	NNDR fit	98	0.9422	[0.9120, 0.9729]
DOPE	matched density	$1n$	Distance MIA AUC	98	0.5132	[0.5012, 0.5249]
Gaussian copula	matched density	$1n$	Distance MIA AUC	98	0.5035	[0.4949, 0.5235]
Chow-Liu	matched density	$1n$	Distance MIA AUC	98	0.5077	[0.4958, 0.5203]
Independent marginals	matched density	$1n$	Distance MIA AUC	98	0.4982	[0.4883, 0.5070]
DOPE	matched density	$1n$	C2ST CatBoost AUC	97	0.6234	[0.5705, 0.8368]
Gaussian copula	matched density	$1n$	C2ST CatBoost AUC	97	0.7296	[0.6596, 0.9162]
Chow-Liu	matched density	$1n$	C2ST CatBoost AUC	97	0.6866	[0.6055, 0.8986]
Independent marginals	matched density	$1n$	C2ST CatBoost AUC	97	0.8694	[0.7656, 0.9749]
DOPE	matched density	$4n$	DCR validation	98	0.1086	[0.0528, 0.1521]
Gaussian copula	matched density	$4n$	DCR validation	98	0.1074	[0.0766, 0.1554]
Chow-Liu	matched density	$4n$	DCR validation	98	0.1262	[0.1020, 0.1874]
Independent marginals	matched density	$4n$	DCR validation	98	0.1308	[0.0995, 0.1841]
DOPE	matched density	$4n$	DCR fit	98	0.0797	[0.0404, 0.1280]
Gaussian copula	matched density	$4n$	DCR fit	98	0.0859	[0.0627, 0.1273]
Chow-Liu	matched density	$4n$	DCR fit	98	0.1042	[0.0853, 0.1642]
Independent marginals	matched density	$4n$	DCR fit	98	0.1124	[0.0884, 0.1569]
DOPE	matched density	$4n$	NNDR fit	98	0.9053	[0.8492, 0.9360]
Gaussian copula	matched density	$4n$	NNDR fit	98	0.9198	[0.8858, 0.9603]
Chow-Liu	matched density	$4n$	NNDR fit	98	0.9221	[0.8883, 0.9644]
Independent marginals	matched density	$4n$	NNDR fit	98	0.9435	[0.9119, 0.9702]
DOPE	matched density	$4n$	Distance MIA AUC	98	0.5052	[0.4919, 0.5160]
Gaussian copula	matched density	$4n$	Distance MIA AUC	98	0.5004	[0.4850, 0.5152]
Chow-Liu	matched density	$4n$	Distance MIA AUC	98	0.5196	[0.5023, 0.5380]
Independent marginals	matched density	$4n$	Distance MIA AUC	98	0.5077	[0.4891, 0.5176]
DOPE	matched density	$4n$	C2ST CatBoost AUC	96	0.6465	[0.5863, 0.8508]
Gaussian copula	matched density	$4n$	C2ST CatBoost AUC	96	0.7348	[0.6902, 0.8804]

Method	Configuration	Size	Metric	n lineages	Median	Hierarchical CI
Chow-Liu	matched density	$4n$	C2ST CatBoost AUC	96	0.7062	[0.6017, 0.9302]
Independent marginals	matched density	$4n$	C2ST CatBoost AUC	96	0.8578	[0.7555, 0.9727]

Table 32: **Stored fidelity from the population ledgers, fit seed 11.** Logistic two-sample AUC, mean marginal Kolmogorov–Smirnov distance, and pair-correlation fidelity; n counts lineages with three finite stored values, with the family-cluster bootstrap interval. Cohorts differ across methods, so the rows are no ranking. The stored detector is logistic, not CatBoost, and TabSyn is absent from these ledgers. No Holm family.

Method	Configuration	Size	Metric	n lineages	Median	Hierarchical CI
DOPE	DOPE profile	$1n$	stored logistic C2ST	98	0.5172	[0.5000, 0.5376]
DOPE	DOPE profile	$1n$	marginal KS	98	0.1216	[0.0950, 0.1680]
DOPE	DOPE profile	$1n$	pair correlation	98	0.9532	[0.9317, 0.9610]
DOPE	DOPE profile	$4n$	stored logistic C2ST	98	0.5172	[0.5000, 0.5376]
DOPE	DOPE profile	$4n$	marginal KS	98	0.1138	[0.0917, 0.1583]
DOPE	DOPE profile	$4n$	pair correlation	98	0.9531	[0.9346, 0.9645]
Gaussian copula	native selected	$1n$	stored logistic C2ST	100	0.5246	[0.5100, 0.5518]
Gaussian copula	native selected	$1n$	marginal KS	100	0.1393	[0.1171, 0.2132]
Gaussian copula	native selected	$1n$	pair correlation	100	0.9614	[0.9516, 0.9652]
Gaussian copula	native selected	$4n$	stored logistic C2ST	100	0.5246	[0.5100, 0.5537]
Gaussian copula	native selected	$4n$	marginal KS	100	0.1336	[0.1137, 0.2115]
Gaussian copula	native selected	$4n$	pair correlation	100	0.9635	[0.9488, 0.9687]
Chow-Liu	native selected	$1n$	stored logistic C2ST	100	0.5423	[0.5040, 0.6292]
Chow-Liu	native selected	$1n$	marginal KS	100	0.1293	[0.0892, 0.2361]
Chow-Liu	native selected	$1n$	pair correlation	100	0.9273	[0.8834, 0.9418]
Chow-Liu	native selected	$4n$	stored logistic C2ST	100	0.5404	[0.5185, 0.6303]
Chow-Liu	native selected	$4n$	marginal KS	100	0.1266	[0.0852, 0.2292]
Chow-Liu	native selected	$4n$	pair correlation	100	0.9283	[0.8906, 0.9465]
Independent marginals	native selected	$1n$	stored logistic C2ST	100	0.5213	[0.5000, 0.5347]
Independent marginals	native selected	$1n$	marginal KS	100	0.1363	[0.0866, 0.2847]
Independent marginals	native selected	$1n$	pair correlation	100	0.8994	[0.8099, 0.9289]
Independent marginals	native selected	$4n$	stored logistic C2ST	100	0.5166	[0.4988, 0.5488]
Independent marginals	native selected	$4n$	marginal KS	100	0.1297	[0.0813, 0.2672]
Independent marginals	native selected	$4n$	pair correlation	100	0.8986	[0.8258, 0.9329]
CTGAN	native selected	$1n$	stored logistic C2ST	22	0.7839	[0.6997, 0.8819]
CTGAN	native selected	$1n$	marginal KS	22	0.2319	[0.1887, 0.2943]
CTGAN	native selected	$1n$	pair correlation	22	0.9091	[0.8313, 0.9406]
CTGAN	native selected	$4n$	stored logistic C2ST	22	0.7839	[0.6997, 0.8821]
CTGAN	native selected	$4n$	marginal KS	22	0.2300	[0.1839, 0.2947]
CTGAN	native selected	$4n$	pair correlation	22	0.9123	[0.8367, 0.9431]
TVAE	native selected	$1n$	stored logistic C2ST	22	0.5918	[0.5616, 0.6392]
TVAE	native selected	$1n$	marginal KS	22	0.1500	[0.1199, 0.2598]
TVAE	native selected	$1n$	pair correlation	22	0.9240	[0.8905, 0.9418]
TVAE	native selected	$4n$	stored logistic C2ST	22	0.6033	[0.5544, 0.6681]
TVAE	native selected	$4n$	marginal KS	22	0.1382	[0.1174, 0.2431]
TVAE	native selected	$4n$	pair correlation	22	0.9258	[0.8913, 0.9451]
ARF	author default	$1n$	stored logistic C2ST	100	0.5321	[0.5156, 0.5442]
ARF	author default	$1n$	marginal KS	100	0.1128	[0.0895, 0.1433]
ARF	author default	$1n$	pair correlation	100	0.9595	[0.9471, 0.9644]
ARF	author default	$4n$	stored logistic C2ST	100	0.5268	[0.5069, 0.5388]
ARF	author default	$4n$	marginal KS	100	0.1035	[0.0846, 0.1340]
ARF	author default	$4n$	pair correlation	100	0.9618	[0.9487, 0.9653]
ARF	native selected	$1n$	stored logistic C2ST	100	0.5255	[0.5072, 0.5449]
ARF	native selected	$1n$	marginal KS	100	0.1122	[0.0869, 0.1445]
ARF	native selected	$1n$	pair correlation	100	0.9599	[0.9418, 0.9626]
ARF	native selected	$4n$	stored logistic C2ST	100	0.5296	[0.5094, 0.5375]
ARF	native selected	$4n$	marginal KS	100	0.1046	[0.0845, 0.1382]
ARF	native selected	$4n$	pair correlation	100	0.9620	[0.9464, 0.9643]
Forest-Flow	native selected	$1n$	stored logistic C2ST	6	0.5187	[0.4139, 0.7000]
Forest-Flow	native selected	$1n$	marginal KS	6	0.1254	[0.0612, 0.1992]
Forest-Flow	native selected	$1n$	pair correlation	6	0.9548	[0.9242, 0.9799]
Forest-Flow	native selected	$4n$	stored logistic C2ST	6	0.5187	[0.4139, 0.7000]
Forest-Flow	native selected	$4n$	marginal KS	6	0.1231	[0.0557, 0.2062]
Forest-Flow	native selected	$4n$	pair correlation	6	0.9691	[0.9321, 0.9811]

G.1 CONFIGURATION-AWARE FIDELITY AND DETECTORS

The side-by-side columns use the common numeric projection and the published pilot KS and pair-fidelity formulas, including the projected target. Alpha-precision and beta-recall use the fixed unembedded equations. The stored logistic detector remains distinct from the shared grouped CatBoost detector: five disjoint row-group folds, the fixed iteration and depth budget, and the

Table 26: **Published ARF matched receipts at $4n$, CatBoost.** The denominator is the paired complete informative lineages in the generated row. No Holm family in this table.

Publication	Lineages	DOPE	Receipt	Difference
ARF, author_default	97	0.940	0.840	0.077
ARF, native_selected	97	0.940	0.826	0.095

Table 27: **Raw auditor medians on the informative lineages, fit seed 11.** The denominator is the informative lineages for that auditor. No Holm family.

Auditor	n	RMSE real	RMSE synth.	training-mean skill, real	training-mean skill, synth.
CatBoost	97	0.053	0.081	0.881	0.792
Linear	92	0.105	0.112	0.585	0.551
MLP	75	0.098	0.083	0.666	0.785

predeclared equal per-class cap. Every metric gives its own complete-three-seed lineage count and family-cluster interval. These marginal summaries use different availability cohorts and establish no cross-method ranking. ARF author-default and native-selected configurations stay separate. Forest-Flow and originally unmeasured size-n alpha/beta entries retain unavailable cells. The retained mixed-type TabSyn alpha/beta cohort remains separate from the common-numeric cohort. No Holm family is applied to these descriptive tables.

H Ablations and replay

Step budget. `features12_steps8192` is the same residual with four times as many AdamW steps. Where both budgets have a complete three-seed group, the CatBoost median paired difference, longer budget minus shorter, is 0.034 (68 lineages); the linear medians are 0.990 and 0.987 (64 lineages; difference -0.000283) and the MLP medians 1.065 and 1.057 (52 lineages; difference 0.00451). Appendix J.2 lists every lineage. **Profile and default.** The main paper's untuned trainer default is a second axis (width and step budget together). The full width, step, readout, screening-size, and byte-budget grid, and the interaction stress suite (XOR, parity, products, many relevant features), are defined in predeclare_v2 and scheduled.

Replay. The scored runs did not store per-step losses; a later logged replay of both step budgets, before the readout refit, did. All 100 lineages finished. Median training loss starts at 0.321 and ends at 0.00274 and 0.000763; median validation loss starts at 0.271 and ends at 0.00443 and 0.00181. The replay files are distinct from the scored artifacts.

I Baseline coverage and retained evidence

Table 36 records which baseline publications exist and on how many lineages. The ARF follow-on supplies complete five-fit measurements on the same validation lineages. TabSyn's full scaled-schedule panel is in the main paper; the earlier eight-lineage TabSyn subset below is a different cohort. Forest-Flow confirmation is distinct from the Forest-Diffusion pilot modes, and no full-population diffusion result is inferred from the Flow subset. Receipt paths and digests are in `docs/whitepaper/generated/baseline-coverage.json` and `docs/whitepaper/generated/retained-evidence.json`.

Retained validation. Tables 37 and 38 add the merged retained-sample evidence. Each lineage averages all three sample seeds before a lineage-bootstrap median, which differs from the median-of-seeds rule of the main tables. Table 37 covers ARF and the copula and Chow–Liu defaults and native configurations; Table 38 includes only the eight complete scaled-schedule TabSyn lineages of an earlier run and the classical methods on identical inputs. Table 39 gives exploratory TabSyn-minus-comparator differences on those lineages, with conservative simultaneous intervals (Bonferroni over four metrics times six configurations), which are unbounded at this count. Table 40 extends the published DOPE–ARF comparison with lineage intervals under its six-test family.

Table 28: **Charged bytes, fit seed 11.** For DOPE artifacts the charge is the encoded model file plus `projection.json`. The DOPE, density, Forest-Flow, and ARF rows count every artifact with a stored charge; the CTGAN and TVAE rows count native-selected fits whose three $4n$ sample cells all succeeded, so main Table 1, which counts every charged lineage, prints different medians for them. No Holm family.

Method	Median bytes	Within cap
DOPE reference	1,895.5	98/100
Gaussian copula	5,464.5	62/100
Chow–Liu	29,865	31/100
Independent marginals	2,549.5	94/100
CTGAN	741,834	0/22
TVAE	317,141.5	0/22
Forest-Flow	325,848,913.5	0/6
ARF (native selected)	1,627,718.5	0/100

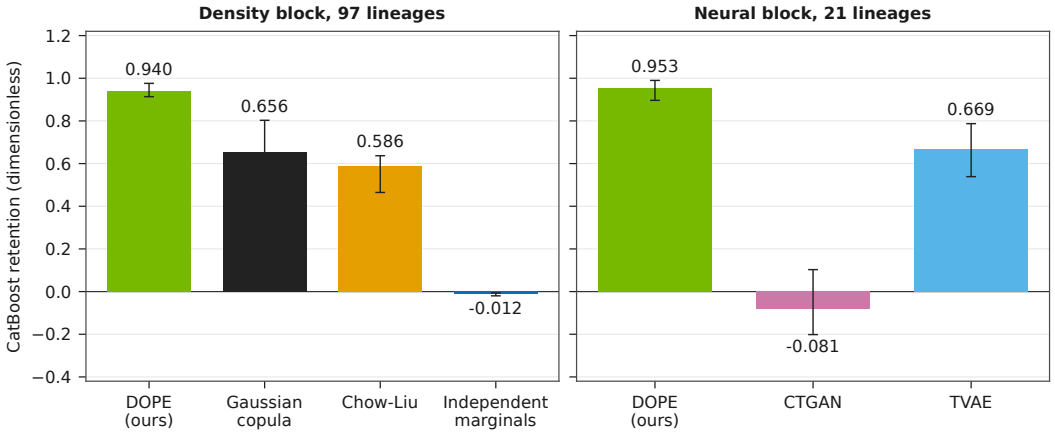


Figure 4: **CatBoost retention at $4n$, fit seed 11.** Each bar is a method's paired-lineage retention median and its lineage-bootstrap interval. Density uses 97 lineages and the neural panel 21; the panels are separate cohorts. No Holm family.

First-batch diagnostics. The canonical-job batch below is incomplete at population and sample-group level, so its values are individual physical evaluations without a population interval or ranking. DOPE jobs in this batch use frozen research profiles that differ from the reported profile. Pearson entries are the pinned source mean, alpha and beta use its pinned unembedded variant, and MIA is an empirical distance attack. Per-column marginals, detection folds, DCR and NNDR, costs, and job digests are in the first-batch source listed in `docs/whitepaper/generated/retained-evidence.json`.

BeyondArena (`2ecfe882ccfb814fc27c4de10a64ceefd5d7655c`) is a separate fit check outside the 100 PMLB lineages. 5 of the 12 families in that check finished inside the cap, and 7 did not prepare (4 outside the projected range, 3 overlapping); the median elapsed times per family are 5.00 seconds for `steps2048` and 14.0 seconds for `steps8192`. The inventory of all 142 families is in Appendix J.3. No retention auditor ran on BeyondArena.

J PER-LINEAGE TABLES

The tables below list every prepared lineage of the fit-seed-11 population, the sufficiency budget, and the BeyondArena inventory. They are the rows behind the medians of the historical block tables, and a blank cell is an unavailable fit or an uninformative auditor rather than a zero. The five-seed v2 cells are in the committed scalar ledger `research/benchmark/results/review-fixes-v2/scalar-cells.jsonl`, which holds one row per fit, sample seed, size, and auditor for every arm.

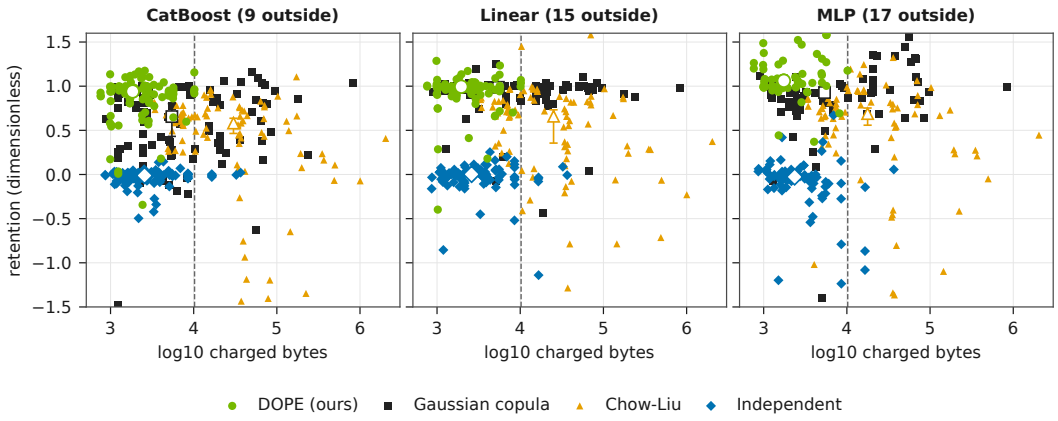


Figure 5: **Density-block retention at $4n$ against charged bytes, per lineage, fit seed 11.** Each point is one lineage's retention level (vertical) against its charged bytes on a log axis. Values outside $[-1.5, 1.6]$ are counted in the panel title and omitted from the frame. The dashed line is the 10,240-byte cap. No Holm family.

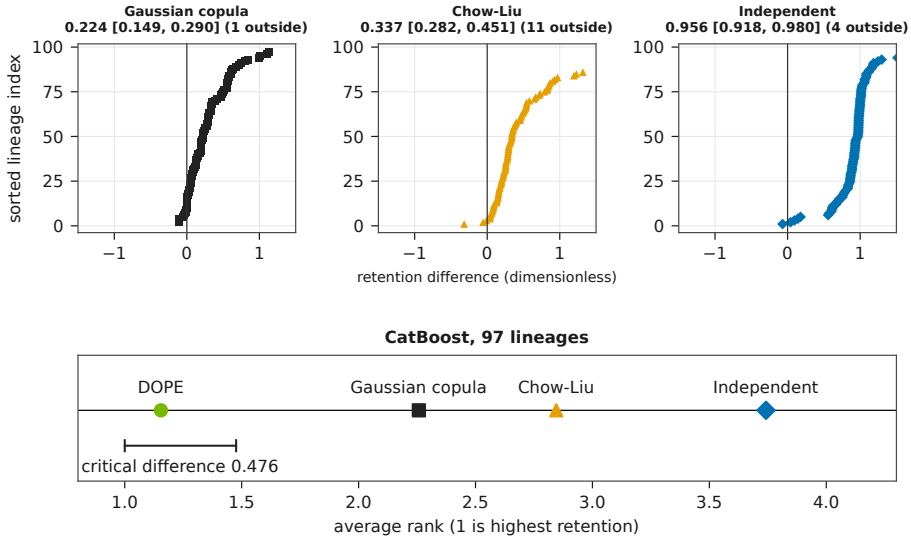


Figure 6: **CatBoost paired differences, sorted, and the Nemenyi scale, fit seed 11.** Rank 1 is the higher retention; the segment length is the critical difference. This display is outside the nine-test Holm family.

J.1 Reported profile, all 100 lineages (fit seed 11)

These rows are the historical fit-seed-11 population of the reported profile. The cap column marks charged bytes at or under 10,240; it is a byte check, not a release certificate. Retention columns are CatBoost at n and $4n$, linear at $4n$, and the MLP at $4n$, each the median of the three sample seeds; the identifier is the lineage's dataset hash prefix and the name is its PMLB name.

Table 43: Displayed profile, all 100 lineages. The cap column is charged bytes at or under 10,240. It is not a release certificate. No Holm family.

id	name	rows	feat.	bytes	cap	CB n	CB $4n$	lin $4n$	MLP $4n$
02a45777900441ce	215_2dplanes	800	10	2,583	pass	0.9930	1.0001	0.9990	1.0477
047e62b52a3f4ede	695_chatfield_4	156	12	3,311	pass	1.0041	1.0155	1.0048	—
04c80944e971cb3e	first_principles_supernovae_zr	157	1	762	pass	0.8753	0.8846	0.9959	1.2627

id	name	rows	feat.	bytes	cap	CB n	CB $4n$	lin $4n$	MLP $4n$
0561d673cb73c27e	666_rmftsa_ladata	339	10	2,921	pass	1.0095	1.1026	1.0338	1.1394
060b9fff8092db17	589_fri_c2_1000_25	667	25	5,717	pass	0.8893	0.9534	0.9911	1.5794
06b0312a585b436e	1196_BNG_pharynx	800	10	2,631	pass	0.8601	0.8964	1.0458	1.7049
0779d3fd5b3133a2	nikuradse_2	241	1	758	pass	0.9096	0.9261	2.1723	2.2082
07f38946316c039b	593_fri_c1_1000_10	667	10	2,876	pass	0.5650	0.6068	0.9617	1.6005
0a8428ef25a69175	537_houses	800	8	2,609	pass	0.6257	0.7036	1.0150	1.0256
0d482c1dd7cc6592	210_cloud	72	5	1,844	pass	1.0374	0.9772	1.0944	1.0239
0db59fd2579600d0	665_sleuth_case2002	98	6	1,962	pass	1.0371	1.1324	1.1345	—
0f86a714ec5f3ac6	1191_BNG_pbc	800	18	4,183	pass	0.8097	0.8761	0.9173	—
167578f5deba4977	547_no2	333	7	2,443	pass	0.8073	0.8513	1.0097	0.9898
19f2c3c2d67defd7	feynman_II_11_27	800	4	1,494	pass	0.9704	0.9890	0.9960	1.0048
19f4780b53b3fa41	678_visualizing_environmental	74	3	1,515	pass	—	—	—	0.4431
1a4456d1cc71a26a	feynman_test_1	800	7	2,191	pass	1.0608	1.1671	0.9816	—
1a803b27c2400dc4	feynman_III_10_19	800	4	1,520	pass	0.9882	0.9952	0.9942	0.9965
1cb97b1d7314d43d	229_pwLinear	133	10	2,587	pass	0.8417	0.9878	0.9104	—
205f4b4982b9ea49	608_fri_c3_1000_10	667	10	2,845	pass	0.5263	0.5671	1.0451	0.8182
26eff843be35d831	581_fri_c3_500_25	333	25	5,647	pass	0.7602	0.9020	1.1297	1.5792
282f8ddb43b5e677	595_fri_c0_1000_10	667	10	2,904	pass	0.8974	0.9234	0.9962	1.0250
297b7688a045c583	603_fri_c0_250_50	167	50	10,171	pass	0.8848	1.1562	1.0700	2.6883
2c8d0985f3557d2b	feynman_II_11_28	800	2	1,026	pass	0.9831	0.9899	1.0031	0.9976
30ca2887550f0d8d	201_pol	800	26	8,145	pass	0.5742	0.5976	0.7015	0.6923
3408a6a513c627a3	strogatz_barmag1	267	2	1,011	pass	0.9750	0.9775	0.9969	1.0058
35a9aa9b88f0e4d1	344_mv	800	10	2,737	pass	0.9985	1.0016	0.9662	1.0347
3d3321e27d14f9aa	599_fri_c2_1000_5	667	5	1,707	pass	0.8359	0.8611	0.9668	1.6797
44eaabe4cb923766	613_fri_c3_250_5	167	5	1,712	pass	0.8671	0.9276	0.8599	—
46b25db9396c8209	227_cpu_small	800	12	3,327	pass	0.8569	0.9072	0.9018	0.9364
46c5a59e5ae42037	1193_BNG_lowbwt	800	9	2,436	pass	0.9635	0.9703	1.0030	0.9871
471f11cc2d38853a	feynman_test_11	800	4	1,481	pass	0.9645	0.9760	0.9975	0.9826
47c581af5de1136b	582_fri_c1_500_25	333	25	5,670	pass	0.7510	0.9088	0.7173	—
48b7a3bf81e5f3e5	605_fri_c2_250_25	167	25	5,692	pass	0.4046	0.6465	1.9138	—
4a354d98c965b01f	597_fri_c2_500_5	333	5	1,713	pass	0.7550	0.8055	0.9814	1.8381
4aa84fff8ad0cd26	feynman_III_13_18	800	4	1,501	pass	0.8748	0.9137	0.8850	0.9348
51730b348b1bbd14	557_analcatdata_apnea1	316	3	1,237	pass	-0.0032	0.0086	—	—
54b221668673164c	feynman_III_14_14	800	5	1,726	pass	1.0346	1.1164	1.0011	1.0611
552665868f8f72ed	strogatz_vdp2	267	2	989	pass	0.9945	0.9963	0.9960	0.9969
585fef2a1cb6b33f	strogatz_lv2	267	2	1,018	pass	0.8286	0.9859	0.7502	—
642738a361f1c63a	feynman_I_12_1	800	2	1,032	pass	0.9892	0.9910	0.9892	0.9933
64c64f0f9983c48f	strogatz_bacres2	267	2	998	pass	0.9793	0.9832	0.9783	0.9862
654566a1f91408aa	4544_GeographicalOriginalofMusic	706	117	22,233	fail	—	—	—	—
693890aae9dc860b	first_principles_supernovae_zg	162	1	761	pass	0.8492	0.8908	1.0066	1.1997
6cacb11611345a24	feynman_I_11_19	800	6	1,947	pass	0.9524	0.9812	0.9780	0.9800
7291b347fe0238fb	feynman_II_11_20	800	5	1,757	pass	0.9928	1.0432	1.0016	1.0674
7423783f7e38bb9b	strogatz_shearflow1	267	2	1,021	pass	0.4900	0.5614	-0.3978	—
7a3d123dd555110d	598_fri_c0_1000_25	667	25	5,668	pass	0.8613	0.9166	1.0046	1.0034
7c8943b160464add	601_fri_c1_250_5	167	5	1,719	pass	0.7859	0.8552	1.0773	—
7e51711baa1b1fb1	feynman_I_12_11	800	5	1,718	pass	0.9267	0.9320	1.0015	0.9860
80589368fa78f21a	218_house_8L	800	8	2,392	pass	0.5835	0.6708	0.9616	1.1040
8164950b155f7f56	feynman_I_12_2	800	4	1,511	pass	0.9702	0.9986	0.9485	0.9753
85a31ae409944a1b	feynman_II_10_9	800	3	1,269	pass	0.9330	0.9508	0.9589	0.9515
8d859cfa7b51a351	feynman_III_15_12	800	3	1,248	pass	0.5172	0.5542	1.0228	1.0749
923483635406b213	1203_BNG_pwLinear	800	10	2,581	pass	1.2079	1.1750	0.9941	1.5230
9312d813229df035	690_visualizing_galaxy	216	4	1,458	pass	0.9725	0.9755	0.9853	1.0870
94a00ea67f49bf84	596_fri_c2_250_5	167	5	1,698	pass	0.7546	0.8072	—	—
94bd4520b8ceb3f0	feynman_test_12	800	5	1,747	pass	0.9688	0.9804	0.9913	1.0067
956d34f13e7af830	586_fri_c3_1000_25	667	25	5,701	pass	0.9325	0.9849	1.2339	2.1457
9a10c15885022034	_deprecated_195_auto_price	106	15	5,025	pass	0.8717	0.9485	0.9598	1.3107
a04f964bc53281e0	strogatz_lv1	267	2	985	pass	-3.8784	-3.4212	—	—
a23f4bd626b76590	feynman_test_13	800	5	1,740	pass	0.9178	0.9605	0.9900	1.0991
a32a69e7c701eb14	588_fri_c4_1000_100	667	100	19,019	fail	—	—	—	—
a5ba682572f53add	602_fri_c3_250_10	167	10	2,900	pass	0.5355	0.8109	0.7988	—
a863cd42be4d05d4	663_rabe_266	80	2	1,014	pass	0.9781	0.9933	0.9967	1.0521
a97065df07a47eb0	592_fri_c4_1000_25	667	25	5,655	pass	0.8791	0.9340	0.9578	1.2714
ab279eded3b3c52b	574_house_16H	800	16	4,035	pass	0.1181	0.1796	0.1795	0.3705
aebf5d861a8365b4	522_pm10	333	7	2,409	pass	0.5082	0.6149	0.4125	2.3076
b6adf79118a58cea	562_cpu_small	800	12	3,353	pass	0.6968	0.7747	0.9834	0.9307
b78d323566930bd7	583_fri_c1_1000_50	667	50	10,098	pass	0.8607	0.9444	0.9960	1.6211
b7f9774e2142a758	strogatz_shearflow2	267	2	1,019	pass	0.4707	0.5439	0.7450	1.8955
bc85856c012bcc1b	607_fri_c4_1000_50	667	50	10,209	pass	0.8856	0.9640	3.9394	—
c107ee39eb333a9d	584_fri_c4_500_25	333	25	5,638	pass	0.8062	0.8619	0.9417	—
c30ff9405c916718	579_fri_c0_250_5	167	5	1,692	pass	0.9475	1.0923	0.9961	0.9783
c4f8e81b70d33a13	590_fri_c0_1000_50	667	50	10,180	pass	0.8839	0.9172	1.0359	1.1762
c649a4cee87c4321	529_pollen	800	4	1,506	pass	1.0356	1.1012	0.9717	1.2459
cb5bf66ef2ee09ed	nikuradse_1	241	2	992	pass	0.9346	0.9373	0.9997	1.1909
cba8393879afd1dc	230_machine_cpu	140	6	2,338	pass	1.0026	1.0891	1.1718	—
ce1a79e2d658eb8a	strogatz_glider2	267	2	1,003	pass	0.6382	0.6625	1.0720	1.3756
ce837ccd24a3a18c	strogatz_bacres1	267	2	1,004	pass	0.7352	0.8309	0.9014	0.9826
cee6f811d0bf199d	feynman_II_11_3	800	5	1,728	pass	0.9385	0.9666	0.8950	1.0149
d255d3a2e1c8dbf4	feynman_III_12_43	800	2	1,028	pass	0.9920	0.9935	0.9966	0.9940
d52bfd2b6ab1c850	feynman_I_10_7	800	3	1,250	pass	0.9920	0.9932	0.9940	0.9959

id	name	rows	feat.	bytes	cap	CB n	CB $4n$	lin $4n$	MLP $4n$
d9a3408a840c9f48	560_bodyfat	168	14	4,364	pass	1.0247	1.0436	0.9930	1.3651
da290fae0d6c072f	feynman_test_10	800	3	1,267	pass	0.9955	0.9990	0.9990	1.0015
da78606950f100cd	1199_BNG_echoMonths	800	9	2,544	pass	0.9962	1.0086	0.9655	1.0694
dd16ac872ffb8673	strogatz_glider1	267	2	991	pass	0.1145	0.1723	0.9123	—
de2868ac837699fd	564_fried	800	10	2,861	pass	0.9477	0.9829	0.9543	1.4707
e08f0b9931f93686	225_puma8NH	800	8	2,439	pass	1.0639	1.1043	0.9951	2.8498
e18350fb51a9d1b0	strogatz_barmag2	267	2	1,027	pass	0.8897	0.9399	0.2855	—
e5a3a7b3668a1cc2	556_analcatdata_apnea2	316	3	1,238	pass	0.0339	0.0341	—	—
e68fdd96cc63271b	573_cpu_act	800	21	4,954	pass	0.8080	0.9045	0.6593	1.1465
ed2efb82fce4b098	strogatz_vdp1	267	2	984	pass	0.5228	0.5812	1.2867	1.2401
edc9d41a298fdf60	612_fri_c1_1000_5	667	5	1,705	pass	0.8224	0.8596	0.9670	1.2891
efd53e9c0f36d608	_deprecated_207_autoPrice	106	15	5,017	pass	0.9020	0.9357	1.0899	1.0398
f0a3d6881458f990	604_fri_c4_500_10	333	10	2,913	pass	0.6868	0.8360	0.8552	—
f24e9383331c43e2	503_wind	800	14	3,846	pass	0.9690	0.9962	1.0017	1.0367
f652ea96eff62654	strogatz_predprey1	267	2	989	pass	0.9964	1.1345	1.0218	1.4876
f8cf362e371e8960	strogatz_predprey2	267	2	1,030	pass	0.9876	0.9847	0.9718	1.0815
fc3812567b5fdd64	197_cpu_act	800	21	5,011	pass	0.7680	0.8922	0.6669	1.1253
fe6d7ac3fcd3129e	1201_BNG_breastTumor	800	9	2,427	pass	-0.6141	-0.3437	—	—

J.2 SUFFICIENCY BUDGET (FIT SEED 11)

These rows are the `features12_steps8192` budget check on the lineages where it produced a charged byte count or a retention; the remaining lineages have neither and are omitted. The columns match the reported-profile table, so the two tables can be read side by side for the step-budget axis of Appendix H. The longer budget is a check on the reported profile and is not the reported model.

Table 44: Sufficiency budget. 69 lineages have a charged byte count or a retention. 31 lineages have neither and are omitted. The cap column is charged bytes at or under 10,240. It is not a release certificate. No Holm family.

id	name	rows	feat.	bytes	cap	CB n	CB $4n$	lin $4n$	MLP $4n$
047e62b52a3f4ede	695_chatfield_4	156	12	3,353	pass	1.0106	1.0287	1.0110	—
04c80944e971cb3e	first_principles_supernovae_zr	157	1	750	pass	0.9372	0.9454	1.0346	1.3271
060b9fff8092db17	589_fri_c2_1000_25	667	25	5,734	pass	0.9237	0.9886	1.0258	1.5430
06b0312a585b436e	1196_BNG_pharynx	800	10	2,681	pass	0.8188	0.8768	1.0632	0.7968
07f38946316c039b	593_fri_c1_1000_10	667	10	2,869	pass	0.9780	1.0222	0.9929	1.8097
0a8428ef25a69175	537_houses	800	8	2,636	pass	0.7243	0.7841	1.0073	1.0334
0d482c1dd7cc6592	210_cloud	72	5	1,855	pass	0.9561	0.8815	1.0056	0.9224
0f86a714ec5f3ac6	1191_BNG_pbc	800	18	4,218	pass	0.8069	0.8576	0.8783	—
167578f5deba4977	547_no2	333	7	2,474	pass	0.8821	0.9276	0.9850	1.0276
1a4456d1cc71a26a	feynman_test_1	800	7	2,185	pass	1.1162	1.3026	0.6721	—
1a803b27c2400dc4	feynman_III_10_19	800	4	1,522	pass	0.9698	0.9777	0.9732	0.9807
1cb97b1d7314d43d	229_pwLinear	133	10	2,598	pass	0.8275	1.0471	0.9727	—
26eff843be35d831	581_fri_c3_500_25	333	25	5,655	pass	0.8660	0.9960	1.0900	1.6067
282f8ddb43b5e677	595_fri_c0_1000_10	667	10	2,913	pass	0.9645	1.0058	1.0029	1.1269
297b7688a045c583	603_fri_c0_250_50	167	50	10,183	pass	0.8257	1.1292	1.0240	2.8999
30ca2887550f0d8d	201_pol	800	26	8,215	pass	0.6214	0.6436	0.7350	0.6603
3d3321e27d14f9aa	599_fri_c2_1000_5	667	5	1,713	pass	0.9820	1.0044	1.0441	1.6148
44eaabe4cb923766	613_fri_c3_250_5	167	5	1,708	pass	0.8605	0.9611	0.9427	—
46b25db9396c8209	227_cpu_small	800	12	3,338	pass	0.7033	0.8730	0.9142	0.9908
46c5a59e5ae42037	1193_BNG_lowbwt	800	9	2,468	pass	0.8393	0.8390	0.8798	0.8103
471f11cc2d38853a	feynman_test_11	800	4	1,472	pass	0.9340	0.9481	0.9730	0.9578
47c581af5de1136b	582_fri_c1_500_25	333	25	5,676	pass	0.8232	0.9820	0.8086	—
48b7a3bf81e5f3e5	605_fri_c2_250_25	167	25	5,696	pass	0.2608	0.5247	1.4268	—
4a354d98c965b01f	597_fri_c2_500_5	333	5	1,715	pass	0.9578	1.0001	1.0224	1.3200
51730b348b1bbd14	557_analcatdata_apnea1	316	3	1,246	pass	0.6556	0.6654	—	—
54b221668673164c	feynman_III_14_14	800	5	1,724	pass	1.0175	1.1226	1.0021	1.0727
585fef2a1cb6b33f	strogatz_lv2	267	2	1,021	pass	0.9239	1.0451	0.8041	—
642738a361f1c63a	feynman_I_12_1	800	2	1,034	pass	0.9936	0.9961	0.9993	0.9989
64c64f0f9983c48f	strogatz_bacres2	267	2	996	pass	0.9743	0.9826	0.9816	0.9923
693890aae9dc860b	first_principles_supernovae_zg	162	1	766	pass	0.9415	0.9642	1.0277	1.1876
6cacb11611345a24	feynman_I_11_19	800	6	1,952	pass	0.9911	1.0176	1.0047	1.0089
7423783f7e38bb9b	strogatz_shearflow1	267	2	1,023	pass	0.4860	0.5973	-4.3352	—
7a3d123dd555110d	598_fri_c0_1000_25	667	25	5,672	pass	0.8570	0.9206	1.0010	1.0168
7e51711baa1b1fb1	feynman_I_12_11	800	5	1,720	pass	0.9780	0.9927	0.9910	1.0559
80589368fa78f21a	218_house_8L	800	8	2,410	pass	0.6661	0.7528	0.6830	1.0728
85a31ae409944a1b	feynman_II_10_9	800	3	1,264	pass	0.9803	0.9961	0.9881	1.0061
8d859cfa7b51a351	feynman_III_15_12	800	3	1,280	pass	0.7428	0.8572	0.8771	1.2306
9312d813229df035	690_visualizing_galaxy	216	4	1,470	pass	0.9770	0.9875	0.9843	1.0661
94a00ea67f49bf84	596_fri_c2_250_5	167	5	1,718	pass	0.8262	0.8934	—	—
94bd4520b8ceb3f0	feynman_test_12	800	5	1,745	pass	0.9677	0.9845	0.9937	1.0157
9a10c15885022034	_deprecated_195_auto_price	106	15	5,038	pass	0.8965	0.9544	0.9901	1.3165

id	name	rows	feat.	bytes	cap	CB n	CB $4n$	lin $4n$	MLP $4n$
a04f964bc53281e0	strogatz_lv1	267	2	986	pass	-3.0531	-2.4502	—	—
a32a69e7c701eb14	588_fri_c4_1000_100	667	100	19,041	fail	—	—	—	—
a5ba682572f53add	602_fri_c3_250_10	167	10	2,902	pass	0.7774	0.9773	0.9471	—
a863cd42be4d05d4	663_rabe_266	80	2	1,014	pass	0.9849	0.9996	1.0018	1.0573
ab279eded3b3c52b	574_house_16H	800	16	4,076	pass	0.0805	0.1211	-0.7472	-0.3199
aebf5d861a8365b4	522_pm10	333	7	2,450	pass	0.6372	0.7959	0.6203	3.4124
b78d323566930bd7	583_fri_c1_1000_50	667	50	10,098	pass	0.8817	0.9737	0.8733	0.9465
b7f9774e2142a758	strogatz_shearflow2	267	2	1,022	pass	0.7388	0.8339	-0.4873	1.5681
c107ee39eb333a9d	584_fri_c4_500_25	333	25	5,641	pass	0.8307	0.9024	0.9623	—
c30ff9405c916718	579_fri_c0_250_5	167	5	1,700	pass	0.9081	1.0750	0.9949	0.9830
c649a4cee87c4321	529_pollen	800	4	1,506	pass	1.0619	1.1198	0.9983	1.2801
cb5bf66ef2ee09ed	nikuradse_1	241	2	990	pass	0.9662	0.9677	0.9970	1.1953
ce1a79e2d658eb8a	strogatz_glider2	267	2	1,004	pass	0.5396	0.6099	1.1474	1.2017
ce837ccd24a3a18c	strogatz_bacres1	267	2	1,004	pass	0.7807	0.8568	0.8858	0.9570
d255d3a2e1c8dbf4	feynman_III_12_43	800	2	1,023	pass	0.9947	0.9964	0.9984	0.9967
d52bfd2b6ab1c850	feynman_I_10_7	800	3	1,248	pass	0.9978	0.9988	0.9994	1.0005
da290fae0d6c072f	feynman_test_10	800	3	1,257	pass	0.9681	0.9718	0.9737	0.9744
da78606950f100cd	1199_BNG_echoMonths	800	9	2,578	pass	0.9473	0.9673	0.9447	1.0028
de2868ac837699fd	564_fried	800	10	2,864	pass	0.9826	1.0201	0.9548	1.5094
e08f0b9931f93686	225_puma8NH	800	8	2,440	pass	1.0168	1.0664	0.9433	2.7529
e5a3a7b3668a1cc2	556_analcatdata_apnea2	316	3	1,241	pass	0.6282	0.6707	—	—
e68fdd96cc63271b	573_cpu_act	800	21	4,958	pass	0.8507	0.9689	0.5661	1.1177
ed2efb82fce4b098	strogatz_vdp1	267	2	991	pass	0.5397	0.6472	-0.1705	1.1154
efd53e9c0f36d608	_deprecated_207_autoPrice	106	15	5,024	pass	0.9035	0.9215	1.0781	1.0481
f0a3d6881458f990	604_fri_c4_500_10	333	10	2,918	pass	0.8895	1.0064	1.0421	—
f24e9383331c43e2	503_wind	800	14	3,877	pass	0.9992	0.9953	0.9918	1.0034
f8cf362e371e8960	strogatz_predprey2	267	2	1,030	pass	0.9865	0.9827	0.9658	1.0750
fc3812567b5fdd64	197_cpu_act	800	21	5,006	pass	0.8164	0.9529	0.6632	1.1726

J.3 BeyondArena families in the dataset bucket

The inventory below lists every BeyondArena family name in the dataset bucket with its task, row count, feature count, recorded license, and split type. Only the families with a cleared license entered the fit check of Appendix I, and none of them was scored for retention. The inventory is a list of candidate tables for later panels and makes no claim about any of them.

Table 45: All 142 BeyondArena families. No Holm family.

name	task	rows	cols	license	split
5g_energy_consumption	regression	92,629	22	MIT	grouped
acquire_valued_shoppers_challenge	binary	160,057	112	not cleared	temporal
airfoil_self_noise	regression	1,503	6	CC-BY-4.0	iid
allstate_claims_severity	regression	188,317	131	not cleared	iid
amazon_employee_access	binary	32,769	10	public-domain	iid
amex_non_iid	binary	1,249,605	191	not cleared	grouped
anes_voting_2026	binary	48,587	319	not cleared	temporal
aps_failure	binary	76,000	171	CC-BY-4.0	iid
asp_potassco_classification	multiclass	1,212	138	not cleared	grouped
audiology_diagnosis	multiclass	199	69	CC-BY-4.0	iid
bad_customer_detection	binary	1,723	14	public-domain	iid
bank_customer_churn	binary	10,000	11	public-domain	iid
bank_marketing	binary	45,211	14	CC-BY-4.0	iid
biogeographical_ancestry_prediction	multiclass	635	105	not cleared	iid
biomechanical_orthopaedic_prediction	multiclass	310	7	CC-BY-4.0	iid
bioresponse	binary	3,751	1777	public-domain	iid
blood_tests_drink_prediction	regression	345	6	CC-BY-4.0	iid
blood_transfusion	binary	748	5	CC-BY-4.0	iid
body_density_prediction	regression	252	14	not cleared	iid
california_house_prices_2020	regression	41,528	42	not cleared	temporal
cardiotocography	multiclass	2,126	24	CC-BY-4.0	grouped
churn	binary	5,000	20	public-domain	iid
cirrhosis_patient_survival_prediction	regression	161	18	not cleared	iid
climate_model_weather_forecasting	regression	1,250,000	101	not cleared	temporal
clock_protein_toxicity	binary	171	1118	CC-BY-4.0	iid
coffee_rating_prediction	regression	2,369	13	not cleared	temporal
coil_2000	binary	9,822	86	CC-BY-4.0	iid
concrete_compressive_strength	regression	1,030	9	CC-BY-4.0	iid
consumer_complaints	multiclass	1,226,140	13	us-gov	temporal
cooking_time	regression	1,250,000	197	not cleared	temporal
covertype	multiclass	512,625	15	CC-BY-4.0	grouped
credit_approval	binary	690	16	CC-BY-4.0	iid
credit_card_clients_default	binary	30,000	24	CC-BY-4.0	iid
credit_g	binary	1,000	21	CC-BY-4.0	iid
customer_satisfaction_in_airline	binary	129,880	22	public-domain	iid

Table 45, continued

name	task	rows	cols	license	split
delivery_eta	regression	1,250,000	226	not cleared	temporal
dementia_prediction	multiclass	370	10	not cleared	grouped
diabetes_130_us	binary	71,518	45	CC-BY-4.0	iid
diamonds	regression	53,940	10	MIT	iid
drug_induced_autoimmunity_prediction	binary	597	178	CC-BY-4.0	iid
early_learning_predictors	regression	18,874	745	CC-BY-4.0	grouped
early_stage_diabetes_risk_prediction	binary	251	17	CC-BY-4.0	iid
ecoli_proteins	multiclass	327	7	CC-BY-4.0	iid
ecommerce_shipping	binary	10,999	11	public-domain	iid
electric_motor_temperature_prediction	regression	1,296,316	111	not cleared	grouped
emscad	binary	17,460	18	CC0-1.0	iid
eryhemato_squamous_disease	multiclass	366	35	CC-BY-4.0	iid
fiat_500	regression	1,538	8	CC0-1.0	iid
fitness_club	binary	1,500	7	public-domain	iid
food_delivery_time	regression	45,451	10	not cleared	iid
forensic_glass_identification	multiclass	214	10	CC-BY-4.0	iid
forest_fires	regression	517	13	CC-BY-4.0	iid
gallstone_disease	binary	319	39	CC-BY-4.0	iid
garments_worker_productivity	regression	1,197	16	CC-BY-4.0	temporal
ghanas_indigenous_intel	multiclass	10,928	11	not cleared	temporal
give_me_some_credit	binary	150,000	11	not cleared	iid
hazelnut_spread_contaminant_detection	binary	2,400	31	not cleared	iid
healthcare_insurance_expenses	regression	1,338	7	not cleared	iid
heart_disease_cleveland	binary	303	14	CC-BY-4.0	iid
heart_disease_hungary	binary	294	14	CC-BY-4.0	iid
heart_disease_va_long_beach	binary	200	14	CC-BY-4.0	iid
heart_failure_followup_survival	binary	299	13	CC-BY-4.0	iid
heloc	binary	10,459	24	not cleared	iid
hepatitis_c_prediction	multiclass	608	13	CC-BY-4.0	iid
hepatitis_survival_prediction	binary	155	20	CC-BY-4.0	iid
hiva_agnostic	binary	3,845	1519	public-domain	iid
home_credit_default_risk	binary	307,507	505	not cleared	iid
home_credit_default_stability	binary	1,224,927	712	not cleared	temporal
homeq_default_prediction	binary	5,708	13	not cleared	iid
homesite_quote_conversion	binary	260,753	296	not cleared	iid
horse_colic_survival	multiclass	344	21	CC-BY-4.0	iid
hotel_booking_demand	binary	81,418	32	CC-BY-4.0	temporal
houses	regression	19,675	9	not cleared	iid
hr_analytics	binary	19,158	13	CC0-1.0	iid
ieee_fraud_detection	binary	590,540	436	not cleared	temporal
immoscout_german_house_prices	regression	10,317	24	not cleared	iid
in_vehicle_coupon_recommendation	binary	12,684	25	CC-BY-4.0	iid
indian_liver_patient_dataset	binary	583	11	CC-BY-4.0	iid
iranian_churn	binary	2,850	14	CC-BY-4.0	iid
jm1	binary	10,885	22	public-domain	iid
kdd_cup_09_appetency	binary	50,000	213	public-domain	iid
kick	binary	72,983	33	public-domain	temporal
kickstarter	binary	187,118	16	not cleared	temporal
labour_inspection_compliance	binary	63,634	377	CC0-1.0	iid
lending_club	binary	1,064,751	97	CC0-1.0	temporal
ljubljana_breast_cancer	binary	286	10	CC-BY-4.0	iid
ljubljana_primary_tumor	multiclass	302	18	CC-BY-4.0	iid
lung_cancer	multiclass	197	12601	not cleared	iid
lung_cancer_epithelial_genexp	binary	187	22216	public-domain	iid
maps_router_eta	regression	1,250,000	989	not cleared	temporal
marketing_campaign	binary	2,240	26	CC-BY-4.0	iid
maternal_health_risk	multiclass	1,014	7	CC-BY-4.0	iid
mercari_price_suggestion	regression	1,250,000	7	not cleared	iid
mercedes_benz_greener_manufacturing	regression	4,204	372	not cleared	temporal
miami_housing	regression	13,776	16	not cleared	iid
mic	multiclass	1,699	112	CC-BY-4.0	iid
mice_protein_trisomy_discriminant	multiclass	1,080	78	CC-BY-4.0	grouped
micro_mass	multiclass	571	1084	CC-BY-4.0	grouped
musk	binary	6,598	168	CC-BY-4.0	grouped
mutual_funds_india	regression	793	13	CC0-1.0	iid
naticusdroid_android_permissions_dataset	binary	7,491	86	CC-BY-4.0	iid
obesity_estimation	regression	498	15	CC-BY-4.0	iid
online_shoppers_purchasing_intention_dataset	binary	12,330	18	CC-BY-4.0	iid
otto_group_product_classification_challenge	multiclass	61,878	94	not cleared	iid
pancreatic_cancer_mouse_detection	binary	181	6773	not cleared	grouped
parkinsons_biomedical_voice_measurements	binary	195	25	CC-BY-4.0	grouped
physiochemical_protein	regression	45,730	10	CC-BY-4.0	iid
polish_companies_bankruptcy	binary	5,790	65	CC-BY-4.0	iid
porto_seguro	binary	595,206	38	not cleared	iid
predict_students_dropout_and_academic_success	multiclass	4,424	37	CC-BY-4.0	iid
prostate_cancer_detection	binary	322	15155	not cleared	iid
pva_revenue_prediction_kddcup98	binary	144,095	478	not cleared	iid
qsar_aquatic_toxicity	regression	546	9	CC-BY-4.0	iid

Table 45, continued

name	task	rows	cols	license	split
qsar_biodeg	binary	1,054	42	CC-BY-4.0	iid
qsar_fish_toxicity	regression	908	7	CC-BY-4.0	iid
qsar_tid_11	regression	5,741	1025	public-domain	iid
regensburg_pediatric_appendicitis	binary	763	52	not cleared	iid
rossmann_store_sales	regression	844,392	16	not cleared	temporal
santander_customer_satisfaction	binary	71,080	308	not cleared	iid
santander_customer_transaction_prediction	binary	200,000	601	not cleared	iid
santander_transaction_value	regression	4,447	541	not cleared	iid
sat11_hand_algo_runtime	regression	2,960	171	not cleared	grouped
sberbank_housing_market_forecasting	regression	27,195	387	not cleared	temporal
sdss_17	multiclass	78,053	12	public-domain	iid
seismic_bumps	binary	2,584	16	CC-BY-4.0	iid
sepsis_prediction	binary	1,228,686	44	not cleared	grouped
sepsis_survival_minimal_clinical_records	binary	110,204	4	CC-BY-4.0	iid
sf_permit_time	regression	116,954	38	not cleared	temporal
south_africa_coronary_heart_disease	binary	462	10	not cleared	iid
splice	multiclass	3,190	61	CC-BY-4.0	iid
student_portuguese_performance	regression	649	31	CC-BY-4.0	iid
superconductivity	regression	21,263	82	CC-BY-4.0	iid
taiwanese_bankruptcy_prediction	binary	6,819	93	CC-BY-4.0	iid
telemonitoring_parkinsons_biomedical_voice_measurements	regression	502	21	CC-BY-4.0	grouped
thyroid_discordant	binary	3,711	27	CC-BY-4.0	iid
tour_travels_churn	binary	954	7	CC0-1.0	iid
video_game_fps_prediction	regression	12,288	40	CC-BY-4.0	grouped
video_transcoding_time_prediction	regression	68,784	20	CC-BY-4.0	grouped
website_phishing	multiclass	1,353	10	CC-BY-4.0	iid
wids_diabetes_mellitus	binary	127,358	182	not cleared	iid
wine_quality	regression	6,497	13	CC-BY-4.0	iid
wine_world_cost	regression	1,279	15	CC0-1.0	iid

K Extended related work

Table 46 groups the generator literature by family and states, for each, what it fits and where DOPE differs. No row is a measured comparison; the measured generators follow below.

What this paper measured. The measured generators are the Gaussian copula, a Chow–Liu baseline and independent marginals written for this comparison, CTGAN, TVAE, ARF at five fit seeds, TabSyn on all 100 lineages with a scaled schedule at five fit seeds (six sample cells of one seed-23 lineage reached the sample time cap and stay unscored, and one other lineage has no stored seed-11 samples), and a historical Forest-Flow block; TabDDPM, TabDiff, a language-model generator, synthpop CART, and SMOTE are scheduled under a common budget in predeclare_v2. This panel scores regression retention.

Utility ratios, fidelity, and privacy. Retention is a null-normalized TSTR/TRTR ratio; Esteban, Hyland, and Rätsch introduced train-on-synthetic, test-on-real and train-on-real, test-on-real (Esteban et al., 2017). Utility reviews separate marginal fidelity from downstream prediction (Snoke et al., 2018; Raab et al., 2018); alpha-precision, beta-recall, SDMetrics, and synthcity are the usual fidelity reports (Alaa et al., 2022; Zhang et al., 2020; Qian et al., 2023). Differentially private Bayesian networks and marginal synthesizers optimize a privacy budget (Zhang et al., 2014; McKenna et al., 2021; 2022; Cai et al., 2021). Membership inference, shadow-model attacks, and holdout-referenced distances form the empirical privacy literature (Shokri et al., 2017; Hayes et al., 2018; van Breugel et al., 2023; Stadler et al., 2022; Platzer & Reutterer, 2021); formal differential privacy is a different claim from an empirical resemblance score (Dwork & Roth, 2014; Abadi et al., 2016).

Compression, distillation, and terminology. Minimum description length and Occam's-razor arguments tie short descriptions to prediction (Rissanen, 1978; Blumer et al., 1987), and the information bottleneck keeps what a compressed variable says about a target (Tishby et al., 2000); this paper uses them as motivation and estimates neither description length nor mutual information. Dataset distillation keeps model accuracy in a few learned examples (Wang et al., 2018), knowledge distillation keeps a model's behavior in a smaller model (Hinton et al., 2015), and algorithm-ranking preservation is an earlier synthetic-data criterion (Jordon et al., 2018). Generative compression of images and video predates this work (Santurkar et al., 2017); the title uses the phrase descriptively for a table-generating program judged by a declared learning loss.

Table 33: **Grouped-validation fidelity and empirical privacy, fit seed 11.** Each cell is a median with its 95% lineage-bootstrap interval. The fidelity denominator is the complete three-seed group count, distinct from the 97 retention lineages. Holm families are 12 density tests (DOPE against the Gaussian copula, Chow–Liu, and independent marginals on each of the four metrics) and 8 neural tests (DOPE against CTGAN and TVAE on each of the four metrics), separate from retention.

Method	α-prec.	β-recall	DCR share (unbalanced)	DOMIAS
Median [interval] (lineages).				
DOPE	0.905 [0.889, 0.918] (98)	0.437 [0.398, 0.473] (98)	0.811 [0.806, 0.814] (98)	0.508 [0.498, 0.520] (94)
Gaussian	0.812 [0.786, 0.832] (100)	0.430 [0.378, 0.453] (100)	0.806 [0.802, 0.811] (100)	0.513 [0.494, 0.527] (94)
Chow–Liu	0.858 [0.825, 0.904] (100)	0.359 [0.312, 0.400] (100)	0.810 [0.806, 0.816] (100)	0.512 [0.495, 0.522] (94)
Indep.	0.885 [0.868, 0.897] (100)	0.284 [0.197, 0.347] (100)	0.801 [0.797, 0.806] (100)	0.507 [0.496, 0.521] (94)
DOPE	0.902 [0.882, 0.940] (22)	0.441 [0.381, 0.489] (22)	0.811 [0.802, 0.819] (22)	0.508 [0.480, 0.538] (22)
CTGAN	0.719 [0.520, 0.856] (22)	0.170 [0.142, 0.267] (22)	0.820 [0.800, 0.842] (22)	0.502 [0.463, 0.534] (22)
TVAE	0.850 [0.731, 0.896] (22)	0.356 [0.283, 0.401] (22)	0.825 [0.813, 0.843] (22)	0.527 [0.487, 0.563] (22)

Contrast	α-prec.	β-recall	DCR share (unbalanced)	DOMIAS
Line 2: Holm p; rank-biserial r; n. DOPE minus comparator.				
DOPE−Gaussian	0.083 [0.045, 0.113] 1.86×10^{-10}; 0.781; 98	0.024 [0.011, 0.039] 0.394; 0.215; 98	-0.0025 [-0.00625, 0.005] 1.000; 0.000206; 98	0.00157 [-0.0069, 0.0074] 1.000; -0.085; 94
DOPE−Chow–Liu	0.019 [0.00782, 0.056] 2.00×10^{-5}; 0.551; 98	0.048 [0.029, 0.072] 1.32×10^{-10}; 0.789; 98	-0.00125 [-0.00562, 0.00125] 0.806; -0.169; 98	0.000581 [-0.00802, 0.011] 1.000; 0.00572; 94
DOPE−Indep.	0.018 [0.0072, 0.028] 5.47×10^{-5}; 0.523; 98	0.104 [0.068, 0.138] 9.71×10^{-15}; 0.937; 98	0.00625 [0.0025, 0.011] 0.005; 0.398; 98	0.00244 [-0.00595, 0.015] 1.000; 0.080; 94
DOPE−CTGAN	0.168 [0.054, 0.368] 0.004; 0.794; 22	0.219 [0.131, 0.256] 3.81×10^{-6}; 1.000; 22	-0.00812 [-0.029, 0.015] 0.532; -0.277; 22	-0.00028 [-0.014, 0.031] 0.610; 0.130; 22
DOPE−TVAE	0.071 [0.00713, 0.141] 0.040; 0.644; 22	0.062 [0.00546, 0.135] 0.069; 0.573; 22	-0.025 [-0.030, -0.00641] 0.059; -0.613; 22	-0.021 [-0.036, -0.00972] 0.108; -0.510; 22

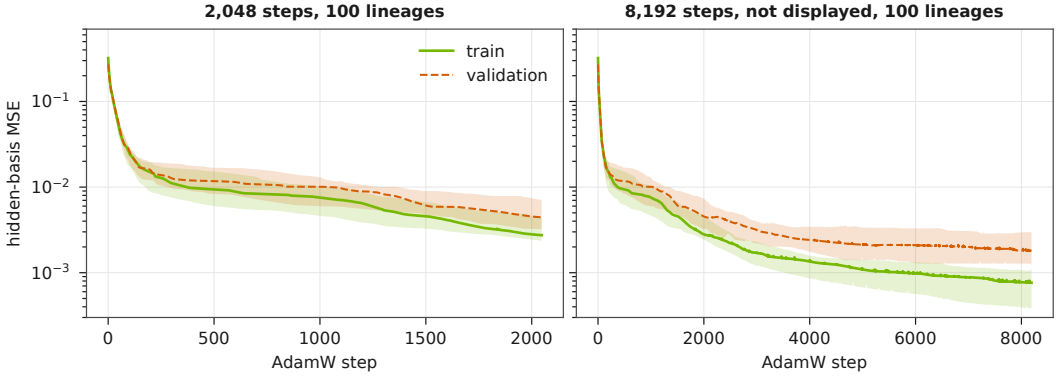


Figure 7: **Logged replay of both step budgets before the readout refit.** The solid line is the training median and the dashed line the validation median; each band is a 95% lineage-bootstrap interval. The longer budget is not the reported model.

Auditors and rank tests. Tree models remain strong tabular predictors, which is why CatBoost is one of the three auditors (Grinsztajn et al., 2022; Prokhorenkova et al., 2018; Shwartz-Ziv & Armon, 2022). TabZilla and TabArena show how brittle a single split can be (McElfresh et al., 2023; Erickson et al., 2026). The regression lineages come from PMLB (Olson et al., 2017; Romano et al., 2022). Comparing several methods on the same lineages needs a rank test and a critical difference, or a Bayesian signed-rank alternative (Demšar, 2006; Benavoli et al., 2017).

REFERENCES

Kjersti Aas, Claudia Czado, Arnoldo Frigessi, and Henrik Bakken. Pair-copula constructions of multiple dependence. *Insurance: Mathematics and Economics*, 44(2):182–198, 2009. doi: 10.1016/j.insmatheco.2007.02.001.

Martin Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC*

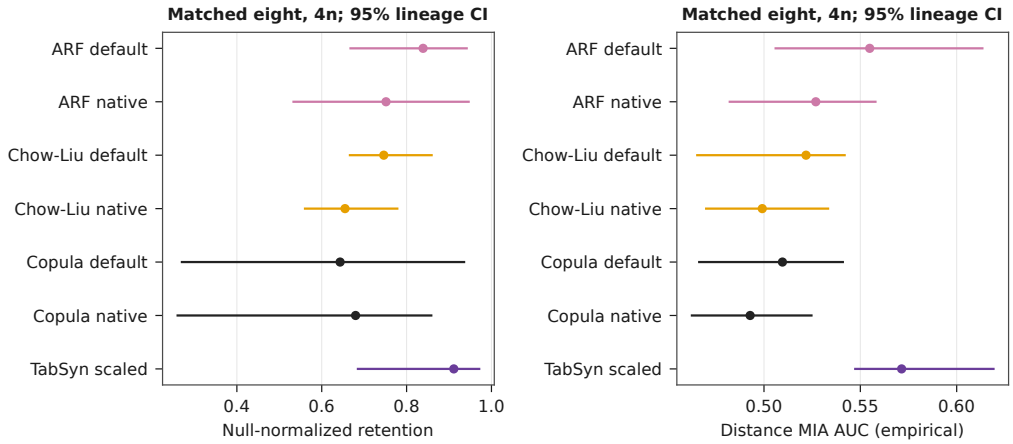


Figure 8: **Earlier eight-lineage TabSyn subset: medians and marginal lineage-bootstrap intervals.** No Holm family. Privacy AUC is an empirical diagnostic.

Conference on Computer and Communications Security, pp. 308–318, 2016. doi: 10.1145/2976749.2978318.

Ahmed Alaa, Boris Van Breugel, Evgeny S. Saveliev, and Mihaela van der Schaar. How faithful is your synthetic data? sample-level metrics for evaluating and auditing generative models. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 290–306. PMLR, 2022. doi: 10.48550/ARXIV.2102.08921. URL https://proceedings.mlr.press/v162/alaa22a.html.

Alessio Benavoli, Giorgio Corani, Janez Demšar, and Marco Zaffalon. Time for a change: a tutorial for comparing multiple classifiers through bayesian analysis. *Journal of Machine Learning Research*, 18(77):1–36, 2017. URL https://jmlr.org/papers/v18/16-305.html.

Anselm Blumer, Andrzej Ehrenfeucht, David Haussler, and Manfred K. Warmuth. Occam's razor. *Information Processing Letters*, 24(6):377–380, 1987. doi: 10.1016/0020-0190(87)90114-1.

Vadim Borisov, Kathrin Seßler, Tobias Leemann, Martin Pawelczyk, and Gjergji Kasneci. Language models are realistic tabular data generators. In *International Conference on Learning Representations*, 2023. doi: 10.48550/ARXIV.2210.06280. URL https://openreview.net/forum?id=cEygmQNOeI.

Vadim Borisov, Tobias Leemann, Kathrin Seßler, Johannes Haug, Martin Pawelczyk, and Gjergji Kasneci. Deep neural networks and tabular data: A survey. *IEEE Transactions on Neural Networks and Learning Systems*, 35(6):7499–7519, 2024. doi: 10.1109/tnnls.2022.3229161.

Kuntai Cai, Xiaoyu Lei, Jianxin Wei, and Xiaokui Xiao. Data synthesis via differentially private markov random fields. *Proceedings of the VLDB Endowment*, 14(11):2190–2202, 2021. doi: 10.14778/3476249.3476272.

N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002. doi: 10.1613/jair.953.

C. K. Chow and C. N. Liu. Approximating discrete probability distributions with dependence trees. *IEEE Transactions on Information Theory*, 14(3):462–467, May 1968. doi: 10.1109/TIT.1968.1054142.

Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7(1):1–30, 2006. URL https://jmlr.org/papers/v7/demsar06a.html.

Cynthia Dwork and Aaron Roth. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9(3–4):211–487, 2014. doi: 10.1561/0400000042.

Nick Erickson, Lennart Purucker, Andrej Tschalzev, David Holzmüller, Prateek Mutalik Desai, David Salinas, and Frank Hutter. TabArena: A living benchmark for machine learning on tabular data. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, pp. 17285–17350, 2026. doi: 10.52202/085713-0519. URL https://openreview.net/forum?id=jZqCqpCLdU.

Cristóbal Esteban, Stephanie L. Hyland, and Gunnar Rätsch. Real-valued (medical) time series generation with recurrent conditional GANs. *arXiv preprint arXiv:1706.02633*, 2017. doi: 10.48550/ARXIV.1706.02633.

Leo Grinsztajn, Edouard Oyallon, and Gael Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems*, volume 35, pp. 507–520, 2022. doi: 10.52202/068431-0037.

Jamie Hayes, Luca Melis, George Danezis, and Emiliano De Cristofaro. LOGAN: Membership inference attacks against generative models. *Proceedings on Privacy Enhancing Technologies*, 2019(1):133–152, 2018. doi: 10.2478/popets-2019-0008.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015. doi: 10.48550/arXiv.1503.02531.

Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326, 2025. doi: 10.1038/s41586-024-08328-6.

Alexia Jolicoeur-Martineau, Kilian Fatras, and Tal Kachman. Generating and imputing tabular data via diffusion and flow-based gradient-boosted trees. In *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pp. 1288–1296. PMLR, 2024. doi: 10.48550/ARXIV.2309.09968. URL https://proceedings.mlr.press/v238/jolicoeur-martineau24a.html.

James Jordon, Jinsung Yoon, and Mihaela van der Schaar. Measuring the quality of synthetic data for use in competitions. *arXiv preprint arXiv:1806.11345*, 2018. doi: 10.48550/arXiv.1806.11345.

Jayoung Kim, Chaejeong Lee, and Noseong Park. STaSy: Score-based tabular data synthesis. In *International Conference on Learning Representations*, 2023. doi: 10.48550/ARXIV.2210.04018. URL https://openreview.net/forum?id=1mNssCWt_v.

Diederik P. Kingma and Max Welling. Auto-encoding variational Bayes. In *International Conference on Learning Representations*, 2014. doi: 10.48550/arXiv.1312.6114.

Akim Kotelnikov, Dmitry Baranchuk, Ivan Rubachev, and Artem Babenko. TabDDPM: Modelling tabular data with diffusion models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 17564–17579, 2023. doi: 10.48550/ARXIV.2209.15421. URL https://proceedings.mlr.press/v202/kotelnikov23a.html.

Chaejeong Lee, Jayoung Kim, and Noseong Park. CoDi: Co-evolving contrastive diffusion models for mixed-type tabular synthesis. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 18940–18956. PMLR, 2023. doi: 10.48550/ARXIV.2304.12654. URL https://proceedings.mlr.press/v202/lee23i.html.

Tennison Liu, Zhaozhi Qian, Jeroen Berrevoets, and Mihaela van der Schaar. GOGGLE: Generative modelling for tabular data by learning relational structure. In *International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=fPVRcJqspu.

Junwei Ma, Apoorv Dankar, George Stein, Guangwei Yu, and Anthony Caterini. TabPFGen – tabular data generation with TabPFN. In *NeurIPS 2023 Second Table Representation Learning Workshop*, 2023. URL https://openreview.net/forum?id=4MkkNsAEmO.

Vikash Mansinghka, Patrick Shafto, Eric Jonas, Cap Petschulat, Max Gasner, and Joshua B. Tenenbaum. CrossCat: A fully Bayesian nonparametric method for analyzing heterogeneous, high dimensional data. *Journal of Machine Learning Research*, 17(138):1–49, 2016. URL https://jmlr.org/papers/v17/11-392.html.

Duncan McElfresh, Sujay Khandagale, Jonathan Valverde, Vishak Prasad C, Ganesh Ramakrishnan, Micah Goldblum, and Colin White. When do neural nets outperform boosted trees on tabular data? In *Advances in Neural Information Processing Systems*, volume 36, pp. 76336–76369, 2023. doi: 10.52202/075280-3337. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/f06d5ebd4ff40b40dd97e30cee632123-Abstract-Datasets_and_Benchmarks.html. Datasets and Benchmarks Track.

Ryan McKenna, Gerome Miklau, and Daniel Sheldon. Winning the NIST contest: A scalable and general approach to differentially private synthetic data. *Journal of Privacy and Confidentiality*, 11(3), 2021. doi: 10.29012/jpc.778.

Ryan McKenna, Brett Mullins, Daniel Sheldon, and Gerome Miklau. AIM: An adaptive and iterative mechanism for differentially private synthetic data. *Proceedings of the VLDB Endowment*, 15(11):2599–2612, 2022. doi: 10.14778/3551793.3551817.

Markus Mueller, Kathrin Gruber, and Dennis Fok. Continuous diffusion for mixed-type tabular data. In *International Conference on Learning Representations*, 2025. doi: 10.48550/ARXIV.2312.10431. URL https://openreview.net/forum?id=QPtoBPn4lZ.

Beata Nowok, Gillian M. Raab, and Chris Dibben. synthpop: Bespoke creation of synthetic data in R. *Journal of Statistical Software*, 74(11):1–26, 2016. doi: 10.18637/jss.v074.i11.

Randal S. Olson, William La Cava, Patryk Orzechowski, Ryan J. Urbanowicz, and Jason H. Moore. PMLB: a large benchmark suite for machine learning evaluation and comparison. *BioData Mining*, 10(1):36, 2017. doi: 10.1186/s13040-017-0154-4.

Noseong Park, Mahmoud Mohammadi, Kshitij Gorde, Sushil Jajodia, Hongkyu Park, and Youngmin Kim. Data synthesis based on generative adversarial networks. *Proceedings of the VLDB Endowment*, 11(10):1071–1083, 2018. doi: 10.14778/3231751.3231757.

Neha Patki, Roy Wedge, and Kalyan Veeramachaneni. The synthetic data vault. In *2016 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, pp. 399–410, 2016. doi: 10.1109/DSAA.2016.49.

Michael Platzer and Thomas Reutterer. Holdout-based empirical assessment of mixed-type synthetic data. *Frontiers in Big Data*, 4:679939, 2021. doi: 10.3389/fdata.2021.679939.

Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, 2018. doi: 10.48550/ARXIV.1706.09516. URL https://papers.nips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html.

Zhaozhi Qian, Rob Davis, and Mihaela van der Schaar. Synthcity: A benchmark framework for diverse use cases of tabular synthetic data. In *Advances in Neural Information Processing Systems*, volume 36, pp. 3173–3188, 2023. doi: 10.52202/075280-0140. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/09723c9f291f6056fd1885081859c186-Abstract-Datasets_and_Benchmarks.html. Datasets and Benchmarks Track.

Gillian M. Raab, Beata Nowok, and Chris Dibben. Practical data synthesis for large samples. *Journal of Privacy and Confidentiality*, 7(3):67–97, 2018. doi: 10.29012/jpc.v7i3.407.

Jorma Rissanen. Modeling by shortest data description. *Automatica*, 14(5):465–471, 1978. doi: 10.1016/0005-1098(78)90005-5.

Joseph D. Romano, Trang T. Le, William La Cava, John T. Gregg, Daniel J. Goldberg, Praneel Chakraborty, Natasha L. Ray, Daniel Himmelstein, Weixuan Fu, and Jason H. Moore. PMLB v1.0: an open-source dataset collection for benchmarking machine learning methods. *Bioinformatics*, 38(3):878–880, 2022. doi: 10.1093/bioinformatics/btab727.

Feras A. Saad, Marco F. Cusumano-Towner, Ulrich Schaechtle, Martin C. Rinard, and Vikash K. Mansinghka. Bayesian synthesis of probabilistic programs for automatic data modeling. *Proceedings of the ACM on Programming Languages*, 3(POPL):1–32, 2019. doi: 10.1145/3290350.

Shibani Santurkar, David Budden, and Nir Shavit. Generative compression. *arXiv preprint arXiv:1703.01467*, 2017. doi: 10.48550/arXiv.1703.01467.

Juntong Shi, Minkai Xu, Harper Hua, Hengrui Zhang, Stefano Ermon, and Jure Leskovec. TabDiff: a mixed-type diffusion model for tabular data generation. In *International Conference on Learning Representations*, 2025. doi: 10.48550/ARXIV.2410.20626. URL https://openreview.net/forum?id=swvURjrt8z.

Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *2017 IEEE Symposium on Security and Privacy (SP)*, pp. 3–18, 2017. doi: 10.1109/sp.2017.41.

Ravid Shwartz-Ziv and Amitai Armon. Tabular data: Deep learning is not all you need. *Information Fusion*, 81:84–90, 2022. doi: 10.1016/j.inffus.2021.11.011.

Joshua Snoke, Gillian M. Raab, Beata Nowok, Chris Dibben, and Aleksandra Slavkovic. General and specific utility measures for synthetic data. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 181(3):663–688, 2018. doi: 10.1111/rssa.12358.

Aivin V. Solatorio and Olivier Dupriez. REaLTabFormer: Generating realistic relational and tabular data using transformers, 2023.

Theresa Stadler, Bristena Oprisanu, and Carmela Troncoso. Synthetic data – anonymisation groundhog day. In *31st USENIX Security Symposium (USENIX Security 22)*, pp. 1451–1468, 2022. URL https://www.usenix.org/conference/usenixsecurity22/presentation/stadler.

Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. *arXiv preprint physics/0004057*, 2000. doi: 10.48550/arXiv.physics/0004057.

Boris van Breugel, Hao Sun, Zhaozhi Qian, and Mihaela van der Schaar. Membership inference attacks against synthetic data through overfitting detection. In *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pp. 3493–3514. PMLR, 2023. doi: 10.48550/ARXIV.2302.12580. URL https://proceedings.mlr.press/v206/breugel23a.html.

Tongzhou Wang, Jun-Yan Zhu, Antonio Torralba, and Alexei A. Efros. Dataset distillation. *arXiv preprint arXiv:1811.10959*, 2018. doi: 10.48550/arXiv.1811.10959.

David S. Watson, Kristin Blesch, Jan Kapar, and Marvin N. Wright. Adversarial random forests for density estimation and generative modeling. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pp. 5357–5375, 2023. doi: 10.48550/ARXIV.2205.09435. URL https://proceedings.mlr.press/v206/watson23a.html.

Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling tabular data using conditional GAN. In *Advances in Neural Information Processing Systems*, volume 32, 2019. doi: 10.48550/ARXIV.1907.00503. URL https://papers.nips.cc/paper/2019/hash/254ed7d2de3b23ab10936522dd547b78-Abstract.html.

David Turtora Zagardo. Geometry-aware tabular diffusion. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pp. 152812–152835. PMLR, 2026. doi: 10.48550/ARXIV.2606.02607. URL https://proceedings.mlr.press/v306/zagardo26a.html.

Hengrui Zhang, Jiani Zhang, Balasubramaniam Srinivasan, Zhengyuan Shen, Xiao Qin, Christos Faloutsos, Huzefa Rangwala, and George Karypis. Mixed-type tabular data synthesis with score-based diffusion in latent space. In *International Conference on Learning Representations*, 2024. doi: 10.48550/ARXIV.2310.09656. URL https://openreview.net/forum?id=4Ay23yeuz0.

Jun Zhang, Graham Cormode, Cecilia M. Procopiuc, Divesh Srivastava, and Xiaokui Xiao. PrivBayes: Private data release via bayesian networks. In *Proceedings of the 2014 ACM SIGMOD International Conference on Management of Data*, pp. 1423–1434, 2014. doi: 10.1145/2588555.2588573.

Kevin Alex Zhang, Kalyan Veeramachaneni, and Carles Sala. SDMetrics: Synthetic data metrics. Open-source library, MIT Data To AI Lab, 2020. URL https://docs.sdv.dev/sdmetrics/.

Zilong Zhao, Aditya Kunar, Robert Birke, and Lydia Y. Chen. CTAB-GAN: Effective table data synthesizing. In *Proceedings of The 13th Asian Conference on Machine Learning*, volume 157 of *Proceedings of Machine Learning Research*, pp. 97–112. PMLR, 2021. URL https://proceedings.mlr.press/v157/zhao21a.html.

Zilong Zhao, Aditya Kunar, Robert Birke, Hiek Van der Scheer, and Lydia Y. Chen. CTAB-GAN+: Enhancing tabular data synthesis. *Frontiers in Big Data*, 6, 2024. doi: 10.3389/fdata.2023.1296508.

Table 34: **Fidelity and detection by configuration and size.** Median, hierarchical interval, and lineage count in each measured cell. Common-numeric cohorts, with separate availability by metric. Stored logistic AUC and grouped CatBoost AUC have distinct columns. AUC alone establishes neither privacy nor release safety. No Holm family.

Method	Configuration	Size	Marginal KS	Pair fidelity	Logistic AUC	CatBoost AUC
ARF	author default	$1n$	0.1128 [0.0895, 0.1433] $n = 100$	0.9595 [0.9471, 0.9644] $n = 100$	0.5321 [0.5156, 0.5442] $n = 100$	0.6159 [0.5751, 0.6928] $n = 100$
ARF	author default	$4n$	0.1035 [0.0846, 0.1340] $n = 100$	0.9618 [0.9487, 0.9653] $n = 100$	0.5268 [0.5069, 0.5388] $n = 100$	0.6120 [0.5696, 0.6877] $n = 100$
ARF	native selected	$1n$	0.1122 [0.0869, 0.1445] $n = 100$	0.9599 [0.9418, 0.9626] $n = 100$	0.5255 [0.5072, 0.5449] $n = 100$	0.6100 [0.5789, 0.7089] $n = 100$
ARF	native selected	$4n$	0.1046 [0.0845, 0.1382] $n = 100$	0.9620 [0.9464, 0.9643] $n = 100$	0.5296 [0.5094, 0.5375] $n = 100$	0.6091 [0.5766, 0.7075] $n = 100$
CTGAN	native selected	$1n$	0.2319 [0.1887, 0.2943] $n = 22$	0.9091 [0.8313, 0.9406] $n = 22$	0.7839 [0.6997, 0.8819] $n = 22$	0.9522 [0.8778, 0.9841] $n = 21$
CTGAN	native selected	$4n$	0.2300 [0.1839, 0.2947] $n = 22$	0.9123 [0.8367, 0.9431] $n = 22$	0.7839 [0.6997, 0.8821] $n = 22$	0.9535 [0.8487, 0.9777] $n = 22$
Chow-Liu	native selected	$1n$	0.1293 [0.0892, 0.2361] $n = 100$	0.9273 [0.8834, 0.9418] $n = 100$	0.5423 [0.5040, 0.6292] $n = 100$	0.6868 [0.6025, 0.8986] $n = 100$
Chow-Liu	native selected	$4n$	0.1266 [0.0852, 0.2292] $n = 100$	0.9283 [0.8906, 0.9465] $n = 100$	0.5404 [0.5185, 0.6303] $n = 100$	0.7061 [0.5910, 0.9369] $n = 99$
DOPE	DOPE profile	$1n$	0.1216 [0.0950, 0.1680] $n = 98$	0.9532 [0.9317, 0.9610] $n = 98$	0.5172 [0.5000, 0.5376] $n = 98$	0.6344 [0.5705, 0.8368] $n = 98$
DOPE	DOPE profile	$4n$	0.1138 [0.0917, 0.1583] $n = 98$	0.9531 [0.9346, 0.9645] $n = 98$	0.5172 [0.5000, 0.5376] $n = 98$	0.6405 [0.5830, 0.8312] $n = 98$
Forest-Flow	native selected	$1n$	0.1254 [0.0612, 0.1992] $n = 6$	0.9548 [0.9242, 0.9799] $n = 6$	0.5187 [0.4139, 0.7000] $n = 6$	—
Forest-Flow	native selected	$4n$	0.1231 [0.0557, 0.2062] $n = 6$	0.9691 [0.9321, 0.9811] $n = 6$	0.5187 [0.4139, 0.7000] $n = 6$	—
Gaussian copula	native selected	$1n$	0.1393 [0.1171, 0.2132] $n = 100$	0.9614 [0.9516, 0.9652] $n = 100$	0.5246 [0.5100, 0.5518] $n = 100$	0.7343 [0.6789, 0.8981] $n = 100$
Gaussian copula	native selected	$4n$	0.1336 [0.1137, 0.2115] $n = 100$	0.9635 [0.9488, 0.9687] $n = 100$	0.5246 [0.5100, 0.5537] $n = 100$	0.7346 [0.6902, 0.8689] $n = 99$
TVAE	native selected	$1n$	0.1500 [0.1199, 0.2598] $n = 22$	0.9240 [0.8905, 0.9418] $n = 22$	0.5918 [0.5616, 0.6392] $n = 22$	0.7479 [0.7221, 0.8859] $n = 22$
TVAE	native selected	$4n$	0.1382 [0.1174, 0.2431] $n = 22$	0.9258 [0.8913, 0.9451] $n = 22$	0.6033 [0.5544, 0.6681] $n = 22$	0.7309 [0.7032, 0.8893] $n = 22$
TabSyn	scaled schedule	$1n$	0.1244 [0.0828, 0.1416] $n = 99$	0.9543 [0.9412, 0.9598] $n = 99$	—	0.6242 [0.5734, 0.7373] $n = 99$
TabSyn	scaled schedule	$4n$	0.1196 [0.0829, 0.1351] $n = 99$	0.9581 [0.9458, 0.9631] $n = 99$	—	0.6168 [0.5866, 0.7772] $n = 98$
Independent marginals	native selected	$1n$	0.1363 [0.0866, 0.2847] $n = 100$	0.8994 [0.8099, 0.9289] $n = 100$	0.5213 [0.5000, 0.5347] $n = 100$	0.8694 [0.7602, 0.9749] $n = 99$
Independent marginals	native selected	$4n$	0.1297 [0.0813, 0.2672] $n = 100$	0.8986 [0.8258, 0.9329] $n = 100$	0.5166 [0.4988, 0.5488] $n = 100$	0.8578 [0.7553, 0.9727] $n = 100$

Table 35: **Alpha-precision and beta-recall.** Fixed unembedded equations, median over complete sample triples, then hierarchical interval and lineage count. Configuration and cohort are explicit; unavailable cells carry no scalar. The retained TabSyn subset remains separate. No Holm family.

Method	Configuration	Size	Cohort	α precision	β recall
ARF	author default	$1n$	common numeric	0.8838 [0.8644, 0.9144] $n = 100$	0.4269 [0.3237, 0.4727] $n = 100$
ARF	author default	$4n$	common numeric	0.8786 [0.8447, 0.9103] $n = 100$	0.4227 [0.3150, 0.4663] $n = 100$
ARF	native selected	$1n$	common numeric	0.8713 [0.8502, 0.9013] $n = 100$	0.4360 [0.3010, 0.4646] $n = 100$
ARF	native selected	$4n$	common numeric	0.8767 [0.8465, 0.9069] $n = 100$	0.4208 [0.3180, 0.4607] $n = 100$
CTGAN	native selected	$1n$	common numeric	unavailable $n = 0$	unavailable $n = 0$
CTGAN	native selected	$4n$	common numeric	0.7193 [0.4848, 0.8751] $n = 22$	0.1695 [0.1140, 0.2667] $n = 22$
Chow-Liu	native selected	$1n$	common numeric	unavailable $n = 0$	unavailable $n = 0$
Chow-Liu	native selected	$4n$	common numeric	0.8585 [0.7403, 0.9238] $n = 100$	0.3595 [0.1793, 0.4230] $n = 100$
DOPE	DOPE profile	$1n$	common numeric	unavailable $n = 0$	unavailable $n = 0$
DOPE	DOPE profile	$4n$	common numeric	0.9051 [0.8751, 0.9254] $n = 98$	0.4366 [0.2566, 0.4863] $n = 98$
Forest-Flow	native selected	$1n$	common numeric	unavailable $n = 0$	unavailable $n = 0$
Forest-Flow	native selected	$4n$	common numeric	unavailable $n = 0$	unavailable $n = 0$
Gaussian copula	native selected	$1n$	common numeric	unavailable $n = 0$	unavailable $n = 0$
Gaussian copula	native selected	$4n$	common numeric	0.8125 [0.7516, 0.8499] $n = 100$	0.4301 [0.2173, 0.4593] $n = 100$
TVAE	native selected	$1n$	common numeric	unavailable $n = 0$	unavailable $n = 0$
TVAE	native selected	$4n$	common numeric	0.8496 [0.7411, 0.8966] $n = 22$	0.3562 [0.2778, 0.3991] $n = 22$
TabSyn	scaled schedule	$1n$	common numeric	0.8714 [0.8033, 0.9182] $n = 99$	0.4807 [0.3965, 0.5177] $n = 99$
TabSyn	scaled schedule	$1n$	retained subset	0.8418 [0.4700, 0.9326] $n = 8$	0.4591 [0.3520, 0.5267] $n = 8$
TabSyn	scaled schedule	$4n$	common numeric	0.8920 [0.7919, 0.9280] $n = 99$	0.4633 [0.4100, 0.5242] $n = 99$
TabSyn	scaled schedule	$4n$	retained subset	0.8262 [0.4467, 0.9352] $n = 8$	0.4466 [0.3420, 0.5686] $n = 8$
Independent marginals	native selected	$1n$	common numeric	unavailable $n = 0$	unavailable $n = 0$
Independent marginals	native selected	$4n$	common numeric	0.8849 [0.8141, 0.9181] $n = 100$	0.2837 [0.0949, 0.3963] $n = 100$

Table 36: **Baseline coverage from committed publications.** Each row summarizes the retained publication for its method, whose path and digest are in `baseline-coverage.json`; that publication is the row's denominator. The TabSyn row is the earlier retained run; main Tables 1 and 2 report the later full scaled-schedule panel. No Holm family.

Method	Published measured coverage	Remaining campaign
TabSyn	49 scaled-default lineages, 143 measured cells at fit11: eight complete n/4n lineages (48 cells), plus 95 n cells on 41 lineages (19 complete/22 partial groups). Native tuning/full population pending.	Pending
TabDDPM	12 retained lineages; 126 physical/132 logical cells; 21 models, fit seed 11. Default/native cohorts differ; native search incomplete.	Pending
Forest-Flow	13 default lineages at five fit seeds; 65 models/390 cells. Separate legacy six-lineage default/native panel; full population pending.	Pending
Forest-Diffusion	Pilot only; separate 100-lineage matched diffusion panel pending.	Pending
ARF	100 default/native lineages at five fit seeds, n/4n, three sample seeds: 6,000 logical cells / 5,892 physical receipts (108 aliases). Between-fit SD available; official-test certification pending.	Pending
Gaussian copula	100 default/native lineages at fit seed 11; 1200 logical cells. Five-fit common coverage pending.	Pending
Chow-Liu	100 default/native lineages at fit seed 11; 1200 logical cells. Five-fit common coverage pending.	Pending

Table 37: **Retained classical validation at $4n$.** Median [95% lineage-bootstrap interval] (informative lineages). DOPE was not scored in this retained harness, so the table has no DOPE row; DOPE minus each of these baselines is in Table 11, and DOPE's detector cells beside theirs are in Table 34. No Holm family in this table.

Method	Config.	CB retention	KS/TV	C2ST CB AUC	Distance MIA AUC
ARF	Default	0.820 [0.787, 0.872] (99)	0.103 [0.097, 0.114] (100)	0.604 [0.582, 0.626] (100)	0.553 [0.538, 0.564] (100)
ARF	Native	0.806 [0.736, 0.855] (99)	0.105 [0.098, 0.114] (100)	0.610 [0.591, 0.627] (100)	0.542 [0.536, 0.559] (100)
Chow–Liu	Default	0.643 [0.552, 0.672] (99)	0.128 [0.106, 0.154] (100)	0.721 [0.652, 0.773] (100)	0.513 [0.503, 0.525] (100)
Chow–Liu	Native	0.566 [0.460, 0.631] (99)	0.126 [0.105, 0.157] (100)	0.702 [0.641, 0.761] (99)	0.513 [0.504, 0.523] (100)
Copula	Default	0.594 [0.427, 0.762] (99)	0.126 [0.094, 0.138] (100)	0.615 [0.591, 0.671] (100)	0.514 [0.503, 0.524] (100)
Copula	Native	0.641 [0.431, 0.803] (99)	0.134 [0.125, 0.158] (100)	0.730 [0.696, 0.779] (99)	0.507 [0.494, 0.521] (100)

Table 38: **Earlier eight-lineage TabSyn subset at $4n$.** Same inputs and real-data controls for every method; median [95% interval] (lineages). A separate cohort from the full TabSyn panel. DOPE was not scored in this retained harness; DOPE minus TabSyn on the full TabSyn panel's matched lineages is in Table 17. No Holm family in this table.

Method	Config.	CB retention	KS/TV	C2ST CB AUC	Distance MIA AUC
ARF	Default	0.839 [0.668, 0.942] (8)	0.129 [0.084, 0.165] (8)	0.623 [0.547, 0.969] (8)	0.555 [0.506, 0.613] (8)
ARF	Native	0.752 [0.534, 0.946] (8)	0.132 [0.084, 0.163] (8)	0.606 [0.583, 0.812] (8)	0.527 [0.482, 0.558] (8)
Chow–Liu	Default	0.746 [0.666, 0.859] (8)	0.186 [0.068, 0.255] (8)	0.756 [0.525, 0.999] (8)	0.522 [0.465, 0.542] (8)
Chow–Liu	Native	0.655 [0.561, 0.778] (8)	0.188 [0.087, 0.248] (8)	0.707 [0.525, 0.999] (8)	0.499 [0.470, 0.533] (8)
Copula	Default	0.643 [0.270, 0.935] (8)	0.146 [0.067, 0.242] (8)	0.687 [0.554, 0.987] (8)	0.510 [0.466, 0.541] (8)
Copula	Native	0.680 [0.260, 0.858] (8)	0.157 [0.097, 0.242] (8)	0.698 [0.642, 0.987] (8)	0.493 [0.463, 0.525] (8)
TabSyn	Scaled default	0.911 [0.685, 0.971] (8)	0.136 [0.120, 0.157] (8)	0.670 [0.597, 0.869] (8)	0.571 [0.547, 0.619] (8)

Table 39: **Exploratory eight-lineage subset differences, CatBoost.** The denominator is the shared lineages in the cell. Holm family size is twenty-four: TabSyn minus each of the six configurations (ARF, Chow–Liu, and the copula, each at its default and its native setting) on each of the four metrics (CatBoost retention, KS/TV, C2ST CatBoost AUC, and distance MIA AUC). The corrected simultaneous intervals are unbounded.

TabSyn minus	n	Median CB difference	Simultaneous median CI	Holm p
ARF/default	8	0.010	Unbounded	1.000
ARF/native	8	0.159	Unbounded	0.469
Chow–Liu/default	8	0.157	Unbounded	1.000
Chow–Liu/native	8	0.149	Unbounded	0.188
Copula/default	8	0.242	Unbounded	1.000
Copula/native	8	0.245	Unbounded	1.000

Table 40: **DOPE–ARF legacy matched receipt at $4n$: exploratory paired differences with conservative simultaneous median intervals.** The denominator is the shared-lineage intersection for that auditor. Holm family size is six: DOPE against default and native ARF under each of the three auditors. Median-of-seed aggregation.

ARF config.	Auditor	n	DOPE minus ARF	Simultaneous median CI	Holm p
Default	CatBoost	97	0.077	[0.019, 0.154]	1.31×10^{-5}
Native	CatBoost	97	0.095	[0.028, 0.175]	5.47×10^{-7}
Default	Linear	92	0.027	[0.0056, 0.090]	8.53×10^{-5}
Native	Linear	92	0.037	[0.00582, 0.113]	1.62×10^{-5}
Default	MLP	75	0.115	[0.043, 0.236]	4.56×10^{-8}
Native	MLP	75	0.143	[0.040, 0.253]	8.01×10^{-8}

Table 41: **First-batch diagnostics, one row per physical job.** Individual evaluations without a population interval. No Holm family.

Job prefix	Method	Pearson diff.	Alpha	Beta	Distance MIA AUC
0002af084a92	Independent marginals	0.805	0.602	0	0.474
000377f02a18	DOPE	0.071	0.836	0.558	0.504
000f66aa8642	DOPE	0.266	0.876	0.187	0.595
001897658278	Chow-Liu	0.194	0.804	0.450	0.632
0021656cfc57	Forest-Flow	0.046	0.780	0.555	0.614
00514b7a27e2	DOPE	0.143	0.875	0.239	0.517
0056ef8c748b	Independent marginals	0.149	0.696	0.361	0.637
006a72169115	Gaussian copula	0.064	0.878	0.383	0.452
006cafa72e1a	DOPE	0.069	0.838	0.725	0.710
006dfe86527f	Independent marginals	0.072	0.805	0.491	0.478
007f4ac594ee	TVAE	0.134	0.944	0.523	0.615
0090be69a85e	Gaussian copula	0.098	0.867	0.480	0.393

Table 42: **BeyondArena families that prepared.** The denominator is the 12-family fit check, distinct from the 142-family inventory. No retention auditor and no Holm family.

Family	Fit rows	2,048 B	8,192 B	Val 2,048	Val 8,192
airfoil	802	804	818	0.010	0.00448
garments	939	2,317	2,372	0.028	0.025
musk	3,738	9,273	9,350	0.146	0.090
telemonitoring	257	1,852	1,890	0.033	0.019
video	35,918	5,054	5,071	0.00247	0.00195

Table 46: **Generator families and how DOPE differs.** Representative work for each family, what that family fits, and the line on which DOPE differs. DOPE implements none of these families' objectives, makes no formal privacy claim, and uses no pretrained prior. No Holm family.

Family	Representative work	What it fits	How DOPE differs
Copula	Patki et al. (2016); pair copulas (Aas et al., 2009)	A joint density of fixed parametric form	Adds an explicit model of the target's conditional mean and charges every byte against a cap
Bayesian network, Chow–Liu	Chow & Liu (1968); PrivBayes (Zhang et al., 2014)	A tree or graph factorization of the joint	Same difference: the target mean is stored, not implied by the factorization
GAN	CTGAN (Xu et al., 2019); TableGAN (Park et al., 2018); CTAB-GAN, CTAB-GAN+ (Zhao et al., 2021; 2024)	Adversarial agreement with the joint table	No adversary; weights orders of magnitude smaller
VAE	Kingma & Welling (2014); TVAE (Xu et al., 2019)	Reconstruction plus a latent regularizer	No latent decoder; a twelve-input residual for the target
Diffusion, score	TabDDPM (Kotelnikov et al., 2023); TabSyn (Zhang et al., 2024); STaSy (Kim et al., 2023); CoDi (Lee et al., 2023); TabDiff (Shi et al., 2025); CDTD (Mueller et al., 2025); GATD (Zagardo, 2026)	A denoising or score objective over the joint table	Optimizes the target's conditional mean directly
Language model	GReaT (Borisov et al., 2023); REaLTabFormer (Solatorio & Dupriez, 2023)	Serialized-row likelihood on a pretrained backbone	Fit on the lineage alone; no pretrained backbone
Forest	ARF (Watson et al., 2023); Forest-Flow (Jolicoeur-Martineau et al., 2024)	Axis-aligned splits or forest flows of the joint	A byte-capped program instead of stored forests
CART, synthpop	Nowok et al. (2016)	Sequential conditional draws that store the trees	Stores a screened residual of fixed width
PFN generator	TabPFGen (Ma et al., 2023); TabPFN (Hollmann et al., 2025)	A prior pretrained on other tables	No pretraining; no external checkpoint, since all sampling bytes are charged
DP synthesis	MST (McKenna et al., 2021); AIM (McKenna et al., 2022); PrivMRF (Cai et al., 2021); PrivBayes (Zhang et al., 2014)	Synthesis under a formal privacy budget	No formal privacy guarantee; privacy evidence is empirical, and the release gate is the MFS contract
Other	GOGGLE (Liu et al., 2023); SMOTE (Chawla et al., 2002); probabilistic programs (Saad et al., 2019; Mansinghka et al., 2016); survey (Borisov et al., 2024)	Graph generators, neighbour interpolation, posterior inference over the joint	A target-objective program judged by learners trained on its samples